STRUCTURE-FUNCTION STUDIES OF AN RTX-ADHESIN FROM *VIBRIO VULNIFICUS* AND ITS ROLE IN BINDING HOST CELL SURFACES

By

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A thesis submitted to the Graduate Program
in Biomedical and Molecular Sciences
in conformity with the requirements for the
Degree of Master of Science

Queen's University
Kingston, Ontario, Canada
August 2026

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Abstract

Numerous Gram-negative pathogenic bacteria employ repeat-in-toxin-adhesins to colonize a niche and initiate biofilm formation. Surface adhesion can occur via specific ligand-binding domains (LBDs) located at the C-terminus of these long, fibrillar adhesin proteins. *Vibrio vulnificus* long adhesin protein 1d (*Vv*1d) is one such RTX-adhesin from *Vibrio vulnificus* (*Vv*), the leading causative agent of seafood-related deaths in North America. LoRe bioinformatic pipeline analysis of *Vv* identified *Vv*1d gene, and subsequent AlphaFold3 modeling of the 0.7 MDa adhesin, indicated 3 unique LBDs at its distal end. One such LBD is a putative calcium-dependent carbohydrate-binding module (CBM). When expressed recombinantly in *E. coli* along with its attached split domain, this CBM demonstrated a strong affinity for dextran-based size exclusion chromatography resin and could be eluted with EDTA-based calcium chelation or competitive binding by free sugars. Competition by N-acetylglucosamine (GlcNAc) appeared to be most effective. Multiple biophysical assays indicated that *Vv*1d CBM bound GlcNAc with low micromolar affinity. Various glycan binding arrays and oligosaccharide binding assays confirmed specificity to oligosaccharides containing terminal GlcNAc and fucose moieties, like chitin and mammalian mucins. Functionally, the recombinant CBM mediated yeast aggregation, bound mammalian endothelial and epithelial cell lines, and bound erythrocytes, causing lysis of the latter. All binding and lytic effects were inhibited by addition of GlcNAc and glycomimetic GlcNAc derivatives. X-ray crystallography outlined the calcium-mediated sugar-binding interaction between GlcNAc ligand and the conserved CBM D*cis*D binding motif. The crystallographic analysis of the CBM-GlcNAc complex and comprehensive characterization of glycan-binding specificity provides a framework for future development of targeted, glycomimetic-based, anti-adhesion strategies against *Vv* and other pathogens using similar binding mechanisms.

Co-authorship & Statement of Collaboration

Excluding the following tasks, all experimental work in this thesis was performed by Mr. Blake P. Soares (BPS) under the guidance of Dr. Peter L. Davies (PLD) in the Department of Biomedical and Molecular Sciences at Queen's University in Kingston, Ontario, Canada. Materials, reagents and equipment were provided by PLD with funding from GlycoNet and CIHR.

Chapter 3.1: Mr. Thomas Hansen and Dr. Laurie Graham (LG) created the bioinformatics pipeline used to identify *Vv*1d and other *Vibrio vulnificus* adhesins. LG additionally analysed and created the initial *Vv* RTX adhesin domain identities and maps, as well as assisting in synthetic creation and verification of expression vectors containing *Vv*1d CBM genes of interest. Content from this chapter section have been deposited to bioRxiv (2026) (https://doi.org/10.1101/2025.09.30.679566).

Chapter 3.2: GlcNAc derivative compounds were formulated and synthesized by Dr. Mark Nitz, Ms. Patricia Stuart and Ms. Juliana Matthews at the Department of Chemistry at the University of Toronto. BPS performed dialysis of *Vv*1d CBM into ITC buffer, while Mr. Matthew Fishman performed with all isothermal titration calorimetry assay runs and performed data processing.

Chapter 3.3: Dr. John Allingham (JA) shared his home source diffractometer for in-house crystal screening and native crystal dataset collection. Additionally, Mr. Mathias Bell and Mr. Kody Klupt performed all Canadian Light Source synchrotron data collection. Synchrotron data processing and crystal structure refinement was performed jointly by BPS and JA.

Chapter 3.4: Dr. Inka Brockhausen and Dr. Corey Stevens collected, purified and shared samples of Ovine Submaxillary Mucin and Human Mucin 2. Mr. Robert Eves (RE) created mucin binding assay protocol and assisted with data collection and processing. Ms. Trina

Dykstra-Macpherson developed the polysaccharide dot blot assay protocol and jointly performed the polysaccharide solubilization and modified dot blot with BPS.

Chapter 3.5: JA provided the fungal strains used for fluorescence microscopy. Dr. Shuaiqi Guo and Mingyu Wang developed the MaxiSorp cell binding assay protocol that was jointly performed with RE and BPS.

Statement of Originality and Disclosure of Use of Artificial Intelligence

In the preparation of this thesis, Large Language Models were used to modify and generate Python-based code for visualization of large glycan array datasets into readable figures via Google Colab. Output code for specific figures was compared to previously generated R scripts to ensure accuracy of calculations and data organization. Final images were created after running the verified code with a collected dataset in Google Colab, which outputted specific image files. No figures were generated by the LLM directly. Additionally, AlphaFold3, an artificial intelligence-based tool, was employed to predict protein structures.

Acknowledgements

First, I would like to extend my deepest gratitude to my supervisor Peter L. Davies for his constant support, gracious patience and unwavering belief in my abilities throughout my time in his lab. I am extremely lucky to have had a supervisor who not only cares deeply about the science his students explore, but the students themselves. I will forever be grateful for the lessons he has taught me that I can carry for the rest of my life. Thank you as well to my advisory committee members Dr. Graeme Howe and Dr. John Allingham for their diligent guidance throughout this process.

Secondly, this project would not have been possible without the help of so many amazing, kind, and patient individuals. Each of them greatly contributed their knowledge and time, but also fostered a friendly and supportive research environment. Thank you to Robert Eves, the powerhouse of our lab, who patiently taught me as a naïve fourth year, answered hundreds of my questions, and welcomed me to the lab with open arms. There is no better research assistant than you. Thank you to Dr. John Allingham, who taught me X-ray crystallography from the ground up with the utmost patience, spent so much time assisting my research, and always showed great hospitality and kindness to me. Thank you to Dr. Laurie Graham, who is a whiz with bioinformatics and synthetic biology, and whose general scientific knowledge is astounding. Thank you to Mathias Bell, who was my undergrad lab teaching partner, is a great board game host and trivia captain, and great friend who was always there to talk to when lab work was frustrating or I forgot a part of some experiment.

Thirdly, I have a few specific shout-outs. To all my lab mates who have been with me since I started, Trina-Dykstra-MacPherson, Jocelyn Lee and Thomas Hansen, thank you for being such welcoming and amazing individuals; you have made this experience truly special.

Also, I want to thank my friends from the sixth floor, both past and present, who made my time in graduate studies unforgettable; filled with support and joy like I have never known. So, thank you (in no specific order) to Madeleine Cook, Isabella Martins, Emma Davison, Kody Klupt, Matthew Fishman, Andrea Petkovic and Byron Hunter. I know I have made life-long friends and am excited to see what the future holds in store for you. Thank you to my Mom, Dad and brothers, who were key in making me who I am today, and have always supported me wholeheartedly. And finally, my deepest thank you to Juliana Vicente, my dear, loving partner, whom I have had the profound honour of growing up with and navigating through undergraduate and graduate studies alongside. You are a brilliant woman who has inspired me to be a better scientist while holding me up through every high and low that life can bring. You will always be my idol and my rock, and I could not have made it through this time of my life without you.

Lastly, I acknowledge the support of the Canadian Institute of Health Research and GlycoNet for funding this project. Also, I acknowledge access to invaluable equipment provided by the National Center for Functional Glycomics at the Beth Israel Deaconess Medical Center, the GlycoNet Integrated Services facility at the University of Alberta, the Canadian Macromolecular Crystallography Facility at the University of Saskatchewan Canadian Light Source, and the Protein Function Discovery Facility at Queen's University. Thank you to Dr. Prameet Sheth, Dr. Mark Ormiston, Dr. Louise Winn, Dr. Inka Brockhausen, Dr. Corey Stevens for the Caco-2 cells, THAEC cells, RTg-W1 cells, ovine submaxillary mucin, and human intestinal mucin 2, respectively.

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List of Abbreviations

AF3: AlphaFold3

At: *Acetivibrio thermocellus*

BSA: Bovine-serum albumin

BIg: Bacterial immunoglobulin-like

CBM: Carbohydrate-binding module

Caco-2: Colorectal adenocarcinoma

Cj: *Cellvibrio japonicus*

EPS: Extracellular polymeric substance

FC: Fold change

FITC: Fluorescein isothiocyanate

GlcNAc: N-acetylglucosamine

6xHis: hexahistidine

IPTG: Isopropyl β-D-1-thiogalactopyranoside

ITC: Isothermal titration calorimetry

K_d: dissociation constant

KRT: Krebs-Ringer-Tris

LBD: Ligand-binding domain

LBR: Ligand-binding region

LiGA: Liquid Glycan array

LoRe: Long Read

MFP: Membrane fusion protein

*Mn*Lap: *Marinomonas nauticus* long adhesion protein

*Mp*IBP: *Marinomonas primoryensis* Ice-binding protein

MW: Molecular weight

NCFG: National Center for Functional Glycomics

NF: Necrotising fasciitis

Ni-NTA: Nickel-nitrilotriacetate

OMP: Outer membrane protein

PA14: Protective antigen 14

PACT: Polyethylene Glycol, Anion, Cation, Tris/pH

pLDDT: Predicted local distance difference test

pAE: Predicted alignment error

PBD: Peptide-binding domain

RBC: Red blood cell

RFU: Relative fluorescence unit

RTX: Repeats-in-Toxins

RTX-DUF: Repeat-in-Toxins Domain of Unknown function

RTg: Rainbow Trout gill

SDS-PAGE: Sodium dodecyl sulfate polyacrylamide gel electrophoresis

TBS-T: Tris-buffered saline with Tween-20

T1SS: Type I secretion system

THAEC: Telomerase-immortalized human aortic endothelial cells

Tm: *Thermotoga maritima*

Vv: *Vibrio vulnificus*

*Vv*1d: *Vibrio vulnificus* adhesin protein 1d

vWFA: von Willebrand factor A-lik

Chapter 1
General Introduction

1.1 Bacteria need to stick together to survive

Bacteria have inhabited our planet for billions of years. During this time, they have been able to diversify and evolve to survive in almost every environment on Earth (1). They can live at enormous pressures and high temperatures in deep ocean thermal vents, in glaciers and sea ice where there is limited free water, at extremes of pH in alkaline lakes and acidic hot springs, and in the anaerobic gastrointestinal tract of mammals bathed in digestive enzymes (2, 3). Another example of bacterial adaptability is the ability of many species to exist in two states: a planktonic, wandering state; or attached to surfaces in large cell aggregates, the latter being a preferred state of existence for the majority of their natural life cycle (4, 5). During this cycle, they can encounter a multitude of attachment disrupting events, such as hydrodynamic flow and shear forces that can displace the bacterium, lowering its survival chances. To avoid this and ensure surface attachment is non-random with prolonged retention time, bacteria produce a vast array of large, adhesive surface proteins. These proteinaceous adhesive molecules facilitate binding to biotic or abiotic surfaces in the bacterial environment. Effective adhesion can lead to subsequent growth, cell division and colonization of a local niche, referred to as a biofilm (6).

In biomedical research, adhesin proteins are described as putative virulence factors that pathogenic bacteria use to recognize distinct cell-surface moieties, enabling the organism to infect target host cells and tissues. For example, uropathogenic *E. coli*, a Gram-negative, mixed-mode transmission pathogen that causes urinary tract infections, produces type I pili containing the terminal fimbrial operon H (FimH) adhesin to mediate bacterial attachment to mannosylated

uroplakin 1 receptors on urinary tract epithelial cells (7). Similarly, the zoonotic, Gram-negative pathogen *Vibrio cholerae* (*Vc*) also uses an adhesin protein, Flagellar Regulated Hemagglutinin A (FrhA), to target fucosylated glycans on various host cells in both aquatic organisms and humans (8). After attachment and subsequent colonization, the bacteria can release a myriad of virulence factors, such as hemolysins, to propagate damage and disease to host cells and extract nutrients. Many of these toxins are released as soluble molecules or injected directly into host cells, thus requiring close proximity to host cell surfaces to be most effective (9, 10). Adherent bacteria gain additional advantages from producing and sticking to large, mucus-like biomasses called biofilms that protect from predation and confer powerful resistance to physical removal by shear forces, attack by innate host immune defenses, and even antibiotic penetration (11).

1.2 The bacterial biofilm lifecycle

For most bacterial pathogens, formation of a biofilm on host cell surfaces occurs directly after initial colonization and largely contributes to their virulence. Biofilms are defined as large aggregates of microbial cells encased by extracellular polymeric substances (EPS), where cells can become irreversibly adhered to host surfaces and change their gene expression profile and down-regulate their growth rates (12). The formation of these large biomasses enables bacteria to be protected from most environmental stressors while providing a key, hydrated environment to derive nutrients and exchange genetic material through horizontal gene transfer (13–15). Compared to single, free-floating cells, large biofilm aggregates assemble large microbial communities that confer a more stress-adapted, biofilm-associated expression profile with a marked increase in virulence factor production (16, 17).

The development and dissolution of bacterial biofilms progress through four specific stages and occurs in a cyclic fashion. These phases include initial reversible attachment, irreversible

attachment, maturation and dispersion (Fig. 1.1A.I-IV) (18, 19). Bacteria use their adhesive surface molecules to reversibly attach via one pole to a surface. Once the bacterium recognizes that this linkage is stable through physiochemical detection, it then irreversibly attaches using its longitudinal axis. This stage includes transitioning into a biofilm-associated state, where expression of motile proteins and capsular polysaccharides are downregulated and expression of adhesins, extracellular polysaccharides and antibiotic resistance genes is upregulated (20). Once irreversibly attached, cells can divide, grow and produce an organized, multicellular superstructure where microcolonies are interspersed with solvent channels (21, 22). As more incoming, free-floating cells adhere to the growing biofilm, the mass grows larger in three-dimensions, accumulating more toxins, nutrients and signalling molecules (23, 24).

However, the growth of the matrix eventually carries these key nutrients away from the bulk of basal cells attached to a host cell surface, stratifying the biofilm and creating subpopulations (25). Most biofilm matrices consist largely of hydrated EPS rather than dense packing of single cells (12). Water in the EPS constitutes as much as 95% of biofilms, acting as a hydrated solvent in which structural components like polysaccharides, proteins and nucleic acid molecules, inhabit (26, 27). These components do not only fill space between the microbes, but can also provide attachment points for new, incoming free-floating cells. Additionally, it has been demonstrated that specific solvent channels are commonly formed between these elements to direct the transport of nutrients, or waste products like antibiotics, in or out of the matrix (28). Once the biofilm reaches critical mass, where certain locations are devoid of sufficient sources of energy, dispersion occurs (29). This dispersion is mediated by a combination of upregulated expression of motility proteins and EPS degrading enzymes or physical factors like shear forces from fluid flow (30, 31). This final step is key to propagating new, free-floating bacteria to find

new surfaces to colonize and restart the cycle (32). Biofilm development and maturation is controlled by complex, multi-faceted pathways and quorum sensing. However, elevated or decreased levels of cyclic-di-GMP is the most significant regulator (33, 34), acting as a molecular switch between sessile and biofilm-associated expression profiles.

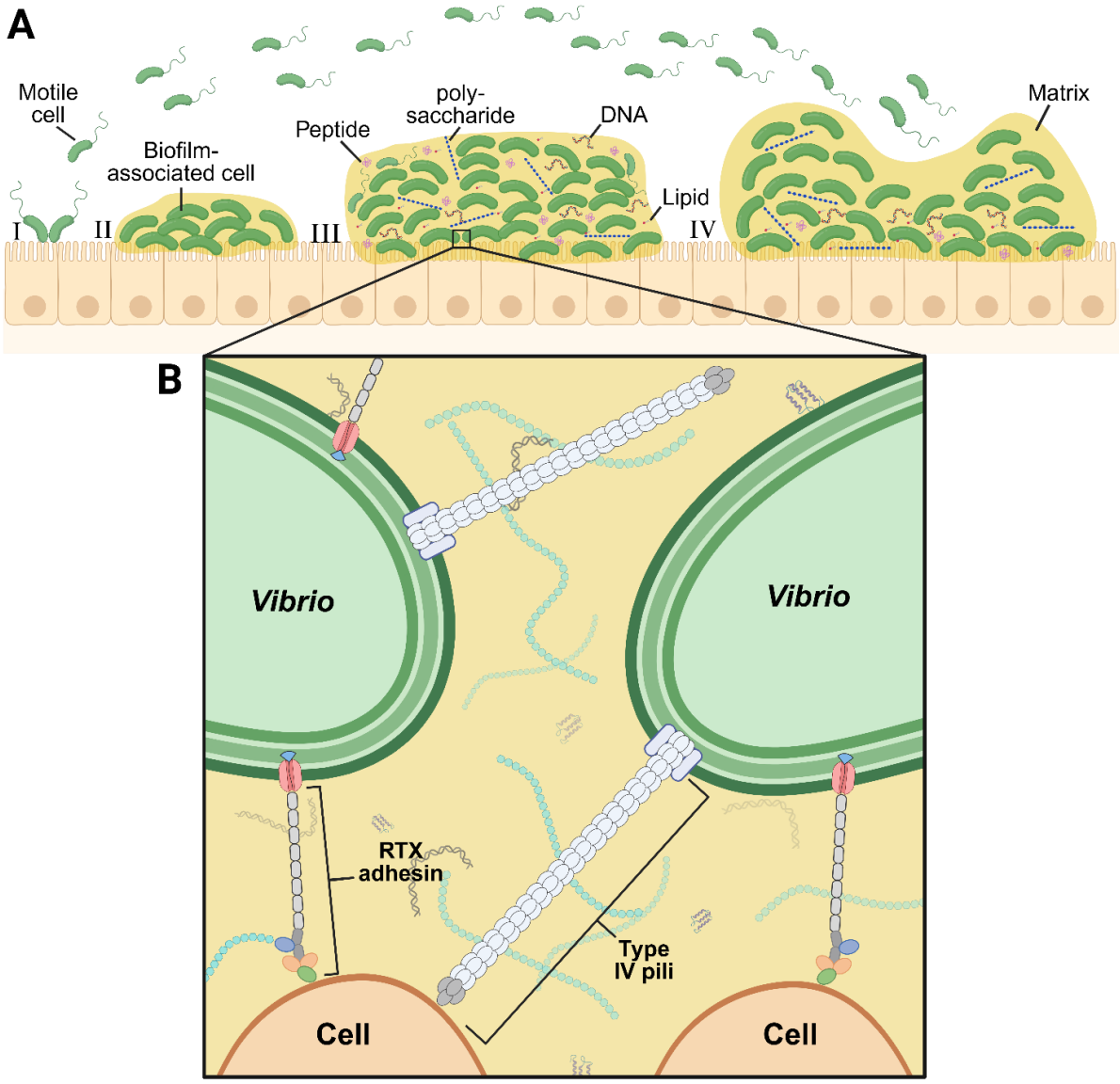


Figure 1.1 Bacterial Biofilm lifecycle. Illustration of the four typical steps in *Vibrio* biofilm formation on intestinal epithelial cells. A.I) Attachment of the bacterial pathogen mediated by surface adhesin interacting with substrates present on host cell surfaces. A.II) Growth of biofilm matrix and initiation of a biofilm-associated expression profile where motile proteins disappear. A.III) Maturation of biofilm and accumulation various of secreted or host extracted vital to bacterial survival and matrix structure. A.IV) Dispersion of nutrient-starved bacteria through autolysis of adhesins before subsequent initiation of free-floating expression profiles include motile proteins and adhesive molecules. B) Inset diagram of *Vibrio* cells using adhesive proteins like Type IV Pili and RTX adhesins to interact with enterocyte microvili surfaces. Figure designed in BioRender.

1.3 Bacterial fibrillar adhesins mediate surface attachment and biofilm formation

Bacteria employ a wide-variety of surface expressed molecules that enable them to interact with their surrounding environment. Many of these appendages can be classified into specific types of proteinaceous structures that allow for DNA transfer, movement or adhesive functions (35–37). Adhesin proteins expressed by bacterial species can be split into two distinct groups (fimbrial and fibriliar) and a myriad of different subgroups depending on their overall fold, surface attachment type and general function. However, all bacterial adhesin proteins follow a common rod-like architecture, with a large, arm-like extender region consisting of repeat units, followed by a terminal adhesive hand-like unit which interacts with surface ligands (38) (Fig. 1.2). The more widely-studied fimbrial adhesins are important adhesive proteins, with a quaternary structure composed of many separately folded subunits which are non-covalently assembled through donor-strand exchange during cell-surface transport (39) (Fig. 1.2A). Adhesins like the Type 3 fimbriae from *Klebsiella pneumoniae* (40) and Type 1 FimH fimbriae from *Escherichia coli* (41, 42) are well-characterized fimbriae that mediate both initial surface attachment and biofilm formation and maturation (43). Oppositely, the recently defined fibrillar adhesins are massive, continuous polypeptide chains that, once surface displayed, fold into discrete repetitive extender and adhesive domains that confer adhesive abilities (44).

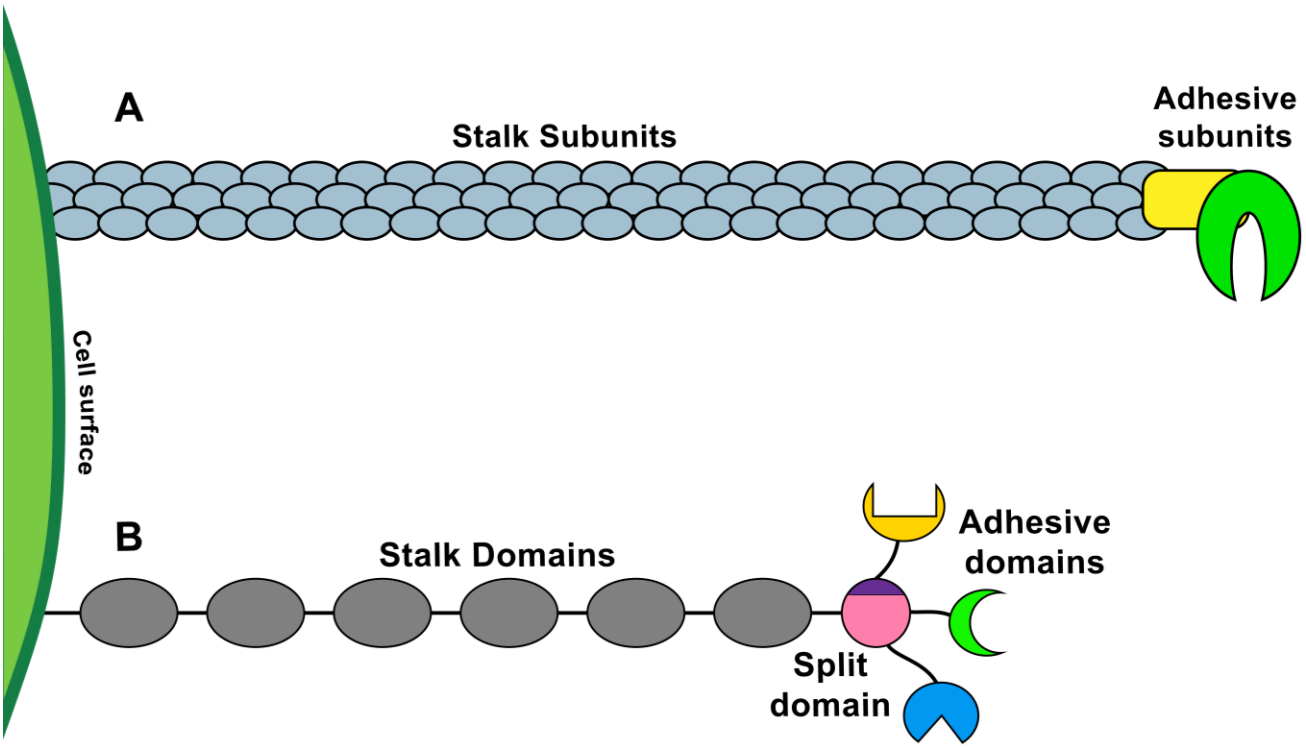


Figure 1.2 Overview of Gram-negative bacterial adhesin groups architectures. A) Diagram of a fimbrial adhesin made up of repeating oligomers of stalk protein subunits (light grey) connected to a distal head region that contains adhesive protein subunits (yellow and green). **B**) Diagram of a fibrillar adhesion, with one continuous polypeptide that discretely fold into specific stalk domains (grey) and into split domains (pink and purple) or adhesive domains (yellow, blue and green) at the distal end. All adhesins retained on the cell surface.

There are different subfamilies of fibrillar adhesins, like the trimeric autotransporter adhesins and biofilm-associated adhesins, which have been shown to interact with ligands on host cell-surface to mediate initial surface colonization and biofilm formation (45, 46). An understudied family, identified through work on the LapA adhesin from *Pseudomonas fluorescens*, is denoted as Repeat-in-toxins (RTX) adhesins (47, 48). These fibrillar proteins have been identified in Gram-negative bacteria and are expressed as large, continuous polypeptides excreted and anchored to bacterial outer membrane surfaces in a different way than other

families. RTX adhesins have been seen to contribute to surface attachment during biofilm formation by Gram-negative bacteria, like *Pseudomonas aeruginosa*, as deletion of their genes from certain bacterial genomes results in decreased virulence and host binding (49, 50).

1.4 RTX-adhesin secretion, retention, domain architecture and binding functions

RTX-adhesins are commonly the largest or second largest proteins encoded by bacterial genomes, and typically share a common domain architecture. The extended polypeptides are typically unfolded in low Ca^{2+}-level cytoplasm, and are directed to the cell surface using a type 1 secretion system signal (T1SS) peptide (Fig. 1.3). Assembly of this transportation system requires formation of an oligomeric complex formed between an ABC transporter (HylB), membrane fusion or adapter proteins (HylD) and an outer membrane protein (OMP). Once the polypeptide passes through the inner membrane translocase complex (HylB/D) and outer membrane protein TolC, rapid folding in the high millimolar Ca^{2+}-containing extracellular medium is nucleated by the Repeat-in-toxins nonapeptide repeats (GGxGxDxUx, where U is any large, hydrophobic amino acid) in the C-terminal beta-roll (51–53). Subsequent folding is transmitted through the rest of the adhesin domains towards the N terminus (54). To be retained at the bacterial surface, a calcium-independent N-terminal retention domain is folded in the cytoplasm, plugging the outer membrane protein TolC and anchoring the adhesin to the periplasmic leaflet of the outer membrane (55–57). The fully folded and retained adhesin consists of a large extender region and a distal ligand-binding region (LBR) which enables binding of the bacterium to environmental surfaces.

The extender region of RTX adhesins constitutes the majority of the protein's mass, consisting of dozens of repeat bacterial immunoglobulin-like (BIg) domains that extend the LBR away from the congested bacterial cell surface (Fig 1.2). Moreover, this region provides some

backbone to the adhesin through Ca^{2+}-binding sites that stiffen interdomain flexible linkers, enabling the extended structures to span such large distances without folding back on themselves. These calcium-binding sites also provide additional strength to the overall fold of each extender domain, increasing resistance to unfolding when in contact with shear forces caused by fluid flow (54, 58). The variation in the number of repeat extender domains seen in various RTX adhesin genes expressed by different bacterial species is not yet understood. However, literature suggests RTX adhesins may have longer or shorter extender regions to allow unhindered binding interactions dependent on the size variation of other species-specific surfaces or surface-expressed molecules (59, 60).

The LBR of the RTX-adhesins is found at this distal C-terminal end and can house a variety of ligand-binding domains (LBDs). Previous research on these adhesins has identified such LBDs as ice-binding domains, peptide-binding domains, von Willebrand factor A-like (vWFA) domains, and carbohydrate-binding modules (CBMs) (Fig. 1.3), with many folds of unknown function that have yet to be characterized (61, 62). A modular combination of these putative LBDs likely allows for surface-specific ligand adherence. However, carbohydrate-binding modules are commonly identified in the LBRs of RTX adhesins. CBMs from environmental or pathogenic bacterial adhesins such as *Marinomonas primoryensis* ice-binding protein*, Marinobacter nauticus* long adhesin protein, *Vibrio cholerae* FrhA and *Aeromonas hydrophila* long adhesin protein have been shown to enable bacteria to bind to various glycans expressed on host cell surfaces and biofilms (8, 59, 63, 64). Moreover, the ubiquity of glycosylation across cell surfaces from all domains of life (65–67) rationalizes how CBMs are key features of adhesin functionality, indicating their importance in binding bacteria to host cells.

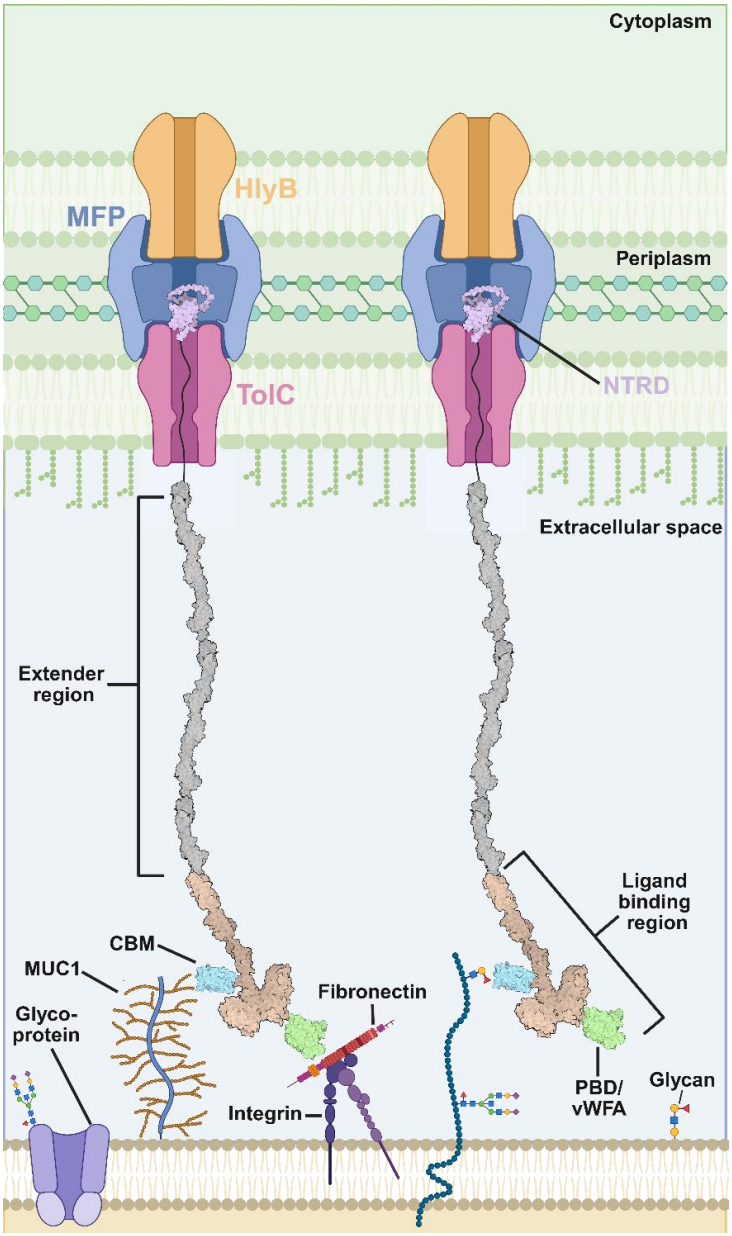


Figure 1.3 RTX adhesin-mediated cell surface binding. Schematic of RTX adhesin bacterial cell surface display, membrane retention and enterocyte surface ligand binding. Components of the Type 1 secretion system (HylA in orange, membrane fusion protein in dark blue, and TolC in pink) combine when an RTX adhesin C-terminal secretion signal is recognized. The unfolded adhesin is translocated through the assembled T1SS, and retained by the N-terminal retention domain (NTRD, in lilac) plugging the HylB pore. The ligand-binding region is projected away from the bacterial cell surface by repeat extender BIg domains (grey) in the extender region. Specialized ligand-binding domains like the carbohydrate binding module (CBM, in light blue), peptide-binding domain or von Willebrand factor A-type domain (vWFA, in green) interact with cell surface glycans, glycoproteins, Mucins (MUC1) or extracellular matrix proteins. Figure designed with BioRender.

1.5 Functionally variable carbohydrate-binding modules are distributed across all kingdoms of life

First discovered in the 20th century, microbial carbohydrate-binding modules can be described as a specific collection of protein domain families that have variable carbohydrate-binding functions. These proteins were first classified as cellulose-binding domains identified as a part of glycosyl hydrolases in fungal species (68), with the term being broadened to carbohydrate-binding modules (CBMs) following the discovery of a myriad of sugar-binding proteins attached to other larger enzymatic structures from different organisms (69, 70). However, CBMs have been mainly identified in prokaryotic and eukaryotic microbial organisms rather than animals or plants. These higher order organisms express similar class of sugar-binding proteins, denoted as Carbohydrate-recognition domains (CRDs). CRDs can have nearly identical functions to CBMs, but are classified differently as they are more commonly attached to non-enzymatic lectins involved in cell-cell recognition, signalling and adhesion (71). However, literature illustrates the blurred lines that separate the stratified organization of sugar-binding protein families, as CRDs and CBMs can possess nearly identical sugar-binding activity and mechanisms, but are simply sorted based on the larger protein units they are attached to. For example, CRDs, like hevein domains, have been classified as both lectins and placed into specific CBM families through multiple sources (72, 73). Interestingly, there are a large variety of sugar-binding proteins that can be described interchangeably with the CBM or CRD moniker, as their fold is distributed across domains of life on both larger enzymatic glycosylhydrolases or lectins. For example, putative PA14 homology domains have been recognized as parts of both glycosyl hydrolase enzymes and yeast flocculant proteins, outlining their equivalent function as both CBMs and CRDs across different orders of life (74, 75).

To date, dozens of CBM families have been described, indexed and verified in literature and on web-based encyclopedias like the Carbohydrate-active enzyme (CAZy) encyclopedia (76). Each family is commonly sorted into three main 'Types' based on their three-dimensional fold and the point of interaction on the target glycan ligand (Fig 1.4A). Type A CBMs have aromatic rich, planar binding surfaces that facilitate binding to multiple discrete sugar moieties at once on crystalline surfaces of polymeric sugar structures like cellulose and chitin. These interactions are mainly stabilized through sugar ring stabilization interactions, with no known divalent cation dependency. This type of interaction is seen with the CBM family 3 domains, like *Acetivibrio thermocellus* (*At*) cellulose-binding domain (CBD), where various aromatic residues create a binding face for crystalline polysaccharides (Fig. 1.4B) (77). Type B, or endo CBMs bind preferentially to four or more sugar units using extended binding pockets or clefts and can contain two subsites where binding occurs, as seen with *Thermotoga maritima* (*Tm*) *CBM27* (Fig 1.4B) (78). Type C CBMs, also designated as exo CBMs, use shallow binding pockets to recognize the reducing or non-reducing ends of their respective glycan targets, normally interacting with three or fewer sugar moieties.

Proteins like vCBM60 from *Cellvibrio japonicus* (*Cj*) are key representatives of Type C CBMs (Fig. 1.4B) (75). However, both Type B and C CBMs can use metal ligands, aromatic, polar and ionizable residues, or a combination of them all to stabilize sugar interactions (79). In RTX-adhesins, CBMs from the conserved protein domain family PA14 are widespread in many different environmental and pathogenic Gram-negative bacterial species (61, 62). PA14 domains appear to function similarly to Type C calcium-dependent CBMs like those seen in CBM family 60, using carboxylates from acidic residues like aspartic acid and a calcium ion to coordinate sugar hydroxyl binding of a terminal glycan moiety (80).

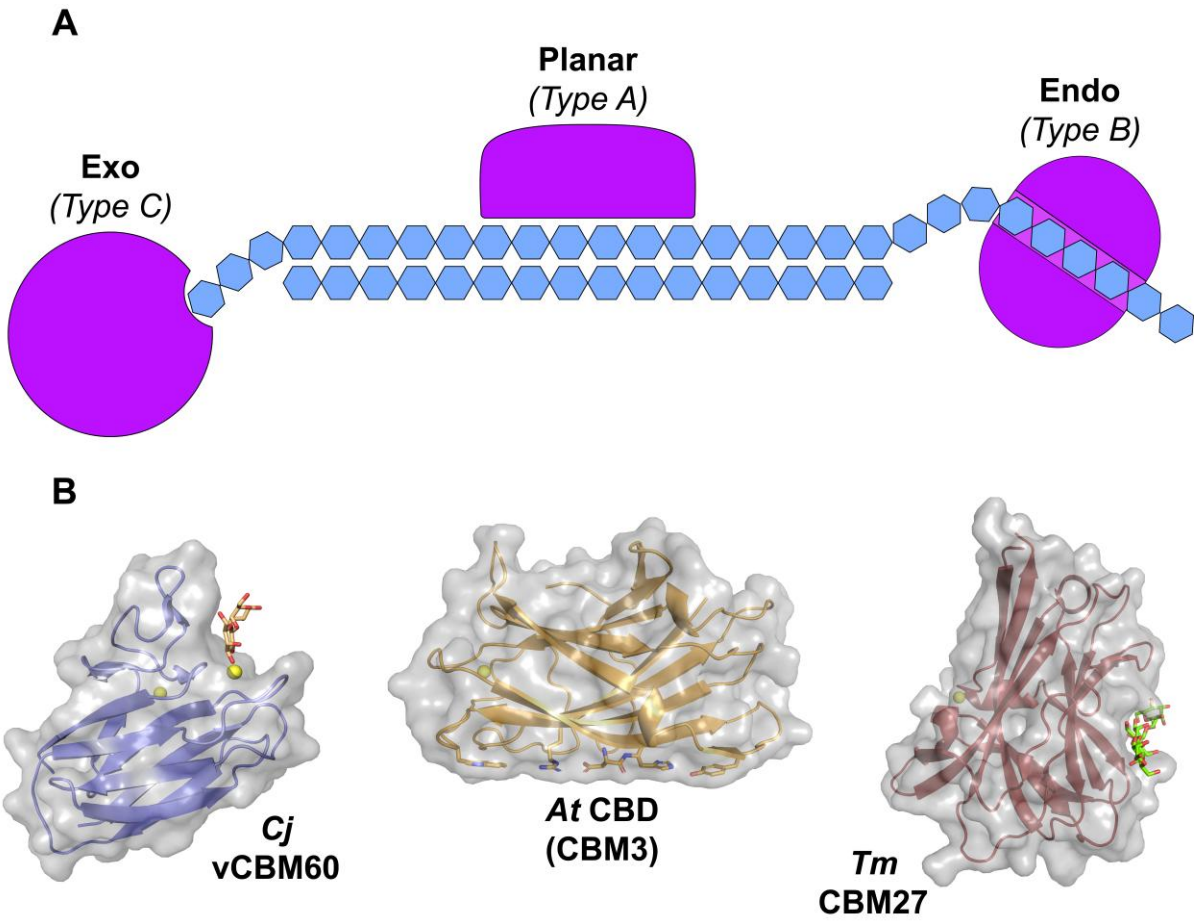


Figure 1.4 CBM type classifications. CBMs are split into three family types dependent on sugar-binding mechanism. **A**) Diagram showing Planar (Type A) Endo (Type B) and Exo (Type C) CBMs (purple) binding to various sections of sugar chains (blue hexagons). **B**) Global folds illustrating a Type C, Type A and Type B CBM (left to right). C*j* vCBM60 (blue) bound to galactobiose (wheat) and calcium (yellow spheres), *At* CBD/CBM3 (orange), and *Tm* CBM27 (red) bound to mannan (green) represented with surface model (grey). Figure adapted from van Bueren, A. *et al* (79).

1.6 PA14 domains are key sugar-binding proteins across many domains of life

PA14 domains, or PA14-homology carbohydrate-binding modules, are sugar-binding proteins that were first identified in the protective antigen of the *Bacillus anthracis* anthrax toxin pro-peptide (81). These PA14 homology domains normally have a conserved all-beta-strand topology that can be detected with hidden Markov models used by databases like InterPro and UniLectin (82, 83). Additionally, structural fold alignments of family members clearly illustrate a

conservation of two core beta sheets that form regardless of extra structural decorations (Fig 1.5A). In the literature and on large classification databases, they have been identified as a part of the functional protective antigen from the anthrax toxin, confer a xyloside-binding function in glycosylhydrolase family enzymes (84, 85), and are connected to yeast adhesins Flo1 and Flo5 that enable recognition of glycan termini on target host cells (86). Additionally, PA14 homology domains are widespread in RTX adhesins from both pathogenic and environmental bacteria, like *Marinomonas nauticus, Marinomonas primoryensis*, and select Vibrio species (62).

To date, PA14 homology domains from the yeast *Candida glabrata* have been crystallized with their preferred sugar ligands lactose or galactose (87), while the fungal and bacterial PA14 homology domains from *Saccharomyces cerevisiae, Komagataella pastoris, M. primoryensis* and *M. nauticus* (formerly *M. hydrocarbonoclasticus*) crystals indicate a preferred interaction with gluco-configured hexoses (63, 80, 88, 89). Sugar ligand-bound crystal structures of *M. nauticus* PA14, *K. pastoris* Flo1, and *S. cerevisiae* Flo5 verify the CBM Type C-like binding mechanism, using two adjacent, cis, carboxylates from aspartate residues to engage in a tri-coordination mechanism with a calcium ion and two hydroxyls from an incoming sugar ligand (Fig 1.5B). Aside from the main binding residues, functional groups from amino acid side chains on nearby loops appear to further stabilize sugar ring structures or other polar contacts.

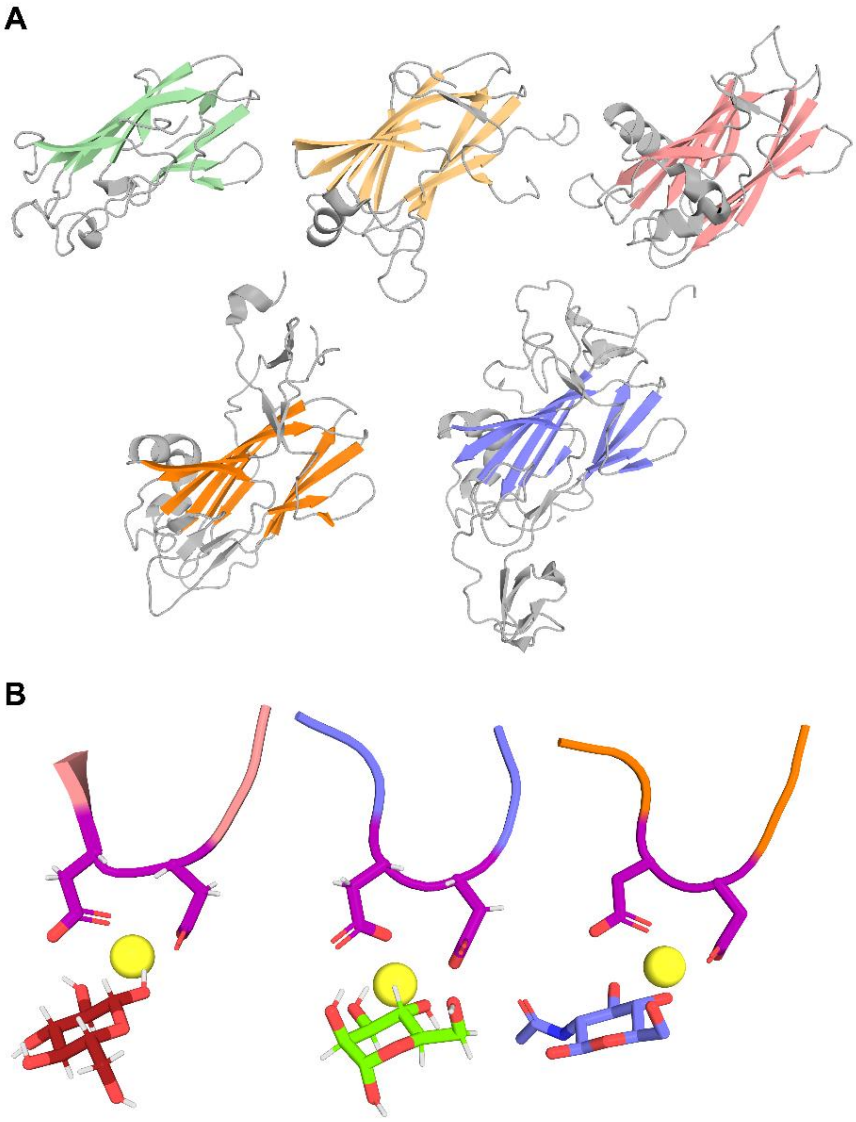


Figure 1.5 Microbial PA14 homology domains and their binding sites. Various PA14 homology structures are shown from different fungal and bacterial species. **A)** Crystal structures outlining the global fold of PA14 domains from *Bacillus anthracis* (PDB: 1T6B, green), *Cellvibrio japonicus* (PDB: 2XVG, wheat), *Marinomonas nauticus* (PDB: 6M8M, salmon), *Komagataella pastoris* (PDB: 5A3L, orange), and *Saccharomyces cerevisiae* (PDB:2XJP, blue). Core strands are coloured with extraneous structural elements coloured in grey. **B)** D*cis*D homolgy motifs (purple) of PA14 domains from *M. nauticus* (salmon), *S. cerevisiae* (blue), *and K. pastoris* (orange) bound to calcium and β-D-glucose, D-mannose and D-GlcNAc, respectively.

Interestingly, bacterial PA14 CBMs show high promiscuity with different hexose and pentose sugar molecules but normally confer high affinity binding to GlcNAc and fucose. The species-

specific variation in preferred ligand suggests that different bacteria who have evolved in different environments may slightly alter PA14 CBM binding pocket structure to best accommodate the target terminal glycan sugar they commonly interact with. Moreover, deletion or sugar-based inhibition of both yeast and bacterial adhesins containing the PA14 CBMs resulted in attenuation of cell binding by the entire microbial cell or recombinant CBM constructs, indicating that they are key to pathogen-host binding function (63, 80, 88).

1.7 *Vibrio vulnificus* infection and adherence dynamics

The opportunistic, zoonotic pathogen *Vibrio vulnificus* is a free-living, halophilic, marine Gram-negative bacterium that causes life-threatening diseases such as septicemia and necrotizing fasciitis as quickly as 24 hours after initial infection, which can be fatal (90). *Vv* has been identified across worldwide water systems and naturally infects various shellfish and finfish (91, 92). Transmission to humans can occur through contact with contaminated water sources, consumption of contaminated seafood or through puncture wounds inflicted by sharp marine animal body components, or even exposing skin abrasions to seawater with high bacterial concentrations (93) (Fig 1.6A). In North America alone, there are over 100 cases of *Vv* infections per annum, with >10% of cases requiring limb amputation and a 50% mortality rate in cases with presentation of septicemia. Global warming has also significantly contributed to a rise in documented *Vv* infection as well as increased geographical expansion to previously unaffected coastal waters with elevated population density (94). While the common treatment for *Vv* is doxycycline mixed with cefotaxime, the increase in infections has produced many virulent isolates that directly evade all human immune defences or harbour multiple antibiotic resistance genes (95). Since the bacterium was first discovered in the 1970s, *Vv* has been categorized into

three different biotypes and further stratified into dozens of strains that each have putative toxin genes and large adhesin protein genes (96).

Like all pathogenic *Vibrionaceae*, *Vv* must adhere to a surface and form a biofilm to successfully colonize a host and propagate a serious infection. In the case of *Vv*, colonization of circulatory system cells or the intestinal epithelium can enable localized and multi-organ infections to occur. Direct access to the bloodstream can be caused by puncture wounds, while ingestion of contaminated food requires the bacterium to evade the harsh, acidic stomach environment before advancement into the intestinal tract. Upon contact with gastric acid, *Vv* signalling pathways induce the expression of lysine decarboxylase CadA and a specific transporter protein (CadB) that work to import lysine from digested peptides, convert it into a cadaverine, and release it into the surrounding environment to block acid from entering through pores (97) (Fig. 1.6B). The lysine-cadaverine conversion also consumes cytoplasmic H^+ ions, stabilizing internal pH during acid stress. Additionally, upregulated manganese superoxidase dismutase works in tandem to reduce oxidative stress (98). This allows *Vv* to survive until gastric emptying, where free-floating cells initiate intestinal tract colonization.

Enteric pathogens like *Vv* have evolved to easily circumvent the established mucosal and epithelial barrier found in the intestinal tract. During initial contact with the intestinal mucosa, *Vv*-specific flagella and capsular polysaccharide production are upregulated to enable the evasion of innate host immune defences and ease of movement through the mucosal matrix that is 10-100-fold thicker than the length of the bacterium (99, 100). Once the bacteria have reached the epithelial surface, *Vv* can start to produce a myriad of toxins to confer cell damage and derive nutrients from their environments. Post-adherence, bacterial cells can induce overexpression of virulence factors, such as the pore forming multi-functional autoprocessing repeats-in-toxin

(MARTX), cytolysin/hemolysin *Vv*hA, elastolytic *Vv*pE, mediated by increased expression of the HlyU transcription factor (101–103). The various exotoxins trigger autophagy, apoptosis, and necrotic cell death to both extract nutrients and weaken cell junctions between intestinal epithelial cells to facilitate bacterial invasion. Then, the bacteria can switch to a biofilm-associated state, expressing genes like cellulose synthase and diguanylate cyclase (VcdR) that cause increased biofilm matrix formation (20, 104). Biofilms are incredibly important for the persistence and successful progression of *Vv*-mediated diseases. Deletion of key biofilm-associated genes like VcdR and BrpR lead to severely decreased biofilm formation (105). These large, protected cell aggregates can accumulate toxin molecules and protect from oxidative stress, causing increased retention and exponential growth of *Vv* populations. This process continues as *Vv* destructively translocate though additional tissue layers until they reach blood vessels in the submucosa, enabling *Vv* to enter the circulatory system and begin to disseminate into other parts of the body (106) (Fig 1.6C).

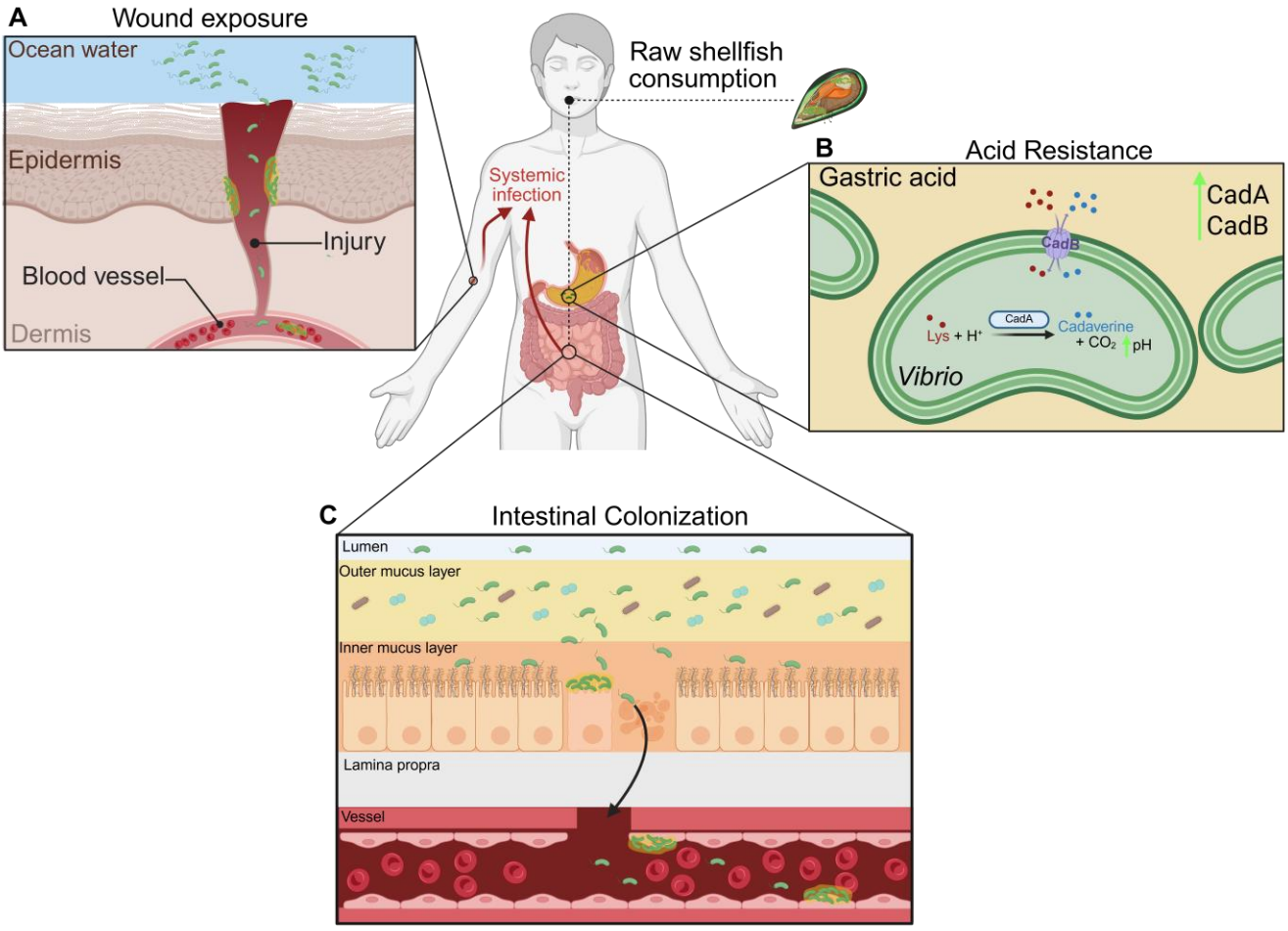


Figure 1.6 General *Vibrio vulnificus* infection mechanism. Diagram of *Vv* transmission routes, resistance mechanisms, and colonization of tissue surfaces with dissemination into the bloodstream and periphery. **A)** Transmission of *Vv* into the bloodstream through an open wound in the dermis connected to a vessel via high bacterial abundance ocean water. **B**) *Vv*-based acid resistance mechanism overview after consumption of raw shellfish. Low pH induces expression of CadA and CadB (purple) resulting in an influx of gastric lysine (Lys, red). This is then converted into cadaverine and CO_2 (blue) by CadA, which uses protons to drive the reaction, raising cytosolic pH. **C**) General overview of intestinal colonization by *Vv* after raw shellfish consumption. *Vv* enter intestinal lumen and penetrates the outer and inner mucus layer to reach epithelial cells. *Vv* release toxins to cause tissue necrosis and death, allow penetration through the lamina propria and others to reach blood vessels. Blood vessel endothelia are colonized and *Vv* is disseminated systemically through the circulatory system. Figure generated in BioRender.

Invasion of the circulatory system is where the mortality rate from *Vv* infections increases to 50%. Once a group of bacteria encounter circulatory system components, like vessel endothelial, red blood and peripheral blood mononuclear cells, they can employ their toxins to produce pores

in cell membranes along with reactive oxygen species that induce cytotoxic damage and necrotic cell death in as little time as 30 minutes (107). This causes the formation of mycotic aneurysms due to rapid breakdown of artery epithelial cells, with blocking of artery flow by dead cell debris. Once arteries rupture, *Vv* can likely further disseminate into soft tissue layers, leading to eventual superficial spread and visible necrotic bullae formation. From this point on, massive septic shock and death can occur (108).

Throughout the entire *Vv* infection mechanism, adhesive, surface-displayed proteins like IlpA, Type IV pili and the mucin-binding protein GbpA must be present to bind to the variety of circulatory system and gastrointestinal tract cell types to ensure bacteria remain attached to their target host surfaces (109). Aside from these well characterized *Vv* adhesive proteins, the bacterium also expresses RTX adhesins, many of which contain carbohydrate-binding modules (62). These adhesins may be a key mediator of bacterial adhesion to the highly glycosylated surfaces of human host cells that enables the progression and persistence of *Vv* related infections.

1.8 Objectives and Hypothesis

Recently, there has been a significant increase in drug-resistant bacteria and a lack of newly developed bacteriostatic or bactericidal antibiotics (110). Thus, the search for alternative strategies like anti-adhesive or biofilm inhibiting drugs is crucial for future global health outcomes. The primary step in most bacterial infections is the initial adhesion of the microbe to either host cell surfaces or the subsequent biofilms they form. Thus, my research investigates the hypothesis that structural and functional characterization of ligand-binding domains used by bacterial adhesins can lead to the development of small molecule mimetic compounds that attenuate attachment to host cell surfaces. The potential for anti-adhesive strategies has been proven by pre-clinical research on glycomimetic antagonistic ligands which blocked binding of

the uropathogenic *E. coli* adhesin FimH (111, 112). Using similar methodology, this could be potentially accomplished for a wide variety of zoonotic pathogens like *Vibrio vulnificus*, allowing for multi-organism infection treatment in greater clinical or aquacultural settings. To accomplish this, pathogenic ligand-binding domains and their respective ligands must be structurally and functionally characterized. The main objectives of this study were to characterize the putative carbohydrate-binding domain within the ligand-binding region of *Vv* long adhesin protein 1d (*Vv*1d) to help develop antagonistic mimetics that may help inhibit bacterial adhesion.

Chapter 2
Materials and Methods

2.1 Bioinformatics and protein modelling

The full length *Vv*1d RTX-adhesin gene was identified from a *Vibrio vulnificus* bacterial genome using the LoRe genome tool (Thomas Hansen, Github). The folds of its 68 domains were predicted via the AlphaFold3 server provided by Google (113). The complete *Vv*1d adhesin model was rendered in Open-source PyMOL software V 3.1.0 (Schrodinger, LLC.) and interpreted to draw a domain map that included putative ligand-binding domains emerging from various split domains.

2.2 Cloning and expression of *Vv*1d constructs

pET-28a (+) expression vectors with genes encoding for *Vv*1d CBM, *Vv*1d vWFA and *Vv*1d RTX-DUF were synthesized by GenScript with codon optimization for expression in *E. coli.* This expression vector enabled N-terminal 6xHis tags to be present on each construct. Protein sequences are found in Appendix Fig. C.

Expression vectors were transformed into BL21 (DE3) *E. coli* cells for IPTG-induced expression. Single colonies of expression vector-containing BL21 (DE3) cells were used to inoculate 25 mL LB broth cultures with 0.1 mg/mL kanamycin and were grown overnight at 37 °C with shaking at 180 RPM. Overnight cultures were used to inoculate 1 L LB broth cultures with the same kanamycin concentration and were grown until an OD_{600} of 0.7 was reached. IPTG was then added to a final concentration of 1mM to induce protein expression. Induced cultures were incubated overnight at 21 °C.

Protein-expressing cultures were then centrifuged in a JS 4.2 rotor (Beckman Coulter) for 30 min at 3300 x g to pellet all bacterial cells. After discarding the supernatant, cell pellets were resuspended in 25 mL of lysis buffer (50 mM Tris-HCl, 150 mM NaCl, 20 mM imidazole) supplemented with protease inhibitor cocktail tablet (Roche). Resuspended cells were then lysed via sonication using a Sonic Dismembrator Model 500 (Fisher Scientific), set for eight rounds of sonication at 70% amplitude for 30 seconds. The sonication probe was cooled for 2 min on ice between each round. The resulting cell lysate was centrifuged at 31000 x g in a JA20 rotor (Beckman Coulter). The lysate supernatant was incubated with HisPur Ni-NTA agarose resin (ThermoFisher) in a beaker with 100 mL of Ni wash buffer (50 mM Tris-HCl, 150 mM NaCl, 20 mM imidazole) for 30 min. Once settled, the protein-bound resin was loaded into a gravity flow column (BioRad) and washed with five column volumes of Ni wash buffer. After the final wash, bound protein was eluted with three aliquots of 10 mL of Ni elution buffer (50 mM Tris-HCl, 150 mM NaCl, 250 mM imidazole). Elution fractions were pooled and dialyzed overnight at 4 °C in Protein Buffer (50 mM HEPES, 150 mM NaCl, 2 mM $CaCl_2$) to induce folding. The dialyzed Ni elution fractions were concentrated six-fold and subjected to size-exclusion chromatography on a HiLoad 16/600 Superdex 75 column (Cytiva). Superdex resin-bound *Vv*1d

CBM was eluted using Protein Buffer supplemented with 5 mM EDTA. Protein purity throughout the purification process was tracked by SDS-PAGE, while protein yield was assessed via UV/Vis spectroscopy. Fractions enriched in the protein of interest were concentrated to 15 mg/mL, aliquoted, flash-frozen, and stored at -80 °C.

2.3 Preparation of fluorescently labelled proteins

To prepare protein for fluorescence microscopy and glycan array analysis, 1-mg aliquots of all constructs were fluorescently-labeled using fluorescein isothiocyanate (FITC) reconstituted in DMSO. To do this, purified protein aliquots were buffer exchanged into a protein buffer containing no free amines (50 mM HEPES, pH 8.0, 150 mM NaCl, 2 mM $CaCl_2$). These aliquots were incubated with 50 µL of 10 mg/mL FITC for 3 h at 4 °C. Unbound salt and FITC-labelled protein were separated on a PD MiniTrap G-10 gravity column (Cytiva).

2.4 Dextran-based competition assay

Dextran-based competition assays performed as per Eves, R. *et al.* (114) with the following modifications: A 300 µL slurry of S200 dextran resin was washed three times with Protein Buffer. After washing, 1 mg of *Vv*1d CBM was loaded onto S200 dextran resin and incubated while nutating for 3 h at 4 °C. After incubation, protein-bound dextran resin was centrifuged at 6600 x g for 3 min in an accuSpin Micro17 Microcentrifuge (Fisher Scientific). To perform incremental elutions, seven tandem 3-µL additions of 0.5 M free saccharide stock solutions were performed, followed by two final 10-µL additions to yield a total of 41 µL or a 20.5 mM final concentration of saccharide applied. Each monosaccharide trial was performed in triplicate, in three different microfuge tubes. An average value of three replicates was used for points plotted with standard deviation calculated via GraphPad Prism 8.0.

2.5 Isothermal titration calorimetry

Isothermal titration calorimetry experiments were conducted using a MicroCal VP-ITC instrument (Malvern) at 30°C in protein buffer (50 mM HEPES pH 8.0, 150 mM NaCl, 2 mM $CaCl_2$). The syringe contained 500 µM GlcNAc which was injected into a calorimetric reaction cell containing 50 µM *Vv*1d CBM. A total of 30 injections of 10 µL increments were taken at 300 s equilibration intervals to obtain ITC data which was performed in duplicate. Thermograms were fit to a one-site binding model using MicroCal Origin 7.0 software.

2.6 Glycan array analyses

A National Center for Functional Glycomics glycan microarray (Version 3.2) was screened with *Vv*Lap*3*CBM by the Consortium for Functional Glycomics (Beth Israel Deaconess Medical Center) at Harvard. The printed glycan array consisted of 155 different biological glycans. The array was incubated with FITC-labeled *Vv*Lap3CBM at 100 µg/mL. The green fluorescence of the labeled protein was used to measure relative fluorescence units (RFU) of glycan-bound protein. Application of sample, scanning, and data analysis were performed as outlined by Luetscher *et al.* and Mehta *et al.* on a fee-for-service basis (115, 116). Additional protocol information is available at https://research.bidmc.org/ncfg/protocols.

A Liquid Glycan Array (LiGA) screening was also performed against immobilized *Vv*1d CBM via Integrated Services provided by GlycoNet at the University of Alberta. A library of 261 unique, barcoded M13 bacteriophages, displaying unique glycans at various densities was used. Different bacteriophages were screened with a varying number of replicates (singlets to nonuplicates). An immobilized Bovine Serum Albumin (BSA) control screening was also

performed to describe fold change in binding of phages to immobilized *Vv*1d CBM. Additionally, eight replicates of bacteriophage displaying no glycans (blank) and fourteen replicates of unreacted azide click linkers (AzOH) to serve as negative control data. Application of sample, bacteriophage library, quantification and data analysis were performed as outlined by Sojitra, M. *et al* (117) on a fee-for-service basis. Final plots of fold change data were created using Python-code based visualization programs Matplotlib (118), NumPy (119) and Pandas (120) in a Google Colab notebook.

2.7 Polysaccharide-binding dot blot assays

All oligosaccharides used and their solubilization conditions are listed in Table 1 of the Appendix, as per literature protocols (121–123) or the manufacturers' recommendation. Rectangular PVDF sample strips were soaked in methanol for 10 min for activation. Activated membranes were then equilibrated in Tris-HCl buffered saline with Tween-20 (TBS-T), which contains 150 mM NaCl, 20 mM Tris-HCl, pH 7.6, 0.02% Tween-20 and were then placed on wet filter paper. Various 10 µL dots of 0.5% w/v oligosaccharides stocks were placed onto outlined spots on the PVDF membrane. Dotted membranes were then dried for 1 h at room temperature. During incubation, dilute *Vv*1d CBM samples (0.01mg/mL) were made up in 5 mL of TBS-T. Dried, dotted membranes were incubated for 15 min with 5 mL of Blocking Buffer (TBS-T supplemented with 3% BSA) at room temperature in a conical 50 mL tube (Falcon). Post-incubation, membranes were washed three times with TBS-T. Dilute *Vv*1d CBM samples were then incubated with blocked membranes for 1 h at room temperature and rinsed with TBS-T three times. Mouse Anti-His-HRP antibody (Cell Signaling Technology) was diluted 1:20000 in 5 mL of TBS-T and incubated with the dotted membrane for 1 h at room temperature, followed by three TBS-T washes. Blotted membranes were then soaked in Immobilon Western

Chemiluminescent HRP Substrate (Millipore) in the absence of light and placed on Radiomat M-Plus Half-Speed Blue X-ray film (Agfa) for a 5-sec exposure. Developed films were imaged by a Cannon EOS 50D camera.

2.8 Mucin binding experiments

Ovine Submaxillary Mucin (OSM) used in this study was donated by Dr. Inka Brockhausen (Queen's University), while human intestinal MUC2 was prepared and donated by Dr. Corey Alfred Stevens (Massachusetts Institute of Technology). Additionally, lab stocks of *Rhagium mordax* antifreeze protein *(Rm*AFP) were used as negative controls. Freeze-dried mucin samples were slowly reconstituted in ddH_2O overnight at 4 °C to a final concentration of 20 mg/mL (2% w/v). Reconstituted mucins were then diluted to 10 mg/mL (1% w/v) in incubation buffer (50 mM Tris-HCl pH 8, 200 mM NaCl, 2 mM $CaCl_2$). For inhibitory samples, 0.1mg/mL *Vv*1d FITC-CBM samples were preincubated with 50 mM GlcNAc for 5 min. Aliquots of *Vv*1d CBM (20 µL of 0.1mg/mL) were then incubated with 20 µL of 10 mg/mL mucin aliquots. After thorough mixing, samples were centrifuged at 17 000 x g for 10 min. Next, the supernatant was removed, and 50 µL of incubation buffer was added to wash the mucin pellet. This wash was repeated two times. After the final wash and centrifugation, the resultant supernatant was removed, and the final mucin pellet was resuspended in 50 µL of incubation buffer for measurement at 520 nm on a fluorimeter (SpectraMAX GeminiXS) to detect bound protein fluorescence.

2.9 Yeast binding experiments

Manipulation and growth of yeast cultures followed common procedures for budding yeast species. All strains were grown and maintained on YPD plates containing 1% yeast extract (BioShop), 2% Peptone (BioShop) and 2% glucose (BioShop). Single yeast colonies were used

to inoculate 10 mL YPD cultures that were subsequently grown at 30 °C overnight. Aliquots of overnight cultures (1 mL) were used to inoculate fresh 10 mL YPD in culture flasks and allowed to grow to mid-logarithmic phase before use in experimentation. Prior to their use, yeast were diluted to 1.0 x 10^7 cells/mL in Tris-buffered saline (TBS, 50 mM Tris-HCl, 150 mM NaCl, pH 7.6). Yeast cells in a 1-mL aliquot were then pelleted and resuspended in 1 mL of TBS containing fluorescently labelled *Vv*1d CBM at 0.1 mg/mL. Each trial was then performed in the presence or absence of 5 mM GlcNAc or functional GlcNAc glycomimetic compounds supplied by Dr. Mark Nitz (University of Toronto). Reactions were incubated at 30 °C overnight and visualized using an Olympus IX83 inverted fluorescence microscope with an Andor Zyla 4.2 plus camera. Fungal strains used are listed in Table 1 of the Appendix and were donated by Dr. John S. Allingham.

2.10 Erythrocyte binding and lysis experiments

Type O whole blood samples were obtained from a human donor (ZenBio) for use in the binding assays. First, 220 µL of the human Type O blood sample was transferred into a 15 mL conical tube, followed by 11 mL of filter-sterilized Krebs-Ringer-Tris (KRT) buffer (120 mM NaCl, 5 mM KCl, 1mM $MgSO_4$, 2 mM $CaCl_2$, 10 mM Tris-HCl, pH 7.6). The solution was then centrifuged at ~210 x g for 10 min. The resultant supernatant was aspirated and discarded, while pelleted erythrocytes were resuspended in 12 mL of KRT for washing, which was repeated three times. After the final wash, the supernatant was aspirated before the addition of 11 mL of KRT to yield a 2% Type O erythrocyte suspension.

For erythrocyte binding experiments, 400 µL of the 2% erythrocyte solution was incubated with 200 µg/mL of fluorescently-labelled *Vv*1d CBM. Additional trials were performed with *Vv*1dCBM in the presence or absence of 5 mM GlcNAc and β-S-GlcNAc. All samples were incubated on a single-speed nutator at 22 °C for 1 h in the absence of light to avoid degradation

of fluorescent labels. Next, protein-bound erythrocytes were pelleted by centrifugation at 800 x g for 3 min in an accuSpin Micro 17 rotor (Fisher Scientific), with the resultant supernatant discarded. The pelleted cells were then washed with 500 µL of KRT buffer, before another centrifugation. This step was repeated three times to capture and discard any unbound proteins or fluorescent tags from the RBC pellet. The final washed pellet was gently resuspended in 400 µL of KRT buffer, with subsequent loading of 10 µL samples onto microscope slide for fluorescence microscopy imaging. Images were obtained using an Olympus IX83 inverted fluorescence microscope with an Andor Zyla 4.2 plus camera.

For hemolysis experiments, various dilutions of *Vv*1d CBM constructs were added to 400 µL of 2% Type O erythrocyte suspensions in KRT buffer. For the positive control, 50 µL of a previously tested, lab stock of hemolytic *Vc*SBD-UKD was added in place of *Vv*1d CBM. For the negative control, 200 µg/mL of lysozyme was added into the blood sample. All samples were incubated at 24 rpm on a nutator at 22 °C for 15 min. For the inhibition sample, 5 mM GlcNAc was added to the highest concentrated protein sample for 5 min before subsequent incubation under the same conditions. After incubation, all samples were centrifuged at 800 x g for 3 min. Absorbance measurements of the resultant supernatant at 414 nm were recorded on a Multiskan GO spectrophotometer (ThermoFisher Scientific) and plotted using Microsoft Excel. Each sample was made in triplicate for data collection. Images of the hemolysis assay samples were taken on an iPhone 13 smartphone.

2.11 MaxiSorp plate cell binding assays

For cell binding plate assays, a variety of cell types were used. The same yeast strains and isolated Type-O erythrocytes from previous sections were re-used for this assay. Additionally, Caco-2 cells were donated by Dr. Prameet M. Sheth, and cultured in Eagle's Minimum Essential

Medium (Gibco) supplemented with 50% fetal bovine serum (FBS). Rainbow Trout gill-W1 (RTg) cell stocks were donated by Dr. Louise M. Winn, and cultured in serum-free Leibovitz's L-15 (Gibco) medium. Telomerase-immortalized human aortic endothelial cells (THAECs) were donated by Dr. Mark L. Ormiston and cultured in Endothelial Cell Basal Medium-2 (Lonza Bioscience) supplemented with EGM-2 endothelial SingleQuots® Kit (Lonza Bioscience).

All MaxiSorp® plate-based cell-binding assays were performed by Robert Eves using the adapted protocol from Dr. Shuaiqi Guo, as described in the thesis of Trina Dykstra-MacPherson, which is available through QSpace (124).

2.12 X-ray crystallography and structure solution of *Vv*1d CBM

The sitting drop vapour-diffusion method was used to screen several different crystallization suites compatible with 2 mM Ca^{2+} (Qiagen). Pure *Vv*1d CBM at 20 mg/mL was mixed with various crystallization solutions (1 µL :1 µL) in 96-well Intelli-plates (Art Robbins Instruments) and left to incubate at 22 °C. The PACT suite solutions F5 (0.2 M $NaNO_3$, 0.1 M Bis-tris propane pH 6.5, 20% w/v PEG 3350), G2 (0.2 M NaBr, 0.1 M Bis-tris propane pH 7.5, 20% w/v PEG 3350), G5 (0.2 M $NaNO_3$, 0.1 M Bis-tris propane pH 7.5, 20% w/v PEG 3350), G6 (0.2 M Na-formate, 0.1 M Bis-tris propane pH 7.5, 20% w/v PEG 3350), and G7 (0.2 M NaAc, 0.1 M Bis-tris propane pH 7.5, 20% w/v PEG 3350) yielded crystals of varying sizes and morphologies. These crystals were soaked in cryoprotectant (three parts synthetic mother liquor: one part 100% ethylene glycol) containing 5 mM GlcNAc for 1-3 min, then screened for proteinaceous diffraction on a home source diffractometer (Rigaku MicroMax – 007HF x-ray source coupled to a R-AXIS IV++ detector) in the lab of Dr. John Allingham. All data was collected at a crystal temperature of 100 K. Crystals that showed diffraction of 3-3.5 Å were

stored in cryoprotectant containing 5 mM GlcNAc, before being analyzed at the Canadian Light Source synchrotron, on beamline 08ID-1 at an X-ray wavelength of 0.95371 Å.

Collected data sets were screened to identify the highest-resolution diffraction dataset for processing. Diffraction images were integrated using AutoXDS, and the data were post-processed with a high-resolution cutoff of 2.28 Å for scaling and merging statistics. The high-resolution limit was selected based on CC½ (0.32 in the highest-resolution shell), following the recommendations of Karplus and Diederichs (125), rather than using a mean I/σ(I) criterion. The structure was solved by molecular replacement in Molrep (126) using a Ca^{2+}-bound, sugar-free PA14 carbohydrate-binding module from *Mp*IBP (PDB: 6X7T) as the search model. The initial model was constructed through iterative manual building in Coot and automated model building with Buccaneer (127). Subsequent refinement was performed using REFMAC5 (128). Crystallographic data collection and refinement statistics were compiled using Phenix Table One (129). The final crystal structure was deposited in the PDB with accession code 36GK.

Chapter 3
Results

3.1 Bioinformatic prediction and AlphaFold modelling of *Vv* RTX-adhesins enables production of stable recombinant constructs

Development of an in-house bioinformatics pipeline, as described in Hansen *et al.* (submitted for peer review) enabled scanning of all open reading frames deposited to the NCBI database to identify those that carry the hallmarks of RTX adhesins. All currently deposited *Vv* genomes on NCBI have two RTX-adhesin gene loci (Fig. 3.1A). Locus 1 is occupied by one of

six RTX genes that differ in their C-terminal ligand-binding regions. Locus 2 is occupied by a gene that encodes an "extender only" RTX-adhesin (*Vv*2) that lacks a C-terminal ligand-binding region. One gene, present at locus 1 in several strains, (NCBI: WP_061058482.1) encodes the full-length, 7749-residue *Vv*1d that is the main subject of this thesis (Fig. 3.1B).

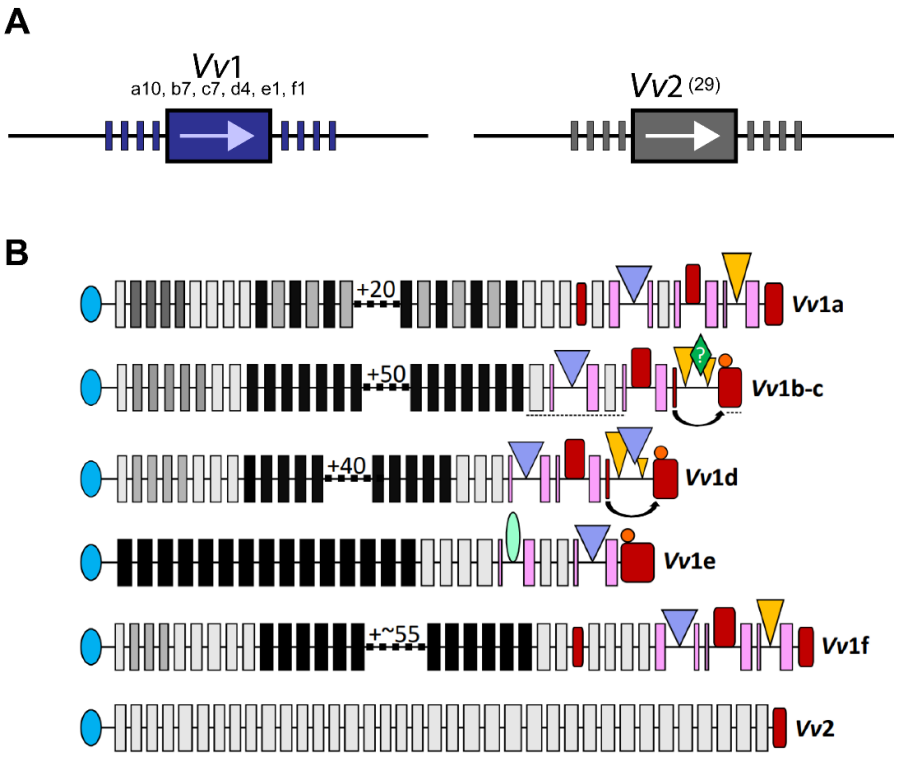


Figure 3.1. Bioinformatic outline of *Vv* RTX-adhesin loci and gene products. A) RTX-adhesin gene loci identified using the LoRe bioinformatic analysis pipeline. Loci are split into two different locations, *Vv*1 and *Vv*2. The *Vv*1 locus can be occupied by any one of six different adhesin genes (a-f) with their abundance out of 30 strains analysed alongside the letter. The *Vv*2 locus is occupied by the same "extender only" RTX-adhesin gene. The four genes on either side of the RTX-adhesin gene locus are represented as smaller rectangles. B) Domain maps for the RTX adhesins encoded within *Vv*. Maps start at the N-terminal retention domain of the adhesin (blue oval), with various extender domains represented as rectangles. Extender domains in the lightest shade of grey are less than 80% identical at the amino acid level to any other extender within the same adhesin. Higher identities are indicated by darker shades of grey to black. Following the extender region, adhesins may contain split extenders (pink rectangles), von Willebrand factor A-like domains (vWFA, yellow triangles), carbohydrate-binding modules (blue triangles), peptide-binding domains (cyan oval) and beta-roll/T1SS domains (red rectangles), alpha-helical bundles (orange circles) and domains of unknown function (green diamonds) that may be split, as indicated by a black arrow. Dotted lines under parts of *Vv*1b-c indicate portions of the LBR that are highly similar between the isoforms. Figure derived from Hansen *et al.* (62).

The entire *Vv*1d adhesin was modeled via AlphaFold 3 (AF3) by splitting the full-length sequence into the N-terminal plus extender region and the C-terminal region. Modeling of the N-terminal region (not shown) indicated a well-conserved Ca^{2+}-independent N-terminal retention domain that was structurally homologous to those in other RTX adhesins (130, 131). The structural prediction for the distal ligand-binding region (Fig. 3.2B) showed most modelled residue positions have low associated predicted alignment error (pAE) and high predicted local distance difference test confidence scores (pLDDT) (Fig. 3.2C). The domain map (Fig. 3.2A) and model (Fig. 3.2B) indicates a C-terminal ligand-binding region architecture starting after final BIg-like extenders (domains 1 and 2) with an Ig-like split domain from which a putative PA14 family CBM is projected (domains 3 and 4, pink and blue respectively). A second Ig-like split domain (domain 5, pink) immediately follows, projecting a beta-roll-containing domain of unknown function (domain 6, red). The latter RTX-like domain has similarities to a putative ligand-binding domain identified in *Ah*Lap (64). The third ligand-binding domain splits off from the C-terminal RTX solenoid (domain 9, red) and is an orthologue of a vWFA domain also seen in *Ah*Lap. However, the vWFA itself also acts as a split domain, projecting a CBM-like domain (domains 7 and 8, green and blue respectively). Overall, the AF3 prediction indicates that *Vv*1d ligand-binding region contains several putative ligand-binding domains that could contribute to adhesion and pathogenesis.

Using the AF3 prediction of the PA14 CBM domain, topography and binding pocket analyses were performed. The strand topology follows a classical PA14 family carbohydrate-binding module pattern, with a conserved D*cis*D homology motif (Appendix Fig. A) that matches those seen in other family members (Appendix Fig. B). The outward projection of the CBM from the split domain, in conjunction with the D*cis*D motif's location at the furthest

domain edge, allows for full exposure of the putative binding pocket away from any steric hinderance from the adhesin backbone, thereby maximizing ligand-binding efficiency.

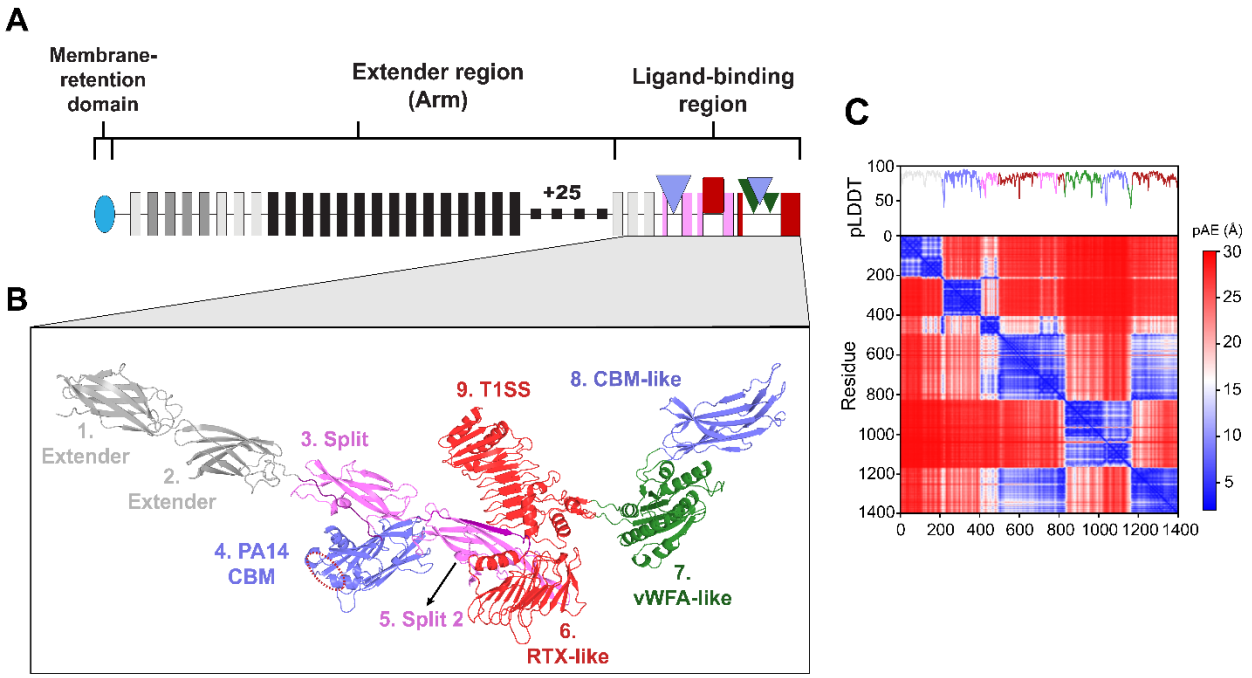


Figure 3.2 Domain architecture and fold prediction of *Vv*1d and its PA14 CBM. A) Full length domain map of *Vv*1d from the N-terminal retention domain (blue oval), through the BIg domains in the extender region (black, grey and dark grey rectangles), on into the ligand-binding region containing two putative CBMs (blue triangles), one vWFA domain (green triangle), split extender domains (pink rectangles) and beta-roll/T1SS domains (red rectangles). **B)** Full AF3 model of C-terminal ligand-binding region of *Vv*1d. Domains are numbered with labels coloured equivalent to their representation in A. **C)** Confidence scores for full length C-terminal ligand binding region fold. pLDDT scores (top) covering every residue modeled, with pAE output heatmap (bottom) indicating alignment error scores in Ångstroms (Å) for every residue modelled. pAE heat map coloured in a red-white-blue pattern, where red is high error, white is medium error, and blue is low error.

Having modelled the domain boundaries for the *Vv*1d CBM and its split domain (*Vv*1d CBM + Split), an *E. coli* codon-optimized gene to code for this protein was designed and ordered from GenScript within the vector pET-28a(+). Production of the 32-kDa *Vv*1d CBM + Split (Appendix Fig. C) construct in BL21 (DE3) *E. coli* was performed as described in Materials and Methods. When this expression vector was induced with IPTG a prominent 32-kDa protein band

of *Vv*1d CBM + Split appeared in lane 3 of the SDS-polyacrylamide gel (Fig. 3.3A). This protein was largely retained in the soluble lysis fraction (lane 4) as compared to the insoluble lysis fraction (lane 5). During binding to the Ni-NTA resin, some protein was lost in the flowthrough and subsequent washing steps (lane 6 and 7). Upon elution by increased concentrations of imidazole, a protein fraction migrating at the expected molecular weight could be seen, with minor contaminating proteins at lower molecular weights (lane 8). Interestingly, all lanes show a closely associated doublet band retained near the expected *Vv*1d CBM + Split band, suggesting a potential degradation product of the target protein (Fig. 3.3A). A subsequent size-exclusion chromatography (SEC) step was performed in attempt to purify His-tagged *Vv*1d CBM + Split to homogeneity. However, *Vv*1d CBM + Split failed to elute at the expected volume and remained bound to the column. A similar observation had previously been made with other PA14 domains (63, 80) where their affinity for certain hexoses caused them to bind to the dextran-agarose beads in the column matrix. Because Ca^{2+} is required for proper folding and function of CBMs in RTX adhesins, a buffer containing the divalent cation chelation agent EDTA was applied to the column, resulting in the elution of the unfolded protein fraction beginning at 230 mL (Fig. 3.3B). The resultant EDTA-containing *Vv*1d CBM + Split elution band (lane 9) (Fig. 3.3A) retained the doublet seen in Ni-NTA fractions. The presence of a stable *Vv*1d CBM + Split minor degradation product suggests that some C-terminal residues have been cut off, since both bands bind to the Ni-NTA resin through their intact N-terminal His-tag. The typical heterologous expression using 2 L *E. coli* cultures produced 65 mg of recombinant *Vv*1d CBM + Split.

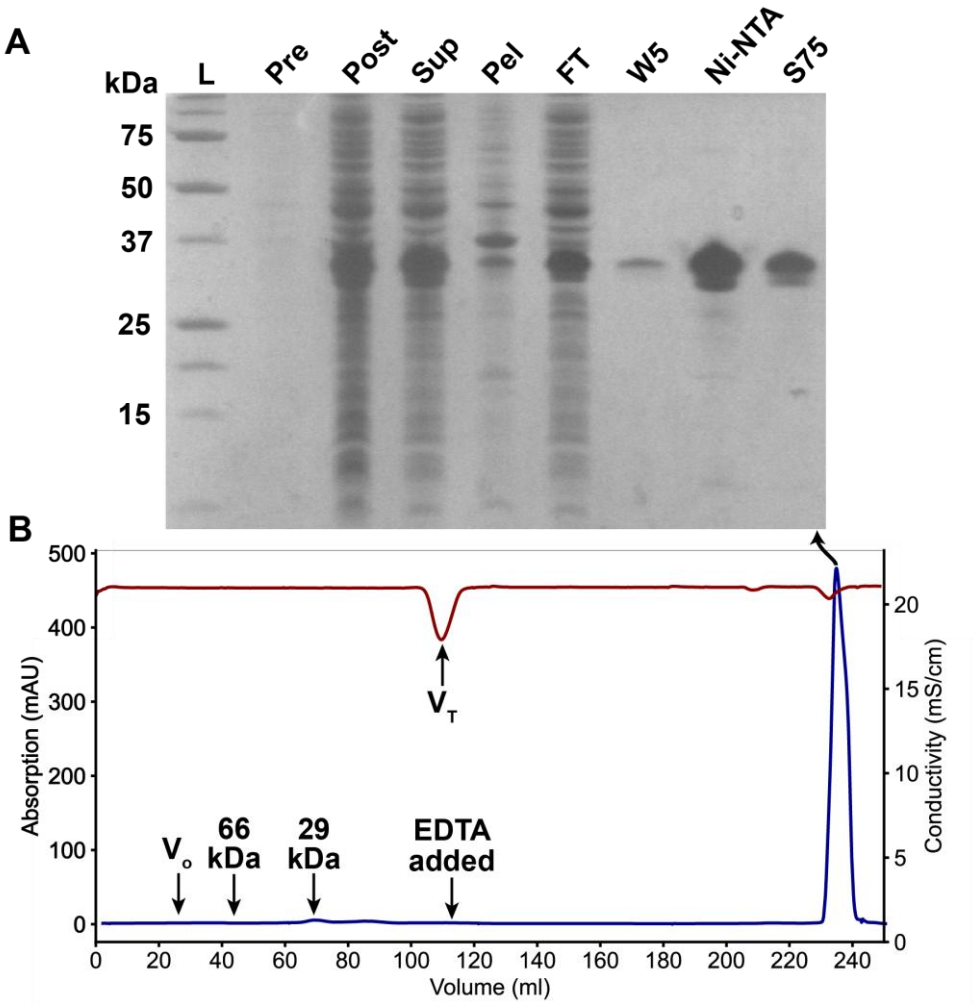


Figure 3.3 Purification of *Vv*1d CBM + Split construct by Ni-NTA chromatography and SEC. A) SDS-PAGE indicating *Vv*1d CBM + Split expression and purification through induction of protein production, Ni-NTA resin affinity chromatography, and dextran affinity chromatography. Lanes 1-9 depict molecular weight standards (L), pre-induction (Pre), post-induction (Post), soluble (Sup) and insoluble (Pel) lysis fractions, Ni-NTA resin flowthrough (FT), a wash fraction (W5), an elution fraction from Ni-NTA resin chromatography (Ni-NTA), and an elution fraction from dextran affinity chromatography (S75). B) Chromatogram showing the elution profile of *Vv*1d CBM + Split from a Superdex 75 size-exclusion column using protein buffer with and without EDTA. The absorbance at 280 nm (blue line plot) was measured over 120 ml of protein buffer flow before and after the addition of EDTA. Absolute conductivity (red line plot) was measured to indicate a full column volume of applied buffer.

3.2 *Vv*1d CBM + Split binds various monosaccharides with a preference for GlcNAc

Dextran binding by *Vv*1d CBM + Split, which was first revealed during the purification process, enabled use of an in-house semi-quantitative Dextran-Based Competition (DBC) assay (114) (Fig. 3.4). Data from the graphed curves indicates a wide range of affinities for different sugars according to their stereochemistry (Appendix Fig. D). GlcNAc was most effective at eluting *Vv*1d CBM + Split from the dextran matrix, which is indicative of preferential binding. Higher concentrations of glucose (Glc), fucose (Fuc), and mannose (Man) were required to equally compete the CBM + Split off the dextran matrix. Xylose (Xyl), galactose (Gal), and N-acetylgalactosamine (GalNAc) were markedly poor competitors. Given that GlcNAc was the best-binding sugar, two C1-modified, non-metabolizable derivative compounds (Appendix Fig. E) were produced by the Nitz lab at the University of Toronto to see how well they functioned as specific competitors. The DBC assay was again employed, but at a much smaller scale to shepherd the amounts of derivative available. The resultant apparent binding curves (Fig. 3.4B) indicated that both the alpha- and beta-derivative compounds bind with the same avidity as native D-GlcNAc, indicating their potential use as non-metabolizable glycomimetic competitors of CBM binding to host glycans.

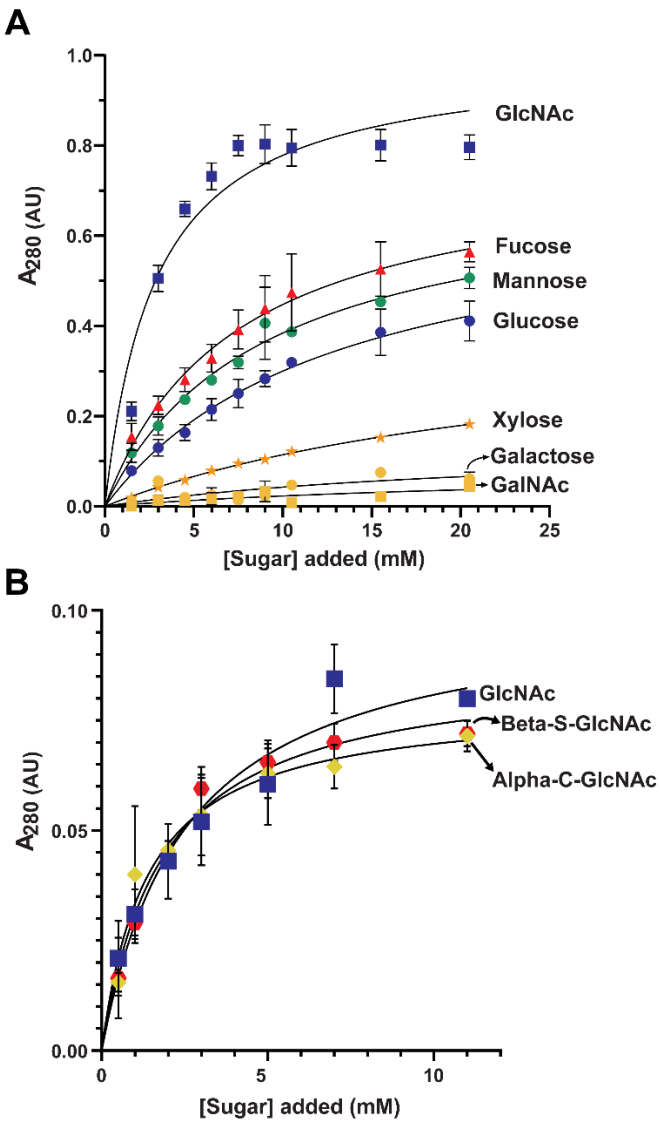
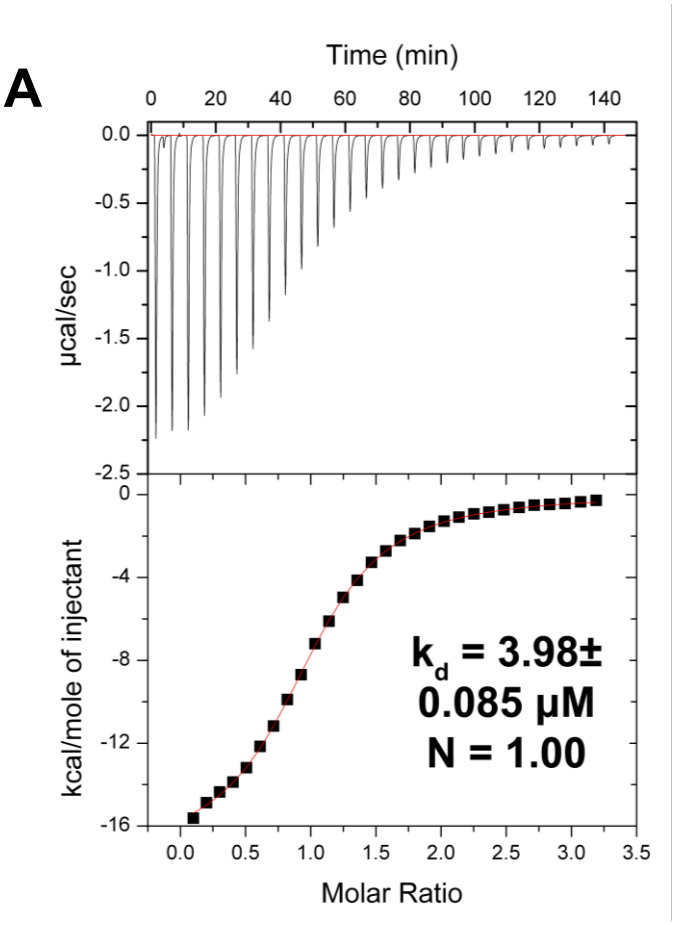


Figure 3.4 Dextran-based competition assay to determine the relative affinities of sugars for *Vv*1d CBM + Split. A) Independent aliquots of dextran-bound *Vv*1d CBM + Split were incubated with the indicated concentrations of each monosaccharide. After equilibration, the amount of protein released into the supernatant was measured by absorbance at 280 nm. Protein elution curves from the addition of GlcNAc (blue squares), fucose (red triangles), mannose (green circles), glucose (blue circles), xylose (orange stars), galactose (yellow circles), and N-acetylgalactosamine (yellow squares) are shown. **B**) Stepwise elution of *Vv*1d CBM + Split from individual aliquots of dextran-bound *Vv*1d CBM with increasing concentrations of free GlcNAc and two glycomimetic derivatives. Increasing amounts of protein released by GlcNAc (blue squares), Beta-S-GlcNAc (red hexagons) and Alpha-C-GlcNAc (yellow diamonds) are shown. Absorbance of free protein eluted into the supernatant was measured at 280 nm. Incremental data points were recorded in triplicate, with error bars denoting standard deviation. Each data set was fitted with non-linear regressions using GraphPad Prism.

Having identified GlcNAc as the best-binding free sugar, isothermal titration calorimetry (ITC) was employed to determine the thermodynamic parameters associated with the *Vv*1d CBM-GlcNAc interaction. Titration of D-GlcNAc into a solution containing *Vv*1d CBM + Split illustrated strong exothermic peaks (Fig 3.5A). Subsequent automated integration of the raw data using a one-site binding model produced a two-sided sigmoidal curve. The final readable thermogram indicated a reaction with an enthalpy change of (ΔH) of -17.3 kcal/mol, a temperature-standardized entropy change of -9.82 kcal/mol (Fig. 3.5B), and a 1:1 binding stoichiometry (N = 1.0, Fig. 3.5A). Moreover, this trial outlines a strong negative change in Gibbs free energy (ΔG), suggesting favourable and spontaneous interactions (Fig 3.5B). Fitting of the data to a one-site binding model yielded a dissociation constant (K_d) of 3.98 ± 0.085 μM for this interaction (Fig 3.5A). An independent replicate trial produced a comparable binding isotherm and K_d, indicating that the measurement was reproducible within experimental error. (Appendix Fig. F).



B

ΔG (kcal mol^{-1})	ΔH (kcal mol^{-1})	TΔS (kcal mol^{-1})
-7.48	-17.3	-9.82

Figure 3.5 ITC data for GlcNAc binding by *Vv*1d CBM + Split. A) Thermogram measuring energetic interactions between *Vv*1d CBM + Split and GlcNAc. The raw thermodynamic (top) and integrated (bottom) data over the duration of titrations are shown, along with calculated k_d and stoichiometry values. Experimental points are represented as filled squares. N represents the measured stoichiometry of binding. **B)** Measured thermodynamic parameters for the binding interactions between the proteins in replicate trials and the D-GlcNAc.

3.3 *Vv*1d CBM-GlcNAc ligand-bound crystal structure illustrates a Ca^{2+}-dependent sugar-binding mechanism

To define the molecular basis of carbohydrate recognition by *Vv*1d, we determined the crystal structure of the *Vv*1d CBM + Split domain in complex with GlcNAc. Following ligand-soaking experiments, a GlcNAc-bound dataset was collected at the CLS synchrotron and the structure was solved by molecular replacement using Ca^{2+}-bound PA14 CBM from the *Mp*IBP (PDB: 6X7T) as a search model. The final model was refined to 2.28 Å resolution and contains five CBM + Split molecules in the asymmetric unit (PDB: 36GK; Table 1). Electron density was well defined for residues P15-N307, whereas the N-terminal affinity tag, thrombin cleavage sequence, and C-terminal S308 residue lacked interpretable density, indicating these regions are disordered.

Three of the five molecules contained bound GlcNAc, whereas the remaining two lacked ligand. Inspection of the crystal lattice showed that the ligand-binding pockets of these latter molecules are occluded by crystallographic symmetry contacts, preventing access of GlcNAc during soaking (Supplementary Fig. G). Importantly, all three GlcNAc-bound molecules exhibited an identical mode of sugar recognition.

Table 3.1 Diffraction data collection and refinement statistics for *Vv*1d CBM + Split

Collection	
Wavelength	0.953710
Resolution range	48.81 - 2.28 (2.31 - 2.28)
Space group	I 2 2 2
Unit cell	120.58 189.49 199.69 90 90 90
Total reflections	1426006 (45192)
Unique reflections	103976 (3413)
Multiplicity	13.7 (13.2)
Completeness (%)	99.97 (99.91)
Mean I/σ(I)	12.37 (0.73)
Wilson B-factor	61.17
R-merge	0.1161 (3.387)
R-meas	0.1207 (3.523)
R-pim	0.03272 (0.9629)
CC1/2	0.999 (0.317)
CC*	1 (0.694)
Refinement	
Reflections used in refinement	103962 (3413)
Reflections used for R-free	5270 (161)
R-work	0.1997 (0.3531)
R-free	0.2267 (0.4263)
Number of non-hydrogen atoms	11128
macromolecules	10747
ligands	76
solvent	305
Protein residues	1439
RMS (bonds)	0.016
RMS (angles)	2.43
Ramachandran favored (%)	94.88
Ramachandran allowed (%)	4.41
Ramachandran outliers (%)	0.70
Rotamer outliers (%)	4.50
Clash score	4.46
Average B-factor	78.51
Macromolecules	78.81
Ligands	97.62
Solvent	63.13

The *Vv*1d CBM adopts the conserved fold observed in homologous PA14 CBM family members (63, 80, 89), consisting of antiparallel beta sheets surrounding a hydrophobic core that is flanked by several short alpha helices (Fig 3.6). Multiple surface loops extend from the beta

sheet core and display elevated B-factors, consistent with increased conformational flexibility. The adjacent Split extender domain adopts a BIg-like fold but makes no substantial contacts with the neighbouring CBM. Similar to the CBM, the N-terminal portion of the Split domain exhibits higher B-factors than the remainder of the domain, suggesting greater local flexibility (Supplementary Fig. H).

Each CBM + Split molecule coordinates six Ca^{2+} ions, with four located within the CBM and two within the Split domain (Fig. 3.6). Two of the CBM-associated Ca^{2+} ions stabilize loops that form the carbohydrate-binding pocket, suggesting that calcium contributes both to the structural architecture of the binding site (D*cis*D binding motif) and to ligand recognition.

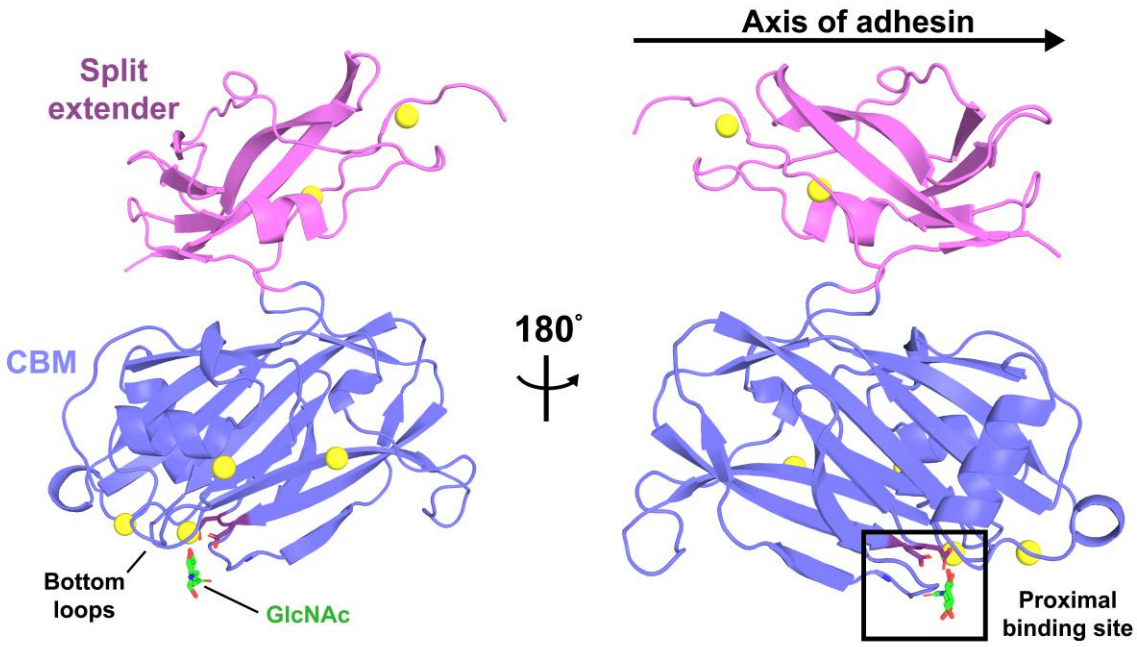


Figure 3.6 Global fold of *Vv*1d CBM + Split. Front and back view of *Vv*1d CBM + Split (light blue) including the adjacent split domain (pink) with the extension axis of the main adhesin arm labelled. Bottom loops that form the proximal binding site containing calcium (yellow) and a bound GlcNAc (green) molecule are outlined.

The GlcNAc-binding pocket is formed by three loops that converge around a central calcium ion (Fig. 3.7 A, B). The bound D-GlcNAc molecule is positioned such that its C3 and C4 hydroxyl groups directly coordinate the Ca^{2+} while simultaneously participating in an extensive hydrogen-bonding network with surrounding protein residues. The coordinating calcium ion is stabilized by residues D137, D138, D181, and the backbone carbonyl between G183 and G184 (Fig. 3.7C). In addition to Ca^{2+} coordination, the C3 hydroxyl forms polar interactions with the D138 side chain, the backbone carbonyl and amide of N185, and the backbone carbonyl between G183 and G184 (Fig. 3.7D). The C4 hydroxyl forms hydrogen bonds with the side chains of D137, D138, and Q182, as well as the backbone carbonyl between G183 and G184. The identical coordination geometry observed for all three GlcNAc-bound molecules indicates that this represents the conserved mode of carbohydrate recognition within the *Vv*1d CBM.

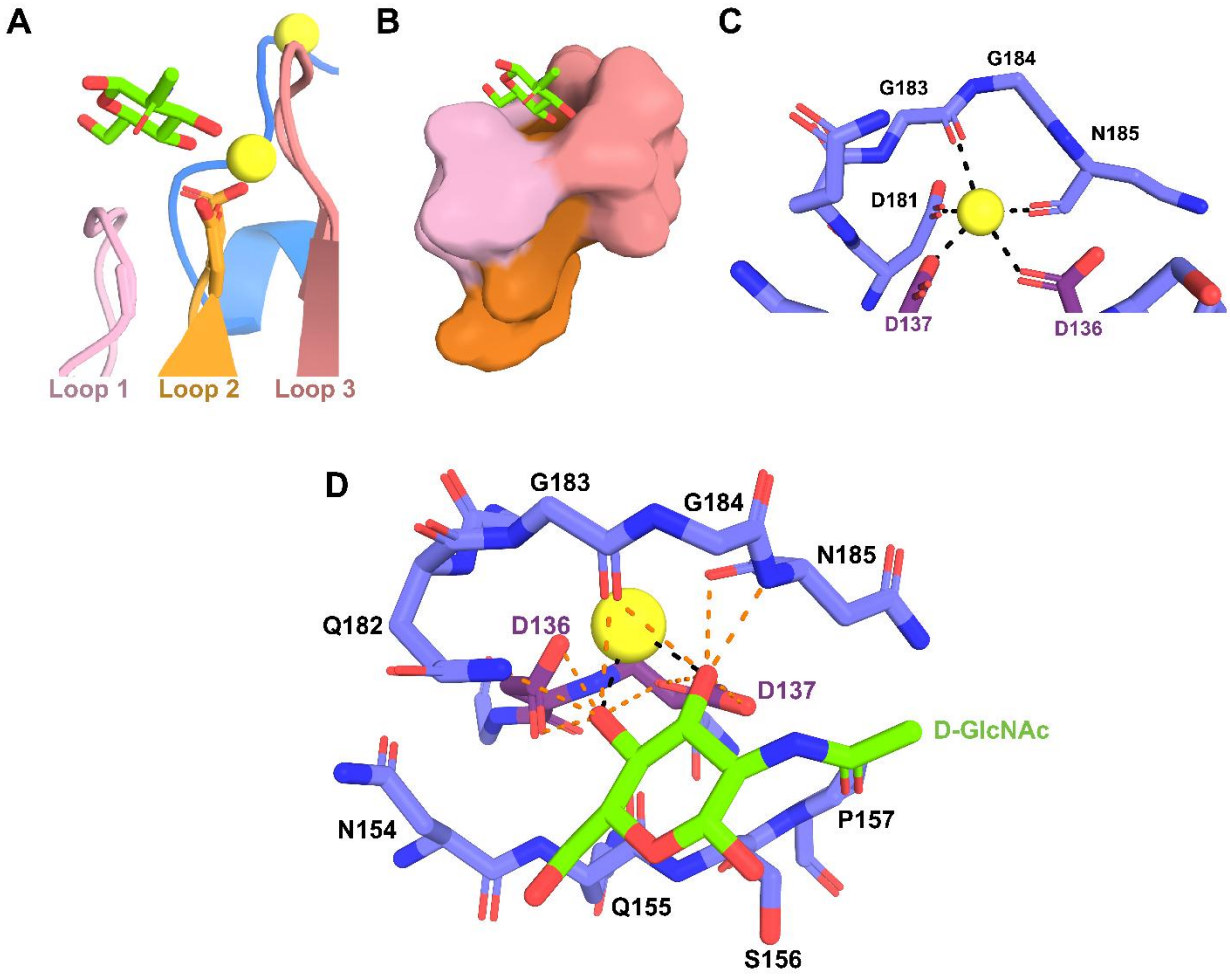


Figure 3.7 Sugar-binding site of *Vv*1d CBM. A) Side view of binding pocket loop 1 (pink), loop 2 (orange) and loop 3 (salmon). **B**) Surface representations of the binding loops (coloured as in A) forming a binding pocket with a bound GlcNAc (green) molecule. **C**) Calcium (yellow) coordinating residues in the *Vv*1d CBM binding pocket. All residues are labelled with their one letter code and sequence position. Metal coordinating interactions represented black dashes. **D**) Important residues contained in the sugar-binding site. Residues are labelled with the one letter code and sequence position. Dashed lines represent metal coordination (orange) between sugar and calcium, or polar contacts (black) between sugar and protein.

3.4 Terminal sugars on complex glycans are key targets for *Vv*1d CBM

Having identified and characterized the binding of various free monosaccharides to *Vv*1d CBM, the presumption was that these sugars were the terminal moieties of larger polysaccharide binding partners and/or cell surface glycans. Thus, to identify glycans that can be bound by *Vv*1d CBM *in vitro*, the purified protein was labelled with fluorescein isothiocyanate (FITC) and

screened against an array of mammalian glycans at the Harvard Center for Glycoscience and the Liquid Glycan Array (LiGA) at the University of Alberta via GlycoNet Integrated Services. Signals from the arrays indicated that *Vv*1d CBM + Split preferentially bound glycans with terminal GlcNAc, fucose, mannose and L-galactose moieties. The first three of these had strongly displaced the CBM + Split from the dextran matrix (Fig. 3.4A). L-galactose was not tested in the DBC assay but is structurally different from the non-binding D-galactose. It has flipped stereocenters that position C2 and C3 hydroxyls as a trans, vicinal, equatorial hydroxyl pair. Although absolute RFU values were low across the microarray, comparison to background control spots revealed reproducible enrichment for a subset of glycans. Two of the top three hits consisted of linear and branched chitin oligosaccharides with terminal β1-2, β1-4 or β1-6 GlcNAc linkages. Additionally, significant binding occurred on a branched xyloglucan with a terminal fucose moiety (Fig. 3.8). The corresponding relative fluorescence units (RFUs) for each the best binding glycans were between 6-8 ± 1 RFU. Interestingly, the 5 of the 6 highest binding signals came from glycans with terminal GlcNAc moieties. Conversely, the various glycans printed on the array that exhibited ≤ 2 RFUs typically contained populations with terminal sialic acid, galactose, or GlcNAc moieties. Moreover, glycans containing core (non-terminal) β1-4-linked GlcNAc units showed no binding. Any glycan binding that measured ≤ 2 RFU was determined to be insignificant, as it was equal to the phosphate buffer blank control with no glycans present (Appendix Table 1).

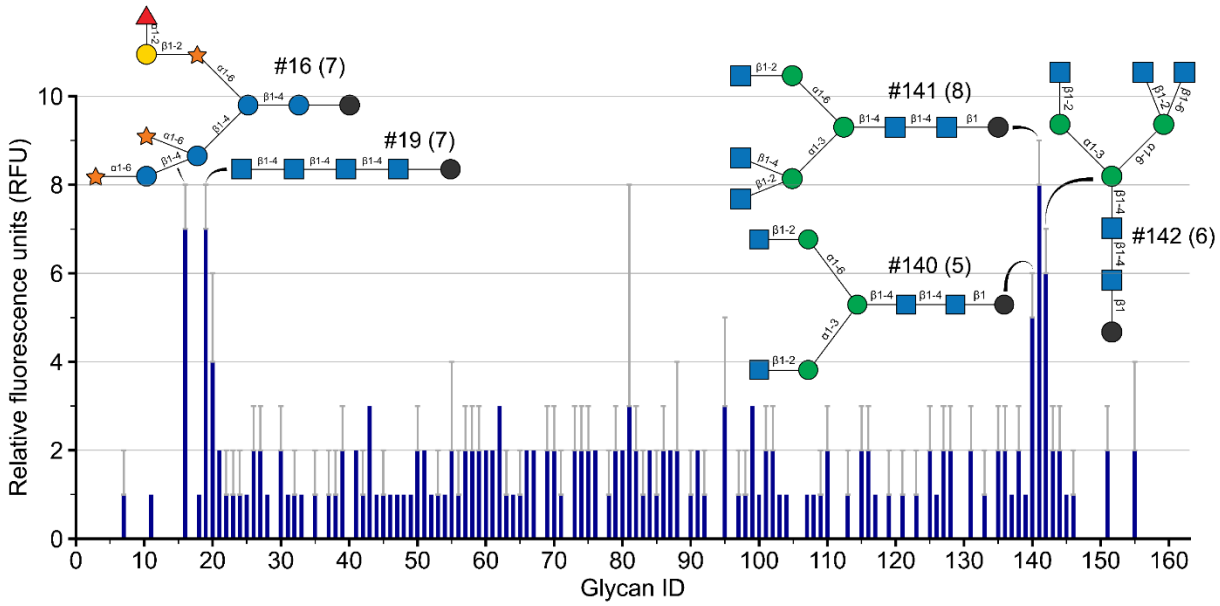


Figure 3.8 Micro array analysis of *Vv*1d glycan-binding partners. A total of 155 N- or O-linked biological and commercial glycans were probed with fluorescently labeled *Vv*1d CBM + Split. Binding is indicated in blue as relative fluorescence units, with standard deviation represented as grey error bars. The top five best binding glycans are shown in schematic representations, with sugar moieties fucose (red triangle), galactose (yellow circle), glucose (blue circles), mannose (green circles), GlcNAc (blue squares), and xylose (orange stars). Glycans are illustrated using GlycoGlyph. Data analysis and plotting was performed in Python 3 using NumPy, Pandas and Matplotlib.

To further verify and categorize glycan binding partners of the *Vv*1d CBM, a LiGA library, comprised of various barcoded phages displaying unique glycans at various densities, was screened against a non-binding bovine serum albumin (BSA) control and CBM + Split immobilized to affinity resin. This array contained a total of 122 unique glycans that could be screened (Appendix Table 2). Any resultant bound phages that survived the wash steps were then sequenced, indicating which specific glycans, at a specific density, bound more favorably to the CBM + Split in comparison to BSA. A minimum cutoff was set at a fold change (FC) value equal to 5, as anything less than that is normally considered nonspecific or background binding. Additionally, all blank and control phages showed a FC of 1 or less, indicating any results with similar FC needed to be completely discounted. LiGA data indicated *Vv*1d CBM + Split binding

to similar glycans that were positive in the Harvard array. Phages displaying glycans with terminal fucose, GlcNAc, mannose and L-galactose moieties showed measurable retention by the CBM + Split as compared to BSA, which varied based on the density of glycans display on each phage (Fig 3.9).

The top 10 hits for binding are outlined in Table 3.2. Due to the variety in density of display for all glycans, it is difficult to say specifically which glycan was bound the tightest by *Vv*1d CBM + Split. However, these results illustrate the promiscuity of binding that the *Vv*1d CBM + Split can perform by showing measurable FC values for various glycans with different terminal monosaccharide moieties. Interestingly, glycans (GM2, 7, 10) with terminal moieties determined to be non-binding by DBC assays, showed greater than 5 FC in binding. These results may indicate an artifact of low level or nonspecific binding. Various glycans that showed FC of between 0.1 and 5 were those with terminal Gal or GalNAc moieties like globosides and gangliosides. However, in this low binding group, there are still many mannosylated, fucosylated or GlcNAc-terminated glycans that show no binding even at high density levels, which was unexpected (Appendix Table 2).

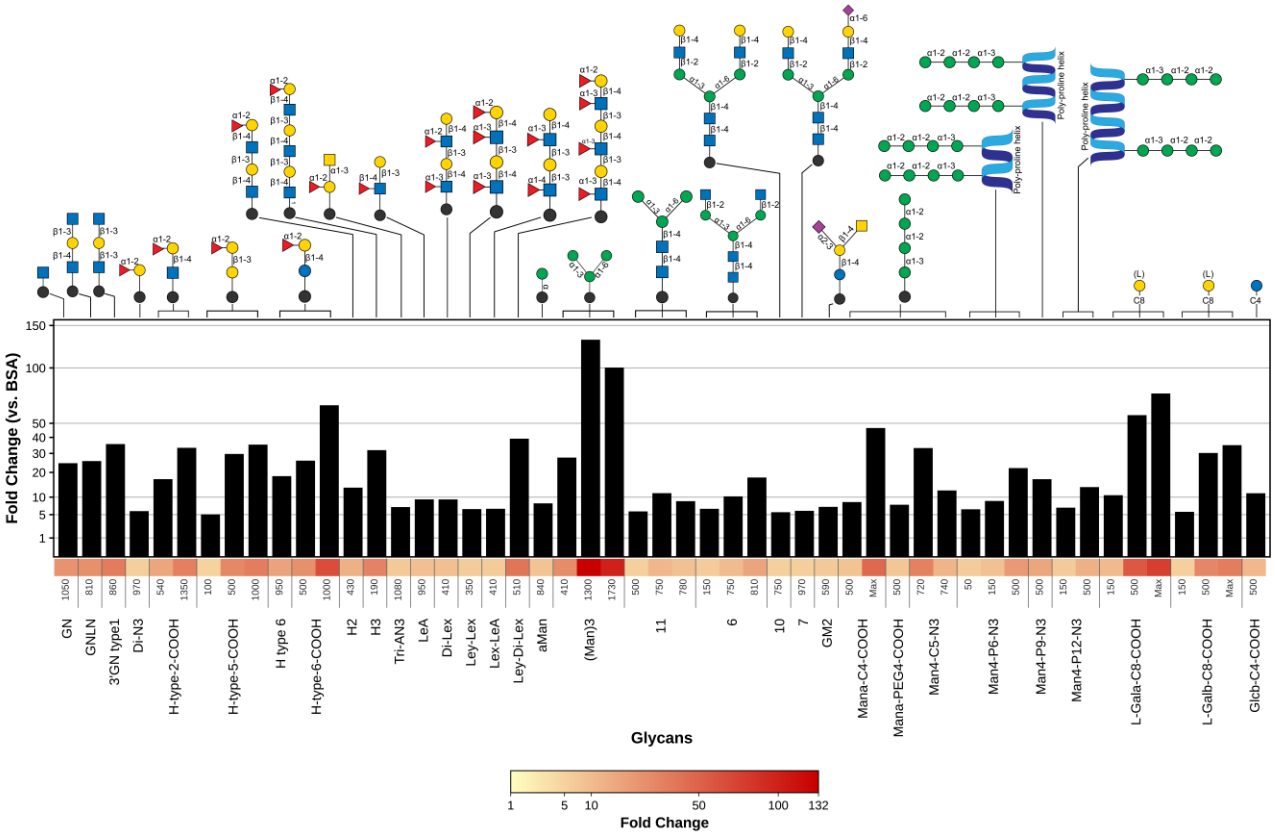


Figure 3.9 Liquid Glycan array binding profile of *Vv*1d CBM + Split. *Vv*1d CBM + Split was screened against a library of barcoded phage-displayed glycans, with binding quantified as a fold change (FC) of phage recovery relative to a BSA reference control. **Top:** SNFG (Symbol nomenclature for Glycans) representations of each glycan probe on the array that showed detectable binding. Each icon column corresponds to a distinct glycan structure conjugated to an M13 phage. **Middle (bar chart):** FC shown as bar height on a log-scaled y-axis. Each bar represents a single glycan probe at one presentation density, with replicates collapsed into one value representing the median log2 fold change. Bars shown for probes with FC ≥ 5. **Bottom (heatmap strip):** The same FC values are also illustrated as a color gradient (yellow to red, range 1-132). Surface density at which each glycan is presented on the page from the glycan library represented as an integer. 'Max' denotes the highest available density of display possible for that specific glycan.

Table 3.2 Top 10 best binding glycans to *Vv*1d CBM + Split via LiGA analysis

Glycan dictionary name	Density	Number of replicates in library	Fold Change (AU)
(Man)3	1300	1	132.0
(Man)3	1700	1	100.4
L-Gala-C8-COOH	Max	5	74.76
H-type-6-COOH	1000	5	64.39
L-Gala-C8-COOH	500	4	56.26
Mana-C4-COOH	Max	9	46.51
Ley-Di-Lex	510	1	39.14
3'GN type1	860	1	35.64
H-type-5-COOH	1000	5	35.28
L-Galb-C8-COOH	Max	9	34.94

Glycan binding array analyses made the connection between *Vv*1d CBM + Split binding free sugars and glycans. Binding occurs when the terminal moieties on glycan targets and free sugars display the same epitopes (vicinal hydroxy groups in a specific orientation) recognized by the CBM.

Additional *in vitro* experimentation was performed to identify other binding partners that might play a role in *Vv* pathogenesis. Digestive tract mucins from mammalian hosts were tested for CBM binding. Mucin types were diluted to a concentration within the representative physiological range (1% w/v or 10 mg/mL) and incubated with FITC-labelled protein samples. As compared to the minimal autofluorescence of the mucins from humans and sheep, the

negative control (*Rhagium mordax* AFP) illustrated no quantifiable increase in RFUs. Conversely, incubation of human and ovine mucins with *Vv*1d CBM + Split showed a considerable increase in RFUs (325 and 54 RFU, respectively) as compared to mucin only controls (10 and 0.345 RFU, respectively) (Fig 3.10A, B). Additionally, preincubation of the CBM + Split with 10 or 50 mM GlcNAc before the addition of mucins resulted in a pronounced decrease in protein bound to human and ovine mucin (26 and 11.1 RFU, respectively), indicating inhibition of CBM-mediated mucin binding through competition by free GlcNAc.

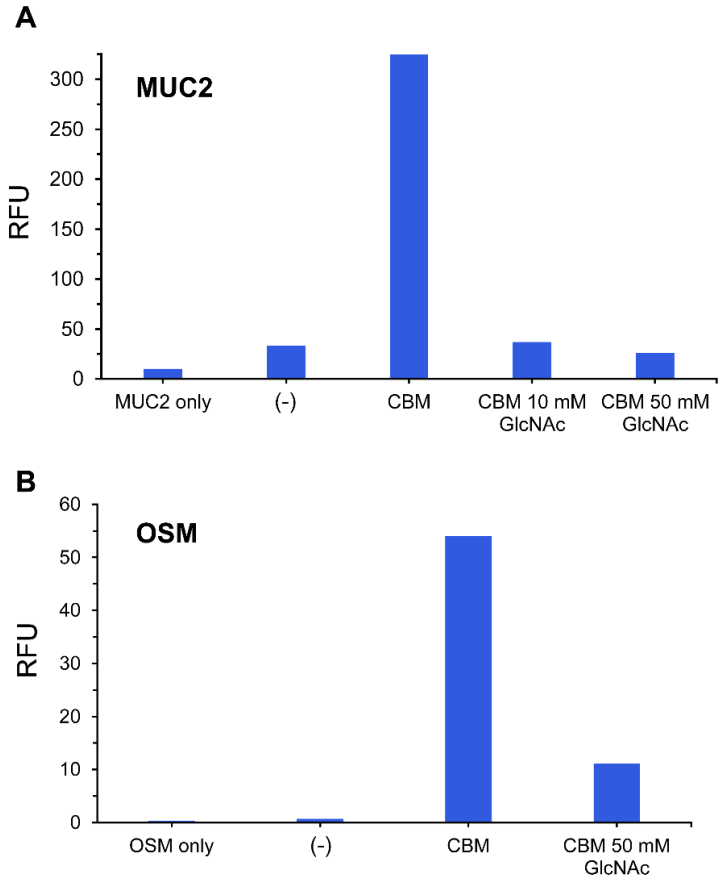


Figure 3.10 Adhesion of *Vv*1d CBM + Split to mucins. A) Bar diagram indicating relative fluorescence detected after human intestinal MUC2 incubation in the presence of, from left to right, no protein, FITC-labelled *Rm*AFP (-), FITC-labelled *Vv*1d CBM + Split, FITC-labelled *Vv*1d CBM + Split with 10 mM GlcNAc, *Vv*1d CBM + Split with 50 mM GlcNAc. **B)** Bar diagram indicating relative fluorescence detected after Ovine submaxillary mucin (OSM) incubation in the presence of, from left to right, no protein, FITC-labelled *Rm*AFP (-), FITC-labelled *Vv*1d CBM + Split, FITC-labelled *Vv*1d CBM + Split with 50 mM GlcNAc.

Other potential *Vv*1d CBM binding partners include various polysaccharides found in aqueous environments such as the shells of certain aquatic animals (chitin), in the cell walls of some aquatic plants (sodium alginate), or even in human skin (hyaluronic acid). Thus, a modified dot-blot assay was performed to determine if *Vv*1d CBM could bind to these polysaccharides. A deacetylated form of chitin (chitosan) was used to ensure this GlcNAc-rich polysaccharide could be solubilized and spotted onto membranes. Screening against the plant polysaccharide pectin was tested as a negative control, because its terminal monosaccharides (galactose and its derivatives like galacturonic acid or pentoses like arabinose and xylose) are not strongly bound by *Vv*d1 CBM + Split. The resulting dot blot indicated that of the polysaccharides tested, *Vv*1d CBM + Split only bound to chitosan (Fig 3.11). The intensity of the dot at the 6xHis control spot appears near equivalent to the chitosan-bound test dot.

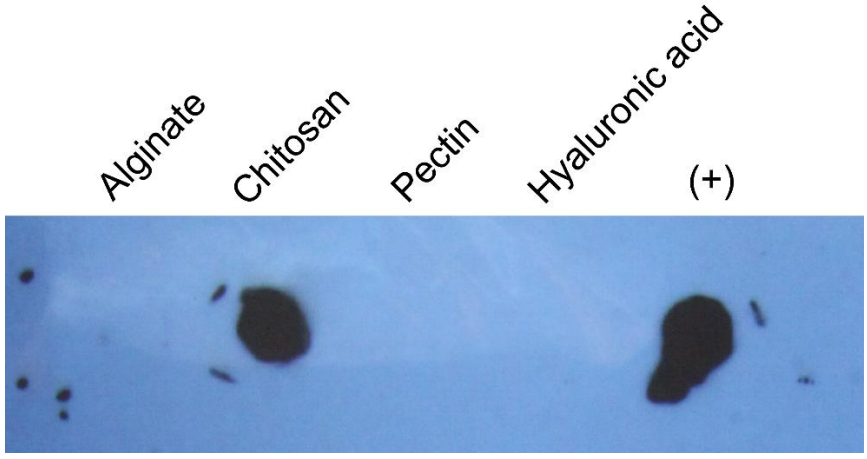


Figure 3.11 Polysaccharide binding specificity of *Vv*1d CBM + Split assessed via dot blotting. Four different polysaccharide solutions (sodium alginate, chitosan, pectin, and hyaluronic acid) were tested for interaction with 6xHis tagged *Vv*1d CBM + Split. The positive control was 6xHis containing protein (+) dotted directly onto the membrane. After incubation with a mouse-anti-His antibody, chemiluminescent detection by X-ray film can be seen as black dots.

3.5 Binding of *Vv*1d CBM to various cell types can be inhibited by GlcNAc and GlcNAc derivatives

To give biological significance to the interaction of *Vv*1d CBM with various monosaccharides and glycans, the CBM + Split was tested for interaction with various cell types that display glycans with terminal sugars that elute the protein from dextran beads (Fig 3.4A). Identifying cell types bound by *Vv*1d CBM could help elucidate the function of the *Vv*1d CBM in the pathogenesis of humans or other animals. At the same time, the ability of D-GlcNAc to block cell binding was tested in these experiments. Additional inhibition trials were done with a non-metabolizable glycoside derivative of D-GlcNAc, beta-S-GlcNAc. This glycoside retained the N-acetyl group functionalized onto the C2 position, and the trans, vicinal, equatorial hydroxyls at the C3 and C4 position that are needed for CBM binding.

Fluorescence microscopy was initially employed to see if FITC-labelled *Vv*1d CBM + Split could interact with cell surface glycans on cells in culture. Due to the evident binding to mannose as outlined by DBC assays and glycan array results, the interaction of *Vv*1d CBM + Split with two extensively mannosylated, human commensal yeast species, *Saccharomyces cerevisiae* and *Candida albicans*, was assessed by incubating the cells with the FITC-labelled *Vv*1d CBM + Split construct. Both yeasts were intensively labelled, but also experienced massive aggregation (Fig. 3.12A.ii, B.ii). For *S. cerevisiae*, this aggregation was independent of the yeast's endogenous aggregative abilities, as this strain tested was non-flocculant and adhesin deficient. Interestingly, similar large, stained cell aggregates were visible for *C. albicans*, where this strain retains its endogenous adhesins and ability to flocculate. Upon preincubation of the yeasts with 5 mM free GlcNAc or the glycoside beta-S-GlcNAc, the binding by *Vv*1d CBM + Split and aggregation of the cells were both eliminated (Fig. 3.12A.iii-iv, Biii-iv) The yeast

became monodispersed and resembled those in the control samples, indicating a successful inhibition of CBM + Split binding by both compounds.

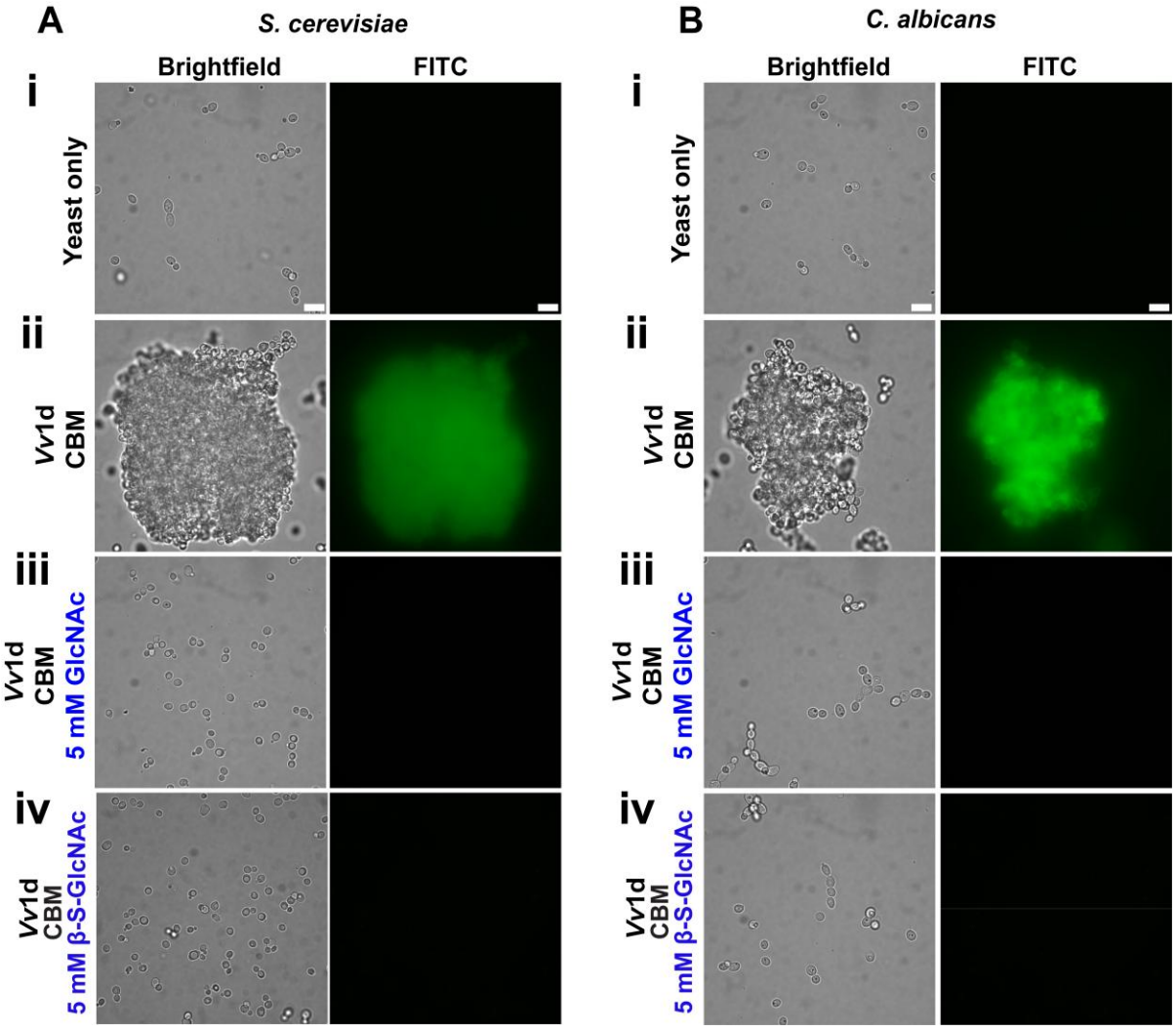


Figure 3.12 Inhibition of *Vv*1d CBM + Split binding to yeast species. A) Fluorescence microscopy images illustrating the binding of the FITC-labelled *Vv*1d CBM + Split construct to *S. cerevisiae* in the absence and presence of GlcNAc or β-S-GlcNAc. **A.i**) Images of yeast only control with no protein added. **A.ii**) Images of *Vv*1d CBM + Split incubated with *S. cerevisiae.* **A.iii**) Images of *Vv*1d CBM + Split incubated with *S. cerevisiae* and 5 mM GlcNAc. **A.iv**) Images of *Vv*1d CBM + Split incubated with *S. cerevisiae* and 5 mM β-S-GlcNAc. B) Binding of the FITC-labelled *Vv*1d CBM + Split to *C. albicans* in the absence and presence of GlcNAc or β-S-GlcNAc. **B.i**) Images of yeast only control with no protein added. **B.ii**) Images of *Vv*1d CBM + Split incubated with *C. albicans.* **B.iii**) Images of *Vv*1d CBM + Split incubated with *C. albicans* and 5 mM GlcNAc. **B.iv**) Images of *Vv*1d CBM + Split incubated with *C. albicans* and 5 mM β-S-GlcNAc. The white scale bar represents 10 μm.

Similarly, as shown by glycan array analysis and DBC assays, *Vv*1d CBM + Split bound free L-fucose and many blood group-like antigens or terminal L-fucose moieties strongly. Human erythrocytes are highly fucosylated on their surface, indicating that they may be a key binding target for the septicemia-causing *Vv* during blood infection and dissemination. To probe for this type of binding interaction, FITC-labelled *Vv*1d CBM + Split was incubated with human erythrocytes for 1 h and imaged by fluorescence microscopy. The resultant images illustrate clumping, distortion, and lysis of erythrocytes, with copious amounts of cell debris as compared to untreated cells (Fig. 3.13A, B). Fluorescent signals were observable on both cell membranes and debris. To again verify the CBM-mediated erythrocyte binding occurred through glycan interactions, 5 mM GlcNAc and beta-S-GlcNAc were added prior to incubation with erythrocytes. This addition completely blocked any CBM + Split binding to cell surfaces (Fig. 3.13 C, D), while also eliminating the lysis seen in the absence of these compounds.

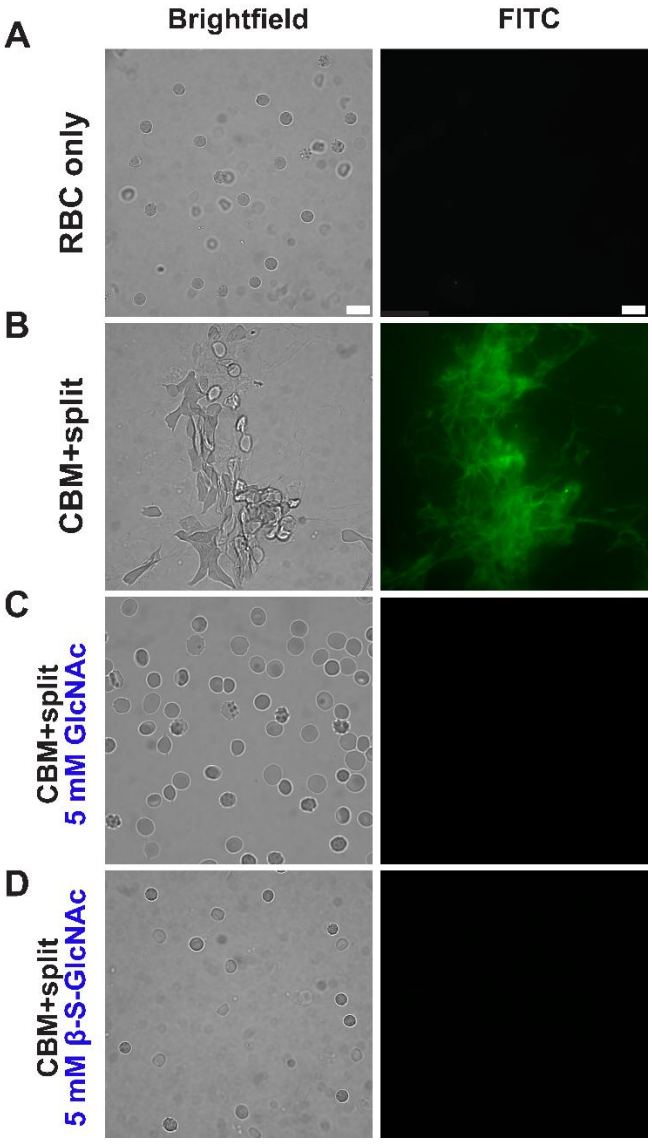


Figure 3.13 Inhibition of *Vv*1d CBM + Split binding to type O human erythrocytes. Panels representing the fluorescence microscopy-derived images of FITC-labelled *Vv*1d CBM + Split constructs incubated with erythrocytes in the presence or absence of GlcNAc or β-S-GlcNAc. **A)** Control, no protein added trial. **B)** *Vv*1d CBM + Split constructed incubated with erythrocytes. **C)** *Vv*1d CBM + Split constructed incubated with 5 mM GlcNAc and erythrocytes. **D)** *Vv*1d CBM + Split incubated with 5 mM β-S-GlcNAc. All experimental trials were visualized in brightfield (left) and green (right) FITC fluorescence channels. The white scale bar in A represents 10 μm.

To examine erythrocyte lysis more closely, the correlation between *Vv*1d CBM + Split concentration and release of hemoglobin was assayed in Eppendorf tubes. After triplicate trials, a linear increase in supernatant red colour due to the release of hemoglobin was visually evident as the CBM + Split concentration increased from 10 to 200 µg/mL. As compared to untreated RBCs, the negative control sample (addition of lysozyme) demonstrated no increase in supernatant colour, whereas treatment with a known hemolytic agent (*Vc* FrhA CBM) substantially increased the red colour. The addition of the highest concentration of *Vv*1d CBM + Split was still much paler than that seen in the positive control (Fig. 3.14A). When the highest concentration trial was pre-incubated with 5 mM GlcNAc and GlcNAc derivatives before addition of the CBM + Split construct, the supernatant remained completely colourless, equivalent to the negative controls. Thus, any quantifiable lysis appeared to be eliminated after these treatments.

To quantify the differences in the amount of lysis in each trial, the absorbance of the supernatant from each sample was measured at 414 nm to detect hemoglobin in the supernatant. The lysis trials were done in triplicate. An increase in CBM + Split concentration was linearly proportional to an increase in absorbance of the supernatant (Fig. 3.14B). Again, any detectable lysis of the erythrocytes was eliminated by the preincubation of these cells with 5 mM GlcNAc and β-S-GlcNAc before addition of *Vv*1d CBM + Split.

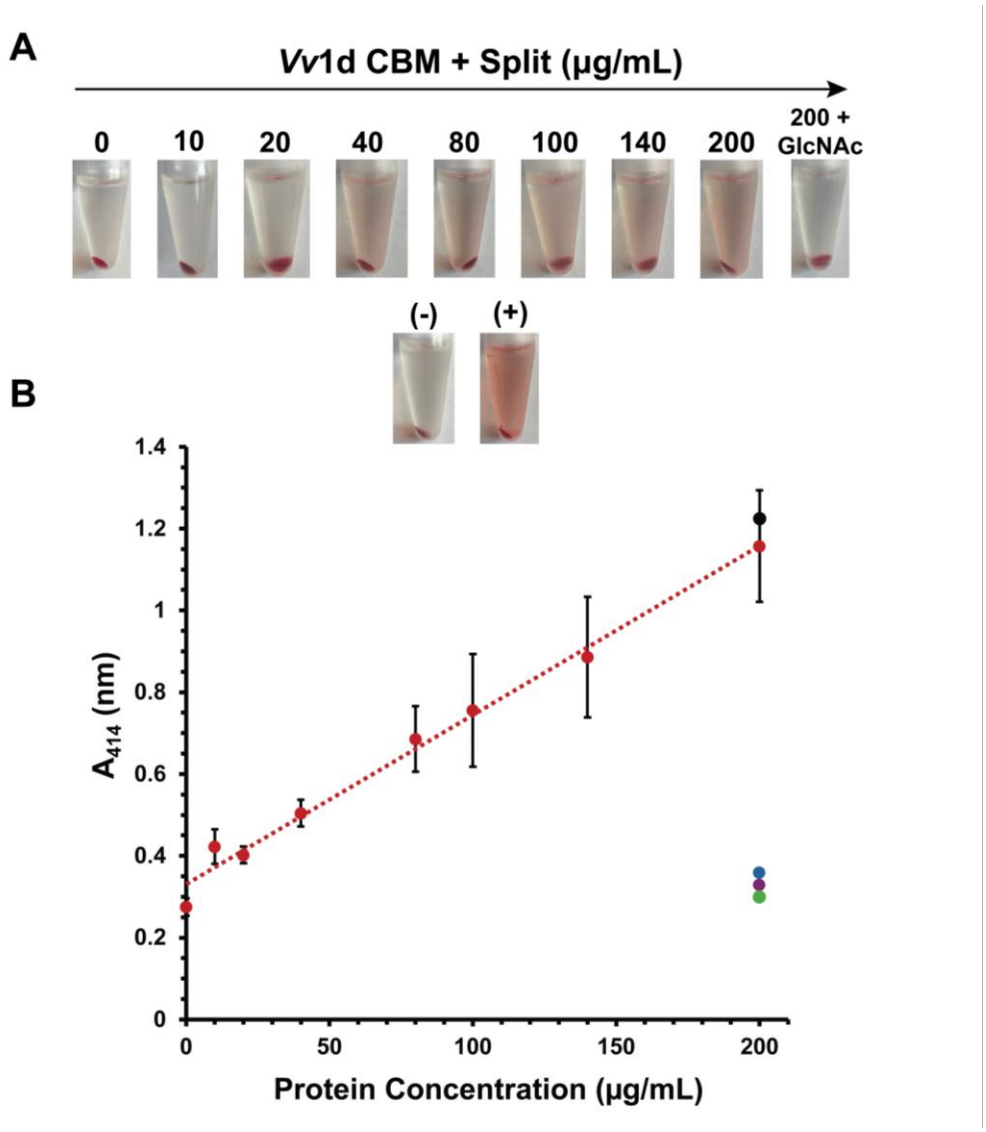


Figure 3.14 Assessing hemolytic ability of *Vv*1d CBM + Split. A) Images of erythrocytes incubated with various dilutions of *Vv*1d CBM + Split and pelleted by centrifugation. Protein concentration (µg/mL) is indicated by the values above each image, with (-) and (+) representing negative (200 µg/mL lysozyme) and positive controls (*Vc* FrhA CBM). Any additional treatments are labelled accordingly. B) Line graph of the absorbances of each sample treated with increasing concentrations of *Vv*1d CBM + Split. Absorbance was measured at 414 nm. Positive control data are represented by the black data point. Negative control data are indicated by the green dot. Trials with 5 mM addition of GlcNAc and β-S-GlcNAc represented with a blue and purple dot, respectively. Error bars above each point represent the standard deviation from triplicates measured for each sample. The linear fit relationship is represented by $y = 0.0041x + 0.3311$ and $R^2 = 0.988$.

After the initial fluorescence microscopy trials, a more rapid, robust, and quantitative cell binding plate assay was performed. *Vv*1d CBM + Split or BSA was absorbed overnight to a 96-well MaxiSorp® microtiter plate, followed by subsequent addition of various cell types to assess their ability to bind to the CBM + Split. Each condition (binding, controls, and inhibitory trials) was performed in triplicate. After an incubation period, unbound cells were washed away and remaining bound cells were fixed, followed by subsequent crystal violet staining to quantify the number of cells bound by measurement of light absorbance at 590 nm.

Initially, two cell types were tested that would help illustrate how *Vv*1d CBM binds to cells that mimic human intestinal epithelial cells (Caco-2) and those derived from human aortic endothelium (THAECs). As compared to the BSA control density (57 AU), Caco-2 cells appear to bind at 7-fold higher density, measured at 363 AU. Addition of 5, 10 or 30 mM GlcNAc caused a consistent, significant decrease in binding, but with a much higher density of cells still bound as compared to the control (Fig 3.15A). In contrast, relatively few THAEC bound *Vv*1d CBM + Split. There was a ~50% increase in cell binding (96 AU) compared to the BSA background (60 AU). Addition of 1, 5, 10 and 30 mM GlcNAc showed no significant decrease in binding by the cells (Fig 3.15B). Another cell line, derived from Rainbow Trout gill cells (RTg), was tested to elucidate if the *Vv*1d CBM + Split could facilitate interactions with the fish gill epithelium. Again, BSA control background binding was similar to other trials (60 AU), with RTg cells binding at a 6-fold higher density in wells coated with *Vv*1d CBM + Split (311 AU). Additionally, binding of the RTg cells decreased three-fold upon addition of 30 mM GlcNAc, showing low density binding in the wells (124 AU) (Fig. 3.15C). Next, re-testing of erythrocyte binding was performed to see if results resembled the fluorescent microscopy assay result. As compared to the BSA control, type O erythrocytes bound at a high density (336 AU), like the

RTg and Caco-2 cells. As seen with the RTg cells, addition of 30 mM GlcNAc significantly decreased the density of bound cells (81 AU), to close to that seen with the BSA control (61 AU) (Fig 3.15D).

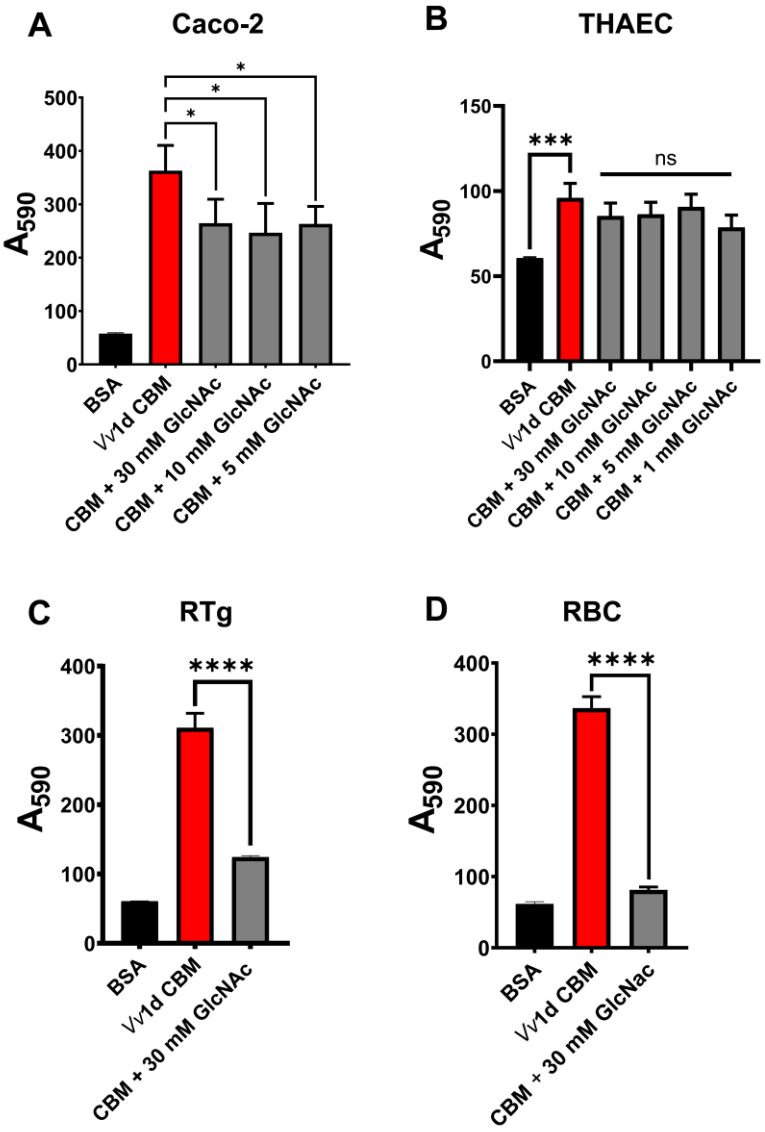


Figure 3.15 Binding and inhibition of *Vv*1d CBM + Split to various cell types. Measurement of cell binding density of various cell types to wells coated with *Vv*1d CBM + Split, in the presence or absence of various GlcNAc concentrations or with BSA only. **A**) Absorbance measurements of Caco-2 cells binding to wells with (from left to right): BSA only, *Vv*1dCBM + Split, *Vv*1dCBM + Split with 30 mM, 10 mM and 5 mM GlcNAc added. **B**) Absorbance measurements of THAEC cells binding to wells with (from left to right): BSA only, *Vv*1dCBM + Split, *Vv*1dCBM + Split with 30 mM, 10 mM, 5 mM and 1 mM GlcNAc added. **C**) Absorbance measurements of RTg cells binding to wells with (from left to right): BSA only, *Vv*1dCBM + Split, and *Vv*1dCBM + Split with 30 mM GlcNAc added. **D**) Absorbance measurements of RBCs cells binding to wells with (from left to right): BSA only, *Vv*1dCBM + Split, and *Vv*1dCBM + Split with 30 mM GlcNAc added. All absorbances measured at 590 nm. Data presented as mean ± SD (n = 3). Statistical significance was assessed by one-way ANOVA followed by Dunnett's multiple comparisons test, with each treatment compared to *Vv*1d CBM + Split + cells trials. *P < 0.05, ***P < 0.001, ****P < 0.0001; ns, not significant. Analyses were performed in GraphPad Prism 8.0.

Chapter 4
Discussion

4.1 AlphaFold3 accurately predicts the architecture and overall fold of RTX-adhesin domains

As recently reported by Graham *et al.* (61), AlphaFold3 can accurately predict the structure of the large, continuous chain of domains that constitute full-length RTX adhesins. Furthermore, it is adept at accurately elucidating the fold of split domains, out of which putative ligand-binding domains are extended from the adhesin axis. The model of over 70+ domains present in the full-length *Vv*1d allowed for the identification of three potential ligand-binding domains with high confidence. Designing reliable expression constructs was made easier using AlphaFold, due to the accuracy of domain boundaries predicted by the model. Past methodologies, like multiple sequence alignments, homology modeling servers, and guess-and-check experimentation led to some unstable protein constructs being produced (8).

4.2 *Vv*1d CBM affinity and specificity for monosaccharides

The presence of a D*cis*D motif in the AF3 *Vv*1d CBM model was an encouragement to try X-ray crystallography experiments with D-GlcNAc. The *trans* GlcNAc hydroxyls at the C3 and C4 positions coordinate the calcium ion found interacting with the homology motif and other residues. The orientation of these hydroxyls appears key to interactions within the calcium ion's coordination sphere, while also forming hydrogen bonds with Q182, G183 and N185 through main chain and side chain functional groups. Positioning of the C3-C4 hydroxyl pair in the binding pocket projects sterically bulky groups, like the C2 N-acetyl and C5 hydroxymethyl of GlcNAc, away from the binding site. L-fucose, the second-best binding sugar, does not contain these groups and still binds with a lesser affinity to *Vv*1d CBM. Thus, there must be a more

complex structural reason as to why D-GlcNAc binds this PA14 CBM more favourably. Like previously characterized PA14 homology CBMs *Mn*PA14 and *Mp*PA14 (63, 80), GlcNAc or glucose residues bind in a nearly identical fashion, illustrating the crucial role that the D*cis*D binding motif and stabilizing polar residues in binding loops 1 and 3 play in sugar interactions.

While only GlcNAc was soaked into the *Vv*1d CBM crystals, the DBC assay demonstrated the potential for *Vv*1d CBM to bind other free monosaccharides. The CBM showed measurable interactions with various hexoses. However, due to the presence of several unknown variables during these assays, the estimated K_d apparent values could not be compared to others derived from DBC assays performed on orthologous proteins (132, 133). Nonetheless, the assay was efficient at screening a multitude of monosaccharides in a short period of time using low sugar concentrations. The best binding partner, GlcNAc, was identified without the need for more complicated methods, all while simultaneously proving the presence of a carbohydrate-binding function within *Vv*1d CBM.

All assay results suggest that *Vv*1d CBM must bind to a specific hexose conformation containing properly oriented sugar hydroxyl pair. Unlike the sugars that bound, galactose and GalNAc do not contain this same sugar orientation at C3 and C4 (Appendix Fig. D). Thus, PA14 CBM sugar selection must be stereospecific and selective for the C3-C4 hydroxyl pairing. Interestingly, L-fucose (or L-6-deoxy-galactose) contains the most divergent structure from the other tested monosaccharides but shows the second-best binding profile. This is due to the conformation of fucose being an enantiomer of galactose, enabling its C2-C3 hydroxyls to be correctly oriented as a trans, vicinal, equatorial pair to interact with *Vv*1d CBM. This is opposite from xylose, which contains a similarly oriented C3-C4 hydroxyl pair to other positive binding

partners but shows weak apparent binding affinity. Further structural research or molecular docking is needed to identify what is causing this unexpected deviation in binding affinity.

Thus, while C3-C4 *trans*, vicinal equatorial hydroxyl orientation is key for sugar binding, further stabilization may depend on three-dimensional sugar orientation. A 5-carbon sugar like xylose, in its dominant pyranose form, has similar conformation alignment to GlcNAc and Glucose (134). However, it may be subtly different in overall 3D geometry, bond angles, or exact C3-C4 positioning, causing weaker binding. The variation in apparent affinity between sugars that have similar *trans*, vicinal, equatorial hydroxyls at various ring positions indicates that specific pyranose ring conformation may be simultaneously required for favourable *Vv*1d CBM binding. Thus, additional ligand-bound crystal structures with fucose, mannose, glucose, xylose, and other deoxy sugar derivatives may help further substantiate difference in sugar specificity as measured by DBCAs.

ITC was used to quantify the binding affinity between *Vv*1d CBM and GlcNAc. The resultant thermograms indicated that GlcNAc was bound with an affinity of ~3.8 μM, lending rationale for the rapid saturation of binding sites and release of protein during the DBC assay with GlcNAc. The results also show enthalpic and entropic contributions to the system during binding. Heat was released via hydrogen bond formation and system disorder largely decreased (135), implying the formation of crucial hydrogen bonding interactions between D*cis*D carboxyl groups and incoming sugar hydroxyls. The thermodynamic parameters of this interaction thus describe a stabilization of the binding pocket and a strong ligand interaction. Previous literature investigating bacterial PA14 CBM homologues have bound fucose with the highest affinity, with GlcNAc as the second-best binding sugar. In contrast, a PA14 homology domain from the yeast *K. pastoris* showed similar, low micromolar binding to GlcNAc (89). These results may illustrate

structure-based evolution dependent on the most common host sugar ligands exposed to the species that express a PA14 CBM. Furthermore, the conserved tight binding to fucose and GlcNAc across genera suggests these two sugars could be used as the foundation for glycomimetic inhibition of PA14 CBM binding. Thus, further *Vv*1d ITC experiments should be performed with other tested monosaccharides and GlcNAc derivatives to quantify the difference in binding affinity. This may be used in conjunction with additional ligand-bound crystal structures to elucidate the true reason for varied sugar specificity seen in this work and in literature.

4.3 *Vv*1d CBM binds favorably to glycans terminated with specific monosaccharides

Glycan array analyses probed which glycans are bound by *Vv*1d CBM. NCFG microarray RFU values, for even the best binding glycans containing these moieties, were low when compared to other array analyses performed on orthologous PA-14 CBMs (63, 80). This might be due to the use of FITC as a fluorescent label, which degrades and quenches easily and may have led to a higher array background. However, the domination of observable glycan binding via terminal GlcNAc moieties is substantiated by the tight binding demonstrated by ITC and DBC assay results. A possible outlier in the high RFU data was seen with a xyloglucan, containing two terminal xylose and a terminal fucose moiety. However, the high-level binding of fucose and low-level binding of xylose to *Vv*1d CBM seen in the DBC assay suggest that contributions to the xyloglucan interactions were via the terminal fucose moieties.

LiGA binding measurements confirmed previous results seen from both the NCFG microarray and DBC assays. The *Vv*1d CBM bound to phages displaying glycans with terminal moieties that were the best binders in other assays. Interestingly, branched mannosylated glycans showed the highest FC values with *Vv*1d CBM. These FC values indicated stronger binding than

many of the GlcNAc and fucose-tipped glycans, even though fucose and GlcNAc were the two best binding sugars as measured in the DBC assay. However, apart from H-type-2-COOH-1350, all other glycans with these key terminal moieties were printed at lower densities and had more replicate phages when compared to (Man)3 phages. This indicates that fold change measurements do not signify a 1:1 correlation with binding affinity. For example, the FC measured for (Man)3 printed at a density of 410 is closer to glycans with terminal fucose and GlcNAc moieties printed at the same or lower density. Additionally, when the (Man)3 branch is added to a chitobiose core (glycan 11), the FC drops to ~10 or less, irrespective of density count. Furthermore, a similar structure with the addition of terminal GlcNAc moieties on each branch does not appear to increase binding, even though the C3-C4 hydroxyls are still accessible. This unexpected result may be caused by glycosidic linkages altering ring or overall sugar conformation (136, 137), which could be influencing *Vv*1d CBM binding. Additionally, dense printing of glycans on the phages may obscure access to the mid-chain sugar hydroxyls.

*Vv*1d CBM binding to both L-fucose and L-galactose on glycans can be rationalized by their shared stereochemistry, as L-fucose is a 6-deoxy derivative of L-galactose. Thus, the presence or absence of a hydroxyl at the C6 position in the L-galactohexopyranose ring must not greatly affect binding by *Vv*1d CBM. This is demonstrated by L-Gala-C8-500 (L-galactose terminated) and H-type-6-COOH-1000 (L-fucose terminated) glycans showing similar FC of binding as compared to the BSA control (giving values of 56.26 and 64.40, respectively). Unexpectedly, lower-level binding (< 10 FC) was seen with glycans containing terminal D-galactose (glycan name 10) or GalNAc and sialic acid (Glycan name 7 and GM2, respectively). Glycan 10 and 7 contain mannoses only connected at the C1, C2 or C6 positions, allowing branch point sugars to retain C3-C4 hydroxyls correctly oriented to confer measurable binding.

Conversely, GM2 has only one previously verified binding moiety, glucose. However, it is internal with its glycosidic attachments occurring through the C4 and C1 positions, eliminating hydroxyls needed for binding. Thus, it is difficult to distinguish what true background or non-specific binding values are on the LiGA array. This is seen through measured p-values of binding to glycans GM2, 7 and 10, verifying that measurements are non-random (Appendix Table 2). However, with previous knowledge that GalNAc and galactose were poor binders via DBC assays, these results can rationally be discounted. Subsequent LiGA screenings with the presence of GlcNAc at varying concentrations may aid in eliminating non-specific or weaker interactions, easing identification of high affinity glycan binding. Additionally, the unexpected variation in these measurements show that LiGA results must be used in conjunction with other binding assays to verify sugar specificity of PA14 CBMs.

Additionally, both the LiGA and NCFG array showed that *Vv*1d CBM binding appeared to accommodate terminal glycan moieties connected via β1-2, β1-4, β1-6, α1-2, α1-3 or α1-4 and α1-6 glycosidic linkages. Unsurprisingly, all termini had C3 and C4 hydroxyls available to interact with the CBM. Moreover, any other glycans which contained GlcNAc moieties attached through β1-3 or β1-4 linkages within the core of the glycan showed weak or absent binding. Seeing that this core linkage would eliminate the availability of the C4 hydroxyl in the 3,4-diol pair, spoiling the interaction with the PA14 CBM D*cis*D motif. Specific binding to the 3,4-diol pair in hexoses indicates that *Vv* can bind sugar moieties within capsular and exopolysaccharides of adjacent *Vv* cell in biofilms or on host cell surfaces. In some studied *Vv* strains, these matrices contain sugars like D-glucosamine, and N-acetyl-D-quinovosamine (138, 139) - a derivative of GlcNAc (140). Binding these sugars would allow for bacterial cell-cell interactions during various stages of the biofilm life cycle via RTX-adhesins. Likewise, GlcNAc binding enables the

bacterium to bind to chitin (a polymer of GlcNAc) a fundamental structural building block in the shells of the bacterium's natural mollusk and crustaceous hosts (141).

4.4 *Vv*1d CBM meditates bacterial adhesion to host surfaces through glycan interactions

Previous data indicated that *Vv*1d CBM interacts favorably with mannose, fucosylated glycans, and N-acetylglucosylated glycans. Thus, binding to cell types with surface glycomes containing these glycans was tested. More specifically, lower fungal eukaryotes like *C. albicans* and *S. cerevisiae* are largely mannosylated via N- or O-linked oligosaccharides on their surface (142, 143). With confirmation that *Vv*1d CBM bound mannose via the DBC assay, fluorescence microscopy showed observable binding of both yeast species, irrespective of the presence of their endogenous adhesive and aggregative functions. Interaction between *Vv*1d CBM and yeasts suggests the ability of *Vv* to co-adhere to commensal yeasts present on host mucosal surfaces in the oral and gastrointestinal tract. Interestingly, *C. albicans* is already known to co-aggregate and co-adhere with several different oral bacterial species (144), which in the case of *Vv*, may inadvertently allow for retention of pathogenic bacteria during infections. Mucin proteins, like MUC2, are also present in the gastrointestinal tract, and have been proven to bind to *Vv*1d CBM in this study. Thus, it can be assumed that this domain enables bacteria to interact favorably with mucosal proteins present on intestinal or oral surfaces to enable the bacteria to colonize said surfaces.

After colonization of intestinal cells, *Vv* routinely invades the blood stream through the gut-blood barrier, causing septicemia and necrotizing fasciitis (NF) (145, 146). This suggests that it must have the ability to colonize the surfaces of the cardiovascular system or cells that flow through it. This hypothesis is supported by glycan array and DBC assay data that indicated the ability of *Vv*1d CBM to bind to L-fucose and fucosylated glycans, commonly displayed on the

surface of erythrocytes, epithelial and endothelial cells (147–151) . Erythrocytes display various terminally fucosylated glycoproteins and glycolipids like Band 3 (152) and Lewis-antigens that are commonly exploited as binding receptors during infections (153). Whilst these glycans can also contain sialyl termini, NCFG glycan array analysis showed no binding to sialylated glycans (Appendix Table 1), indicating this interaction with erythrocyte surfaces occurs solely through terminal fucose epitopes.

Cytotoxicity is a function rarely reported in other CBM-containing RTX adhesins, besides that of the *Vibrio cholerae* Sugar-binding domain with an unknown domain from FrhA (8). While the cause of aggregation and lysis remains elusive, it may contribute to the pathogenesis of NF by inducing thrombosis through blood clots (154). However, MaxiSorp® plate cell binding assays showed no lysis of red blood cells during any trials, indicating that the *Vv*1d CBM must be free in solution to cause quantifiable lysis. Thus, in fluorescent microscopy assays, *Vv*1d CBM may be aggregating other protein units in solution, causing large scale lysis due to friction between aggregated cells. Further testing needs to be performed with the LBR of *Vv*1d CBM to see if any potential aggregation between CBMs can occur with other adjacent LBDs attached to a larger construct.

For infections in aquaculture, *Vv* likely uses a similar route of binding to fucosylated gastrointestinal fish cells and the respective mucin coating them to allow for multi-organ colonization. Cell-binding assays illustrated RTg cells densely binding to *Vv*1d CBM coated wells, indicating the presence of a specific glycan preferred by the protein. Gill epithelial fish cells are decorated with *O*-glycans containing accessible fucosylated glycans (155), suggesting a target moiety that can be exploited during pathogenic *Vv* infections in fish. Similarly, shellfish contain a wide variety of mannosylated, fucosylated, or GlcNAcylated glycans in their shell

matrix (156), in addition to chitin polymers. Modified-dot blot assays showed that *Vv*1d CBM binds specifically to chitosan (de-acetylated chitin), outlining that *Vv* can use chitin as a binding target during pathogensis. Interestingly, oysters and other mollusks are key environmental targets of *Vv* (157), indicating why CBMs produced in RTX adhesins from the species may have evolved a preference for GlcNAc binding.

4.5 Inhibiting host cell surface binding by bacteria through CBM targeting

These results and recent literature (8) show various RTX adhesins expressed by Gram-negative bacteria contain CBMs that bind to complex glycans on different cell types (64). Thus, developing sugar derivative inhibitors could enable the saturation of RTX adhesin CBMs, potentially attenuating their ability to interact with host surface glycans. To test this hypothesis, GlcNAc and a non-metabolizable, glycoside derivate, β-S-GlcNAc, were employed as inhibitory molecules during binding experiments. Low millimolar concentrations of both compounds (5 mM) consistently inhibited protein binding to various cell types, without causing any visual damage or changes to cell morphology when compared to controls. Furthermore, binding of Caco-2, RTg and RBCs to *Vv*1d CBM was significantly inhibited by addition of 30 mM GlcNAc during cell-binding assays, without any observable morphological damage (not shown). The inhibition appeared more effective against RTg and RBCs as compared to Caco-2 cells, indicating that variation in cell surface glycan type and density may make sugar-based inhibition difficult with certain cell types. The preliminary trials used 30 mM concentrations which are quite elevated, but still suggest that sugar-based inhibition is feasible. Moreover, the affinity of the inhibitor compounds can likely be improved beyond their native forms. This may be possible through addition of C1 and C6 functional groups that allow interactions with polar residues on loops 1 and 3 of the PA14 CBMs.

Native sugar and glycomimetic-based inhibition of bacterial adhesion have already been reported in pre-clinical settings. For example, specific c-glycosides have been used to inhibit *Pseudomonas aeruginosa* carbohydrate-binding proteins LecA & B. This inhibition attenuated any ability of the bacterium to form biofilms and consequently lowered its infectious capability (158). Additionally, development of D-mannose derivatives that inhibit the FimH mannose-binding protein have proven efficacious against pathogenic urinary *E. coli* infections (111, 112), and are currently in pre-clinical phases of drug testing.

While the GlcNAc-CBM ligand-bound crystal solved in this research help elucidate key ligand-binding parameters, further structural studies with GlcNAc derivates should be performed to clarify how addition of non-standard functional groups may affect inhibitor binding. Furthermore, additional ligand-bound crystal studies can be used for structure-based drug design trials. These can aid in the development of inhibitory compounds with higher stability and affinity.

4.6 Conclusion

This thesis focused on the *Vv*1d CBM of *V. vulnificus.* Studying this domain, and other putative RTX-adhesin ligand-binding domains will help to better understand colonization and biofilm maturation dynamics conferred by *Vv*. This work outlines the ability of *Vv*1d CBM to bind a variety of mono- and polysaccharides, with glycans on host cell surfaces acting as key receptors for *Vv* binding. Validation of a GlcNAc and non-metabolizable GlcNAc-derivative-based inhibition of host cell adhesion illustrates how anti-adhesive strategies may be useful in attenuating bacterial infections. Additionally, clinical success of other sugar-derivative inhibitor compounds shows that consecutive development of mannose, fucose, or better GlcNAc derivatives could reasonably potentiate adhesive inhibition across various bacterial species.

Lastly, crystallographic analysis of GlcNAc-bound *Vv*1d CBM agrees with the PA14 family binding mode while suggesting a potential structure-based specificity for sugar ligands that varies between organisms.

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Appendix: Chapter 3 Supplemental Information

Table 1. Fungal strains

Strain	Organism	Genotype	Parental Source
CF026	*Candida albicans* (159)	*MATa his1⁻ / his1⁻ leu2⁻ / leu2⁻ arg4⁻ / arg4⁻*	RBY1133
YMD963	*Saccharomyces cerevisiae* (160, 161)	*MATa leu2d0 ura3d0 his3-1 met15d0 LYS2+*	S288C

Supplementary information for Chapter 3.1

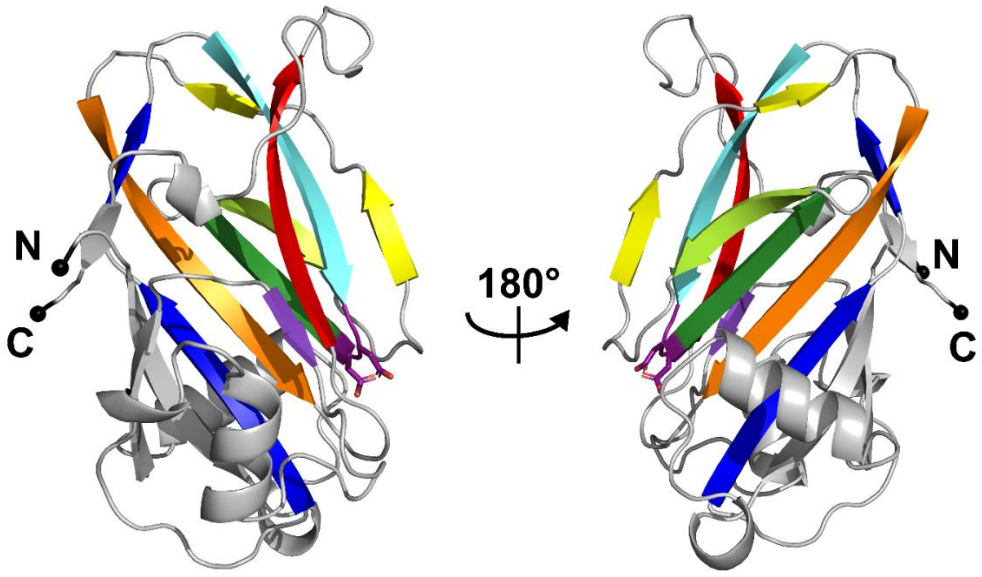


Figure A. Global fold of *Vv*1d CBM and binding motif. Front and back view of the AlphaFold3-predicted tertiary structure of *Vv*1d CBM. A core set of 8 beta strands are coloured in a rainbow pattern from N to C terminus (light purple, blue, cyan, forest green, lime, yellow, orange, and red, respectively), with termini illustrated as black spheres. Other secondary structure insertions or extraneous beta strands coloured grey. D*cis*D homology motif can be seen with residues in stick representation (dark purple).

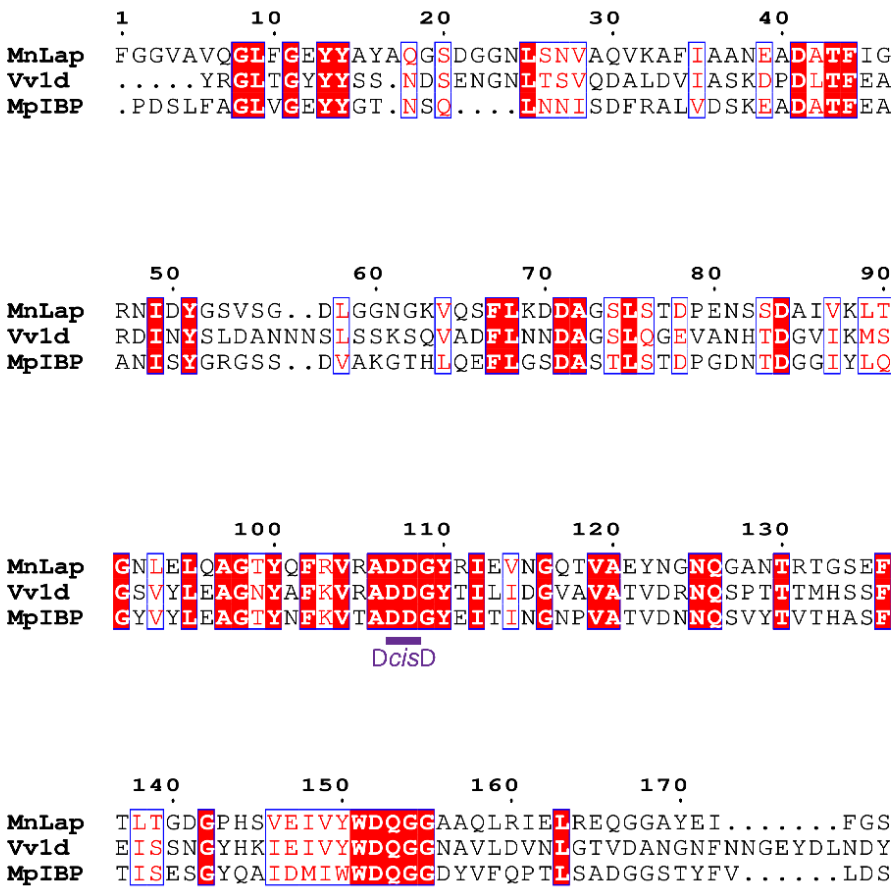


Figure B. Sequence alignments of PA14 family domains from Gram-negative bacteria. *Vv*1d CBM amino acid sequence aligned with amino acid sequences from structurally homologous PA14 family CBMs from *Marinomonas primoryensis* Ice-binding protein and *Marinomonas nauticus*. D*cis*D binding motif outlined in purple. Conservation across protein sequences is represented by strict amino acid similarities (red box), similarities across groups (red character), and similarity in the same amino acid group (blue frame), or no similarity (black). Alignment output visualized using Clustal Omega and ESPript.

> *Vv*1d CBM+split

MGSSHHHHHHSSGLVPRGSHMSVAPVAQDDTVSVYRGLTGYYYSSNDSENGNLTSVQDALDVIA
SKDPDLTFEARDINYSLDANNNSLSSKSQVADFLNNDAGSLQGEVANHTDGVIKMSGSVYLEAG
NYAFKVRADDGYTILIDGVAVATVDRNQSPTTTMHSSFEISSNGYHKIEIVYWDQGGNAVLDVN
LGTVDANGNFNNGEYDLNDYPLIGDTLTTKEGSELVLNSTLLLGNDYDLNDGDTVSIVADGFSS
NDGAVSYDAATGEIIFTPNEGVTGSASFTYTIVDSTGKTDTATVTTLVTPNS

Construct MW (Da)	Theoretical pI	Sequence length
32368.87	4.20	305

Figure C. Recombinant *Vv*1d CBM protein parameters. Sequence of the recombinant *Vv*1d CBM construct, made up of a hexahistidine tag (grey), thrombin cut site (orange), split strand (purple), PA14 CBM (blue), and remaining split extender (pink). Molecular weight (MW, daltons), theoretical pI and sequence length are listed.

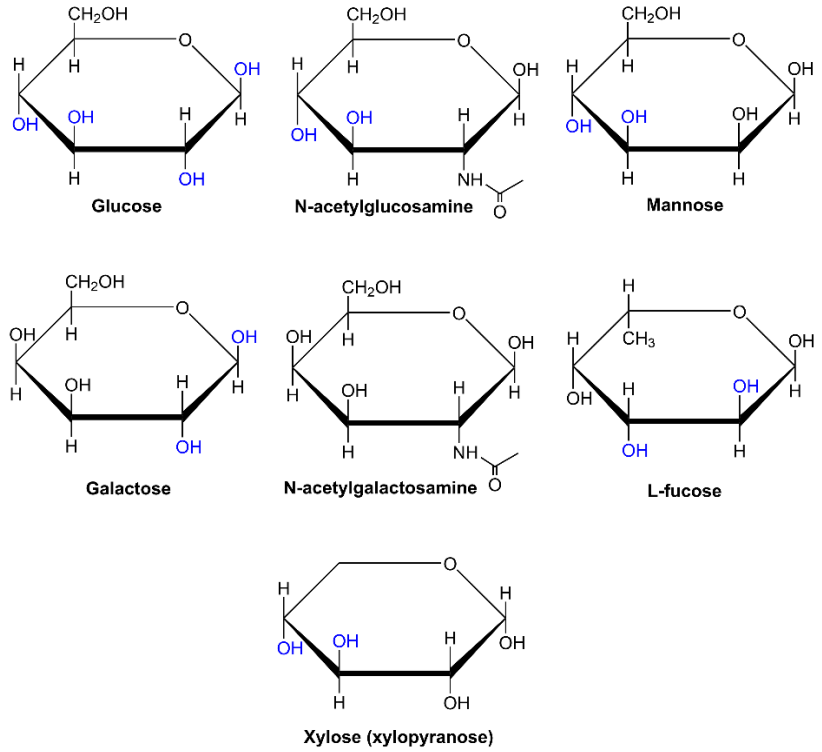


Figure D. Two-dimensional representation of tested monosaccharides. Hayworth projections for Glc, Fuc, Man, Gal, GalNAc, Fuc and Xyl with trans, vicinal, equatorial hydroxyl pairs labelled (blue). Structures shown are

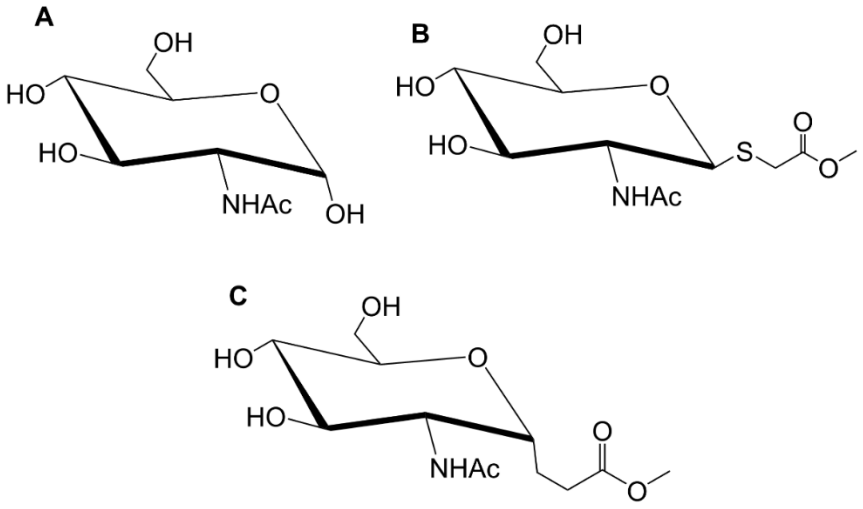


Figure E. Chair conformations of GlcNAc and GlcNAc derivatives. Chair conformation projections of D-GlcNAc (A), β-S-GlcNAc (B), and α-C-GlcNAc (C). NHAc is representative of an N-acetyl functional group.

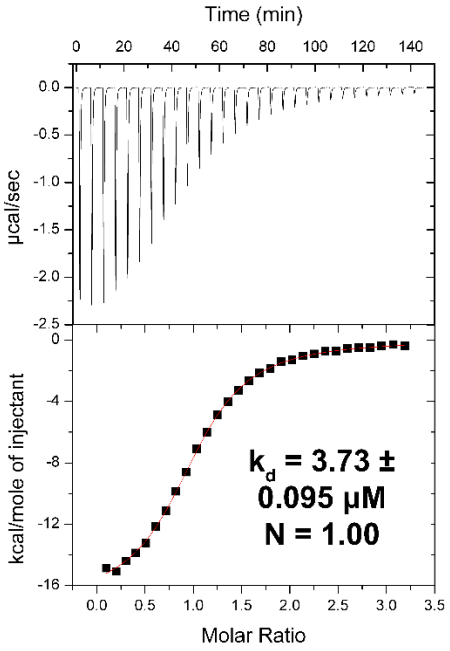


Figure F. ITC data for GlcNAc binding by *Vv*1d CBM. Replicate thermogram measuring energetic interactions between *Vv*1d CBM and GlcNAc. The raw thermodynamic (top) and integrated (bottom) data over the duration of titrations are shown, along with calculated k_d and stoichiometry values. Experimental points are represented as filled squares. Stoichiometry of reaction measured by N.

Supplementary information for Chapter 3.3



Figure G. *Vv*1d CBM crystal morphologies. Representative microscopy image of *Vv*1d CBM crystal morphologies in successful crystallization trials after < 1 month of growth.

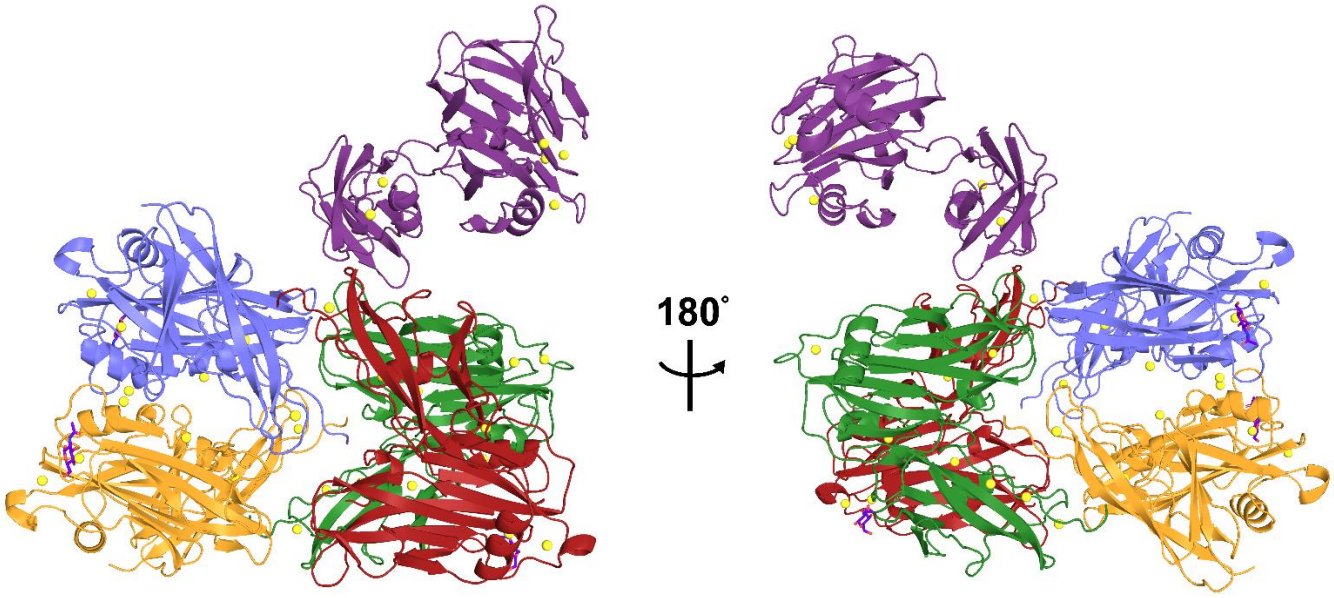


Figure H. *Vv*1d CBM crystal asymmetric unit. Overall packing of the *Vv*1d CBM crystal asymmetric unit, consisting of 5 *Vv*1d CBM subunits (purple, blue, green, light orange, red) with three bound GlcNAc molecules (dark blue) and calcium atoms (yellow).

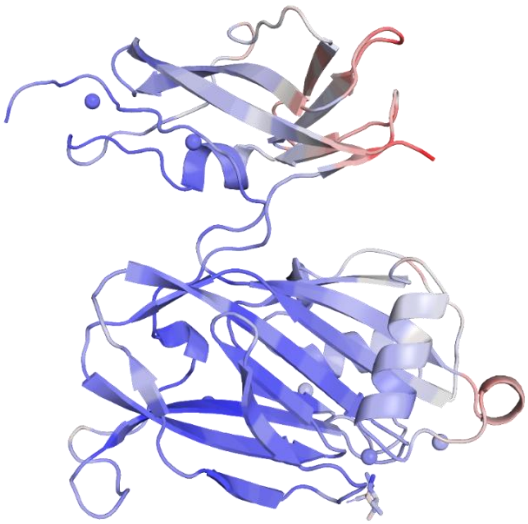


Figure I. *Vv*1d crystal subunit disorder analysis. Measured b-factor of a D-GlcNAc and calcium-bound *Vv*1d CBM asymmetric unit subunit. B-factors are outlined within a spectrum from low (37.5, blue), medium (light grey), and high (red, 173).

Supplementary information for Chapter 3.4

Table 2. Polysaccharides used for modified dot blot assays

Polysaccharide	Supplier: Product No.	Solvent	Final concentration (% w/v)
Chitosan	Sigma-Aldrich: 448869	1% v/v acetic acid	0.5
Pectin	Sigma-Aldrich: P9135	ddH_2O	0.5
Alginate	Sigma-Aldrich: 180947	ddH_2O	0.5
Hyaluronic Acid	Thermo scientific: J66993.03	ddH_2O	0.5

Table 3. Output of NCFG microarray results for *Vv*1d CBM

ID	Glycan class	Name	Glycan string	Average	StDev
1	Maltodextrin	Maltotriose DP3	Glca1-4Glca1-4Glc-AEAB	0	0
2	Maltodextrin	Maltotetraose DP4	Glca1-4Glca1-4Glca1-4Glc-AEAB	0	1
3	Maltodextrin	Maltopentaose DP5	Glca1-4Glca1-4Glca1-4Glca1-4Glc-AEAB	0	0
4	Maltodextrin	Maltohexaose DP6	Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glc-AEAB	0	0
5	Maltodextrin	Maltoheptaose DP7	Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glc-AEAB	0	0
6	Maltodextrin	Maltooctaose DP8	Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glc-AEAB	0	1
7	Maltodextrin	Maltononaose DP9	Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glc-AEAB	1	1
8	Maltodextrin	Maltodecaose DP10	Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glca1-4Glc-AEAB	0	0
9	Cellodextrin	Cellotriose DP3	Glcb1-4Glcb1-4Glc-AEAB	0	0
10	Cellodextrin	Cellotetraose DP4	Glcb1-4Glcb1-4Glcb1-4Glc-AEAB	0	0
11	Cellodextrin	Cellohexaose DP6	Glcb1-4Glcb1-4Glcb1-4Glcb1-4Glcb1-4Glc-AEAB	1	0
12	Cellodextrin	Celloheptaose DP7	Glcb1-4Glcb1-4Glcb1-4Glcb1-4Glcb1-4Glcb1-4Glc-AEAB	0	0
13	Cellodextrin	Cellooctaose DP8	Glcb1-4Glcb1-4Glcb1-4Glcb1-4Glcb1-4Glcb1-4Glcb1-4Glc-AEAB	0	0
14	Xyloglucan	XFG xyloglucan oligosaccharide DP7	Fuca1-2Galb1-2Xyla1-6(Xyla1-6Glcb1-4)Glcb1-4Glc-AEAB	0	0
15	Xyloglucan	XLLG xyloglucan oligosaccharide DP9	Galb1-2Xyla1-6(Galb1-2Xyla1-6(Xyla1-6Glcb1-4)Glcb1-4)Glcb1-4Glc-AEAB	0	0
16	Xyloglucan	XXFG xyloglucan oligosaccharide DP9	Fuca1-2Galb1-2Xyla1-6(Xyla1-6(Xyla1-6Glcb1-4)Glcb1-4)Glcb1-4Glc-AEAB	7	1
17	Xyloglucan	XLFG xyloglucan oligosaccharide DP10	Fuca1-2Galb1-2Xyla1-6(Galb1-2Xyla1-6(Xyla1-6Glcb1-4)Glcb1-4)Glcb1-4Glc-AEAB	0	0
18	Chitin	Chitin oligosaccharides DP3 / Chitintriose	GlcNAcb1-4GlcNAcb1-4GlcNAc-AEAB	1	0
19	Chitin	Chitin oligosaccharides DP4 / Chitintetraose	GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAc-AEAB	7	1
20	Chitin	Chitin oligosaccharides DP6 / Chitinhexaose	GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAc-AEAB	4	2
21	Chitin	Chitin oligosaccharides DP7 / Chitinheptaose	GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAcb1-4GlcNAc-AEAB	2	0
22	Core Structures Type 1	Core type 1 triaose	Galb1-3GlcNAcb1-3Gal-AEAB	1	1
23	Core Structures Type 1	Lacto-N-Triaose / LNT2	GlcNAcb1-3Galb1-4Glc-AEAB	1	1

24	Core Structures Type 1	Lacto-N-tetraose / LNT	Galb1-3GlcNAcb1-3Galb1-4Glc-AEAB	1	1
25	Core Structures Type 1	Para-Lacto-N-hexaose analogue/p-LNH analogue	Galb1-3GlcNAcb1-3Galb1-3GlcNAcb1-3Galb1-4Glc-AEAB	1	0
26	Core Structures Type 2	Core type 2 triaose	Galb1-4GlcNAcb1-3Gal-AEAB	2	1
27	Core Structures Type 2	Galacto-N-neoPentaose / DiLacNAc-b3Galactose	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Gal-AEAB	2	1
28	Core Structures Type 2	Lacto-N-neotetraose / LNnT / neo-LNT	Galb1-4GlcNAcb1-3Galb1-4Glc-AEAB	1	0
29	Core Structures Type 2	Para-Lacto-N-neohexaose / Para-LNnH	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4Glc-AEAB	0	1
30	Core Structures Type 2	Di-N-Acetyl-D-Lactosamine / DiLacNAc	Galb1-4GlcNAcb1-3Galb1-4GlcNAc-AEAB	2	1
31	Core Structures Type 2	Tri-N-Acetyl-D-Lactosamine / TriLacNAc	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAc-AEAB	1	0
32	Core Structures Type 2	Tetra-N-Acetyl-D-Lactosamine / TetraLacNAc	Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAc-AEAB	1	1
33	Globo-series analogues	Globopentaose analogue type 1 / Gb5 analogue type 1	Galb1-3GlcNAcb1-3Gala1-4Galb1-4Glc-AEAB	1	0
34	Globo-series analogues	Globopentaose analogue type 2 / Gb5 analogue type 2	Galb1-4GlcNAcb1-3Gala1-4Galb1-4Glc-AEAB	0	1
35	Globo-series analogues	SSEA-4 hexaose analogue type 1	Neu5Aca2-3Galb1-3GlcNAcb1-3Gala1-4Galb1-4Glc-AEAB	1	1
36	Globo-series analogues	Sialylated-SSEA-4 hexaose analogue type 1	Neu5Aca2-8Neu5Aca2-3Galb1-3GlcNAcb1-3Gala1-4Galb1-4Glc-AEAB	0	1
37	Type 1	Sialylated tetraose type 1 / Sialyl Lewisa precursor	Neu5Aca2-3Galb1-3GlcNAcb1-3Gal-AEAB	1	1
38	Type 1	LS-Tetrasaccharide a / LSTa / Sialyl-lacto-N-tetraose a	Neu5Aca2-3Galb1-3GlcNAcb1-3Galb1-4Glc-AEAB	1	1
39	Type 1	Disialylated pentaose type 1	Neu5Aca2-8Neu5Aca2-3Galb1-3GlcNAcb1-3Gal-AEAB	2	1
40	Type 2	Sialylated tetraose type 2	Neu5Aca2-3Galb1-4GlcNAcb1-3Gal-AEAB	0	0
41	Globo-series	Sialylated globotriose / 3-Sialyl-Gb3	Neu5Aca2-3Gala1-4Galb1-4Glc-AEAB	2	0
42	Stage-Specific Embryonic Antigens	SSEA-4 tetraose / Stage-Specific Embryonic Antigen-4	Neu5Aca2-3Galb1-3GalNAcb1-3Gal-AEAB	1	1
43	Stage-Specific Embryonic Antigens	SSEA-4 hexaose / Stage-Specific Embryonic Antigen-4	Neu5Aca2-3Galb1-3GalNAcb1-3Gala1-4Galb1-4Glc-AEAB	3	0
44	Lacto-series	3'Sialyl-3Fucosyllactose / F-SL	Neu5Aca-3Galb1-4(Fuca1-3)Glc-AEAB	1	0
45	Type 2	Sialylated triaose type 2	Neu5Aca2-3Galb1-4GlcNAc-AEAB	1	1
46	Type 2	LS-Tetrasaccharide d / LSTd / Sialyl-Lacto-N-tetraose d	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4Glc-AEAB	1	0
47	Ganglio-series	GM3 Ganglioside sugar / 3'Sialyllactose / 3'SL / N-Acetylneuraminoyllactose	Neu5Aca2-3Galb1-4Glc-AEAB	1	0
48	Ganglio-series	GD3 Ganglioside sugar / Disialyl lactose / DS-L	Neu5Aca2-8Neu5Aca2-3Galb1-4Glc-AEAB	1	0
49	Ganglio-series	GT3 Ganglioside sugar	Neu5Aca2-8Neu5Acb2-8Neu5Aca2-3Galb1-4Glc-AEAB	1	0
50	Ganglio-series	GM2 Ganglioside sugar	GalNAcb1-4(Neu5Aca2-3)Galb1-4Glc-AEAB	2	1
51	Ganglio-series analogues	GM2 Ganglioside sugar analogue	Galb1-4(Neu5Aca2-3)Galb1-4Glc-AEAB	2	0
52	Ganglio-series	GD2 Ganglioside sugar	GalNAcb1-4(Neu5Aca2-8Neu5Aca2-3)Galb1-4Glc-AEAB	1	0
53	Ganglio-series	GM1a Ganglioside sugar	Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glc-AEAB	1	1
54	Ganglio-series	GM1b Ganglioside sugar	Neu5Aca2-3Galb1-3GalNAcb1-4Galb1-4Glc-AEAB	1	0
55	Ganglio-series	GD1a Ganglioside sugar	Neu5Aca2-3Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glc-AEAB	2	2
56	Ganglio-series	Fucosyl GM1 Ganglioside sugar	Fuca1-2Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glc-AEAB	1	1
57	Ganglio-series	GD1b Ganglioside sugar	Galb1-3GalNAcb1-4(Neu5Aca2-8Neu5Aca2-3)Galb1-4Glc-AEAB	2	1

58	Blood Group H antigens (Blood group O)	Blood group H antigen triaose type 1	Fuca1-2Galb1-3GlcNAc-AEAB	2	1
59	Blood Group H antigens (Blood group O)	Blood group H antigen tetraose type 1	Fuca1-2Galb1-3GlcNAcb1-3Gal-AEAB	2	1
60	Blood Group H antigens (Blood group O)	Blood group H antigen pentaose type 1 / Lacto-N-fucopentaose I / LNFP I	Fuca1-2Galb1-3GlcNAcb1-3Galb1-4Glc-AEAB	2	0
61	IsoGlobo-series analogues	Isoglobo-H analogue type 1	Fuca1-2Galb1-3GlcNAcb1-3Gala1-3Galb1-4Glc-AEAB	2	0
62	Globo-series	Globo-H hexaose / Stage Specific Embryonic Antigen 3b / SSEA-3b	Fuca1-2Galb1-3GalNAcb1-3Gala1-4Galb1-4Glc-AEAB	3	0
63	Globo-series analogues	Globo-H analogue type 1	Fuca1-2Galb1-4GlcNAcb1-3Gala1-4Galb1-4Glc-AEAB	1	1
64	Blood Group H antigens (Blood group O)	Blood group H antigen triaose type 2	Fuca1-2Galb1-4GlcNAc-AEAB	1	0
65	IsoGlobo-series analogues	Isoglobo-H analogue type 2	Fuca1-2Galb1-4GlcNAcb1-3Gala1-3Galb1-4Glc-AEAB	1	1
66	Globo-series analogues	Globo-H analogue type 2	Fuca1-2Galb1-4GlcNAcb1-3Gala1-4Galb1-4Glc-AEAB	2	0
67	Blood Group H antigens (Blood group O)	Blood group H antigen tetraose type 4 Globo H tetraose	Fuca1-2Galb1-3GalNAcb1-3Gal-AEAB	2	0
68	Blood Group H antigens (Blood group O)	Blood group H antigen triaose type 5 / 2-Fucosyllactose / 2-FL	Fuca1-2Galb1-4Glc-AEAB	0	1
69	Blood group oligosaccharide analogues	2'2 Difucosyllactose	Fuca1-2Galb1-4(Fuca1-2)Glc-AEAB	2	1
70	Lacto-series	Lactodifucotetraose / DF-L / Difucosyllactose	Fuca1-2Galb1-4(Fuca1-3)Glc-AEAB	2	1
71	Blood group A antigens	Blood group A antigen triaose	GalNAca1-3(Fuca1-2)Gal-AEAB	1	1
72	Blood group A antigens	Blood group A antigen tetraose type 1	GalNAca1-3(Fuca1-2)Galb1-3GlcNAc-AEAB	0	1
73	Blood group A antigens	Blood group A antigen pentaose type 1	GalNAca1-3(Fuca1-2)Galb1-3GlcNAcb1-3Gal-AEAB	2	1
74	Blood Group A antigens	Blood group A antigen hexaose type 1 / PII	GalNAca1-3(Fuca1-2)Galb1-3GlcNAcb1-3Galb1-4Glc-AEAB	2	1
75	Blood group A antigens	Blood group A antigen tetraose type 2	GalNAca1-3(Fuca1-2)Galb1-4GlcNAc-AEAB	2	1
76	Blood group A antigens	Blood group A antigen pentaose type 2	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-3Gal-AEAB	2	0
77	Blood Group A antigens	Blood group A antigen hexaose type 2	GalNAca1-3(Fuca1-2)Galb1-4GlcNAcb1-3Galb1-4Glc-AEAB	-1	1
78	3fucosyllactose core	3-Fucosylated Blood group A tetraose	GalNAca1-3(Fuca1-2)Galb1-4(Fuca1-3)Glc-AEAB	1	1
79	Blood Group A antigens	Blood group A antigen pentaose type 4	GalNAca1-3(Fuca1-2)Galb1-3GalNAcb1-3Gal-AEAB	2	1
80	Blood group A antigens	Blood group A antigen tetraose type 5 I	GalNAca1-3(Fuca1-2)Galb1-4Glc-AEAB	2	0
81	Blood group oligosaccharide analogues	Fucosylated A antigen type 5	GalNAca1-3(Fuca1-2)Galb1-4(Fuca1-2)Glc-AEAB	3	5
82	Globo-series	Globo-A heptaose	GalNAca1-3(Fuca1-2)Galb1-3GalNAcb1-3Gala1-4Galb1-4Glc-AEAB	2	1
83	Blood group B antigens	Blood group B antigen tetraose type 1	Gala1-3(Fuca1-2)Galb1-3GlcNAc-AEAB	1	1
84	Blood group B antigens	Blood group B antigen pentaose type 1	Gala1-3(Fuca1-2)Galb1-3GlcNAcb1-3Gal-AEAB	2	0
85	Blood group B antigens	Blood group B antigen hexaose type 1	Gala1-3(Fuca1-2)Galb1-3GlcNAcb1-3Galb1-4Glc-AEAB	1	1
86	Blood group B antigens	Blood group B antigen tetraose type 2	Gala1-3(Fuca1-2)Galb1-4GlcNAc-AEAB	2	1

87	Blood group B antigens	Blood group B antigen pentaose type 2	Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-3Gal-AEAB	2	0
88	Blood group B antigens	Blood group B antigen hexaose type 2	Gala1-3(Fuca1-2)Galb1-4GlcNAcb1-3Galb1-4Glc-AEAB	2	2
89	Blood group B antigens	Blood group B antigen pentasaccharide type 4	Gala1-3(Fuca1-2)Galb1-3GalNAcb1-3Gal-AEAB	0	1
90	Globo-series	Globo-B heptaose	Gala1-3(Fuca1-2)Galb1-3GalNAcb1-3Gala1-4Galb1-4Glc-AEAB	1	1
91	Blood group B antigens	Blood group B antigen tetraose type 5	Gala1-3(Fuca1-2)Galb1-4Glc-AEAB	2	0
92	Blood group oligosaccharide analogues	Fucosylated B antigen type 5	Gala1-3(Fuca1-2)Galb1-4(Fuca1-2)Glc-AEAB	1	1
93	3fucosyllactose core	3-Fucosylated Blood group B tetraose	Gala1-3(Fuca1-2)Galb1-4(Fuca1-3)Glc-AEAB	0	0
94	Forssman antigens	Forssman antigen triaose	GalNAca1-3GalNAcb1-3Gal-AEAB	0	1
95	Forssman antigens	Forssman antigen pentaose	GalNAca1-3GalNAcb1-3Gala1-4Galb1-4Glc-AEAB	3	2
96	Globo-series	Globotetraose / Gb4 / P antigen	GalNAcb1-3Gala1-4Galb1-4Glc-AEAB	0	1
97	Isoglobo-Series	Isoforssman antigen pentaose	GalNAca1-3GalNAcb1-3Gala1-3Galb1-4Glc-AEAB	1	1
98	Globo-series	Globotriaose / Gb3 / Pk antigen	Gala1-4Galb1-4Glc-AEAB	1	1
99	Globo-series	P1 antigen / Globotriaose analogue type 2	Gala1-4Galb1-4GlcNAc-AEAB	3	0
100	Lewis Antigens	Sialyl LewisX (SLeX) tetraose	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAc-AEAB	1	0
101	Lewis Antigens	Sialyl LewisX (sLeX) pentaose	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-3Gal-AEAB	2	1
102	Lewis Antigens	LewisX (LeX) triaose / SSEA-1 / CD15	Galb1-4(Fuca1-3)GlcNAc-AEAB	2	1
103	Lewis Antigens	LewisX (LeX) tetraose	Galb1-4(Fuca1-3)GlcNAcb1-3Gal-AEAB	1	0
104	Lewis Antigens	LewisX (LeX) hexaose / Lacto-N-neoDiFucohexaose / LNnDFH	Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)Glc-AEAB	1	0
105	Lewis Antigens	LewisY (LeY) tetraose	Fuca1-2Galb1-4(Fuca1-3)GlcNAc-AEAB	0	1
106	Lewis Antigens	LewisY (LeY) pentaose	Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-3Gal-AEAB	0	1
107	Lewis Antigens	Lewisa (Lea) tetraose	Galb1-3(Fuca1-4)GlcNAcb1-3Gal-AEAB	1	0
108	Lewis Antigens	Lewisb (Leb) tetraose	Fuca1-2Galb1-3(Fuca1-4)GlcNAc-AEAB	1	0
109	Lewis Antigens	Lewisb (Leb) pentaose	Fuca1-2Galb1-3(Fuca1-4)GlcNAcb1-3Gal-AEAB	1	1
110	Blood group A antigens	Blood group A Lewisb antigen pentaose type 1	GalNAca1-3(Fuca1-2)Galb1-3(Fuca1-4)GlcNAc-AEAB	2	1
111	Lewis Antigens	Blood group A Lewisyantigen pentaose type 2	GalNAca1-3(Fuca1-2)Galb1-4(Fuca1-3)GlcNAc-AEAB	0	1
112	Lewis Antigens	Blood group B Lewisb antigen pentaose type 1	Gala1-3(Fuca1-2)Galb1-3(Fuca1-4)GlcNAc-AEAB	0	1
113	Lewis Antigens	Blood group B LewisY antigen pentaose type 2	Gala1-3(Fuca1-2)Galb1-4(Fuca1-3)GlcNAc-AEAB	1	1
114	Lacto-series	3 Fucosyllactose / 3-FL	Galb1-4(Fuca1-3)Glc-AEAB	0	1
115	Lacto-series	Lacto-N-neofucopentaose / LNnFP-V / LNFP VI	Galb1-4GlcNAcb1-3Galb1-4(Fuca1-3)Glc-AEAB	2	1
116	Lacto-series	Lacto-N-fucopentaose V / LNFP V	Galb1-3GlcNAcb1-3Galb1-4(Fuca1-3)Glc-AEAB	2	1
117	Isoglobo-Series	Isoglobotriaose / iGb3 / 3'-Galactosyl-lactose / Linear B-6 trisaccharide	Gala1-3Galb1-4Glc-AEAB	1	0
118	Galili & Xeno antigens	Xeno antigen type 1	Gala1-3Galb1-3GlcNAc-AEAB	0	1
119	Galili & Xeno antigens	Xeno antigen type 2 / Galili antigen triaose / Gala1-3 epitope / Linear B-2 trisaccharide	Gala1-3Galb1-4GlcNAc-AEAB	1	1
120	Galili & Xeno antigens	Xeno Lewisx / aGal-Lewisx	Gala1-3Galb1-4(Fuca1-3)GlcNAc-AEAB	0	0

121	3fucosyllactose core	a3Galactosyl-3Fucosyllactose	Gala1-3Galb1-4(Fuca1-3)Glc-AEAB	1	1
122	Galili & Xeno antigens	Galili antigen heptaose	Gala1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4Glc-AEAB	0	0
123	Galili & Xeno antigens	Galili antigen pentaose	Gala1-3Galb1-4GlcNAcb1-3Galb1-4Glc-AEAB	1	1
124	Galili & Xeno antigens	Galili antigen undecaose	Gala1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4Glc-AEAB	-1	1
125	Di-Tri-Tetrasaccharides	Gala1-4Gala1-4Galb1-4Glc	Gala1-4Gala1-4Galb1-4Glc-AEAB	2	1
126	Di-Tri-Tetrasaccharides	Gala1-4Gala1-Gb3	Gala1-4Gala1-4Gala1-4Galb1-4Glc-AEAB	1	0
127	Glucuronylated oligosaccharides	Glucuronyl-lactose / HNK 1 epitope precursor	GlcAb1-3Galb1-4Glc-AEAB	2	1
128	Glucuronylated oligosaccharides	Glucuronyl-LNT / Glucuronyl-Lacto-N-tetraose	GlcAb1-3Galb1-3GlcNAcb1-3Galb1-4Glc-AEAB	2	1
129	Isoglobo-Series	Isoglobotetraose / iGb4 / Cytolipin R	GalNAcb1-3Gala1-3Galb1-4Glc-AEAB	0	0
130	Isoglobo-Series	Isoglobopentaose / iGb5	Galb1-3GalNAcb1-3Gala1-3Galb1-4Glc-AEAB	0	0
131	IsoGlobo-series analogues	Isoglobopentaose analogue type 1 / iGb5 analogue type 1	Galb1-3GlcNAcb1-3Gala1-3Galb1-4Glc-AEAB	2	1
132	IsoGlobo-series analogues	Sialylated isoglobopentaose analogue type 1 / Sialylated iGb5 analogue type 1	Neu5Aca2-3Galb1-3GlcNAcb1-3Gala1-3Galb1-4Glc-AEAB	0	1
133	IsoGlobo-series analogues	Sialylated isoglobopentaose analogue type 2 / Sialylated iGb5 analogue type 2	Neu5Aca2-3Galb1-4GlcNAcb1-3Gala1-3Galb1-4Glc-AEAB	1	1
134	Ganglio-series analogues	GalGalNac-aGM1 (GalGalNAc asialo GM1) Ganglioside sugar	Galb1-3GalNAcb1-4Galb1-3GalNAcb1-4Galb1-4Glc-AEAB	-1	1
135	Globo-series	Globopentaose (Gb5) / Stage Specific Embryonic Antigen 3a / SSEA-3a	Galb1-3GalNAcb1-3Gala1-4Galb1-4Glc-AEAB	2	1
136	Core Structure	Core type 4 / Gb5 Triaose core structure	Galb1-3GalNAcb1-3Gal-AEAB	2	1
137	Ganglio-series	aGM1 / Ganglioside sugar asialo GM1 / GA1	Galb1-3GalNAcb1-4Galb1-4Glc-AEAB	1	0
138	Ganglio-series	aGM2 / Ganglioside sugar / asialo GM2 / GA2	GalNAcb1-4Galb1-4Glc-AEAB	2	1
139	N-glycans	Man5-AEAB	Mana1-6(Mana1-3)Mana1-6(Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-AEAB	1	0
140	N-glycans	A2G0-Asn	GlcNAcb1-2Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	5	1
141	N-glycans	224A3G0-Asn	GlcNAcb1-2Mana1-6(GlcNAcb1-4(GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	8	1
142	N-glycans	226A3G0-Asn	GlcNAcb1-6(GlcNAcb1-2)Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	6	1
143	N-glycans	A2G2-Asn	Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	2	1
144	N-glycans	NA2-AEAB	Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-AEAB	2	1
145	N-glycans	224A3G3-Asn	Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-4(Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	1	0
146	N-glycans	226A3G3-Asn	Galb1-4GlcNAcb1-6(Galb1-4GlcNAcb1-2)Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	1	1
147	N-glycans	23SA2-A2G2-Asn	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	0	0
148	N-glycans	23SA3-224A3G3-Asn	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-4(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	0	0
149	N-glycans	23SA3-226A3G3-Asn	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-6(Neu5Aca2-3Galb1-	0	0

			4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn		
150	N-glycans	26SA2-A2G2-Asn	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	0	0
151	N-glycans	26SA2-A2-AEAB	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-AEAB	2	1
152	N-glycans	26SA3-224A3G3-Asn	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-4(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	0	0
153	N-glycans	26SA3-226A3G3-Asn	Neu5Aca2-6Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb1-Asn	0	0
154	O-Glycans	Tri-GalNAc	AcNH-{GalNAca1-T}{GalNAca1-T}{GalNAca1-T}-CONH(CH2)3NH2	0	0
155	Negative Control	PHOSPHATE BUFFER (pH 8.5)	PHOSPHATE BUFFER (pH 8.5)	2	2

Table 4. Full LiGA enrichment for BPS1 vs. BSA control.

Barcode	Linkage chain	Test (CPM ± SD)	Reference (CPM ± SD)	log2FC	FC	p-value	q-value	order	n
a-GalNAc-COOH-[135]	GalNAca-COOH	117.5 ± 88.6	152.4 ± 99.9	-0.53	0.69	0.501	0.700	1	9
a-GalNAc-COOH-[540]	GalNAca-COOH	117.2 ± 85.1	191.6 ± 122.2	-0.81	0.57	0.347	0.571	2	7
a-GalNAc-COOH-[1350]	GalNAca-COOH	318.0 ± 383.8	308.0 ± 326.7	0.12	1.08	0.535	0.716	3	9
b-GalNAc-Az-[10]	GalNAcb-PEG3-Az	136.1 ± 181.7	235.5 ± 233.6	-0.63	0.64	0.493	0.570	4	5
b-GalNAc-Az-[20]	GalNAcb-PEG3-Az	167.0 ± 175.4	247.5 ± 271.5	-0.33	0.80	0.623	0.727	5	5
b-GalNAc-Az-[50]	GalNAcb-PEG3-Az	347.1 ± 311.5	315.3 ± 251.1	0.06	1.04	0.653	0.842	6	5
b-GalNAc-Az-[100]	GalNAcb-PEG3-Az	241.7 ± 128.0	519.3 ± 306.3	-1.10	0.47	0.083	0.175	7	4
b-GalNAc-Az-[500]	GalNAcb-PEG3-Az	934.6 ± 379.4	700.0 ± 207.9	0.38	1.30	0.310	0.500	8	5
b-GalNAc-Az-[1000]	GalNAcb-PEG3-Az	863.7 ± 783.7	642.0 ± 567.5	-0.06	0.96	0.317	0.512	9	5
b-GalNAc-COOH-[10]	GalNAcb-COOH	107.7 ± 70.2	172.9 ± 116.3	-0.54	0.69	0.544	0.670	10	5
b-GalNAc-COOH-[20]	GalNAcb-COOH	126.9 ± 49.2	179.4 ± 58.8	-0.54	0.69	0.417	0.648	11	4
b-GalNAc-COOH-[50]	GalNAcb-COOH	121.3 ± 97.0	170.8 ± 102.8	-0.62	0.65	0.614	0.718	12	4
b-GalNAc-COOH-[135]	GalNAcb-COOH	35.2 ± 47.2	19.3 ± 26.2	0.31	1.24	0.428	0.647	13	7
b-GalNAc-COOH-[540]	GalNAcb-COOH	116.9 ± 222.0	119.5 ± 208.8	0.30	1.23	0.653	0.829	14	8
b-GalNAc-COOH-[1350]	GalNAcb-COOH	388.6 ± 324.1	327.5 ± 295.8	0.22	1.16	0.639	0.800	15	7
Tri-GalNAc-COOH-[100]	(_-D-GalNAc-sp)3-NHC(O)-PEG5-COOH	237.2 ± 51.6	282.9 ± 79.2	-0.23	0.85	0.551	0.755	16	5
Tri-GalNAc-COOH-[500]	(_-D-GalNAc-sp)3-NHC(O)-PEG5-COOH	80.2 ± 80.8	106.6 ± 96.8	-0.29	0.82	0.723	0.846	17	5
Tri-GalNAc-COOH-[1000]	(_-D-GalNAc-sp)3-NHC(O)-PEG5-COOH	193.1 ± 188.7	240.0 ± 228.6	-0.44	0.74	0.589	0.794	18	5
GN-[590]	GlcNAc(_-Sp	30.6 ± 46.6	7.3 ± 10.3	2.06	4.18	0.134	0.337	19	1
GN-[1050]	GlcNAc(_-Sp	5905.6 ± 1063.7	240.3 ± 53.8	4.62	24.58	1.3e-26	6.7e-25	20	1

Lac-peg4-[1080]	Gal(_1-4)Glc(_-P4	1402.2 ± 194.3	1832.5 ± 124.2	-0.39	0.77	0.108	0.292	21	1
Lac-[1030]	Gal(_1-4)Glc(_-Sp	65.9 ± 25.1	78.8 ± 31.2	-0.26	0.84	0.762	0.961	22	1
Lac-[1240]	Gal(_1-4)Glc(_-Sp	120.1 ± 65.0	109.4 ± 37.1	0.13	1.10	0.849	1.000	23	1
LacNAc, LN-[970]	Gal(_1-4)GlcNAc(_-Sp	2053.1 ± 563.9	1656.7 ± 362.8	0.31	1.24	0.201	0.452	24	1
LacNAc, LN-[1110]	Gal(_1-4)GlcNAc(_-Sp	11.5 ± 24.7	10.3 ± 11.7	0.17	1.12	0.916	1.000	25	1
Lec-[680]	Gal(_1-3)GlcNAc(_-Sp	12391.1 ± 1486.2	2885.0 ± 245.9	2.10	4.29	4.5e-14	8.4e-13	26	1
Galili-tri-[1000]	Gal(_1-3)Gal(_1-4)Glc(_-Sp	310.2 ± 170.6	278.4 ± 62.2	0.16	1.11	0.741	0.948	27	1
Pk-[860]	Gal(_1-4)Gal(_1-4)Glc(_-Sp	125.9 ± 79.9	167.1 ± 41.5	-0.41	0.75	0.528	0.775	28	1
Gala3-type1-[350]	Gal(_1-3)Gal(_1-3)GlcNAc(_-Sp	47.0 ± 48.3	131.7 ± 46.2	-1.49	0.36	0.083	0.236	29	1
B2 tri-[350]	Gal(_1-3)Gal(_1-4)GlcNAc(_-Sp	4247.9 ± 664.3	2064.8 ± 410.6	1.04	2.06	1.5e-05	1.3e-04	30	1
P1 tri-[620]	Gal(_1-4)Gal(_1-4)GlcNAc(_-Sp	3580.8 ± 492.9	2297.4 ± 354.8	0.64	1.56	0.005	0.025	31	1
LacDiNAc-[50]	GalNAc(_1-4)GlcNAc(_-Sp	4859.9 ± 914.0	46145.8 ± 2719.2	-3.25	0.11	2.0e-20	5.1e-19	32	1
LNT-2-[430]	GlcNAc(_1-3)Gal(_1-4)Glc(_-Sp	627.8 ± 97.2	15.3 ± 16.7	5.36	41.07	9.7e-12	1.7e-10	33	1
GNLN-[810]	GlcNAc(_1-3)Gal(_1-4)GlcNAc(_-Sp	7453.8 ± 1977.6	289.9 ± 203.3	4.68	25.71	4.5e-12	7.9e-11	34	1
3'GN type1-[860]	GlcNAc(_1-3)Gal(_1-3)GlcNAc(_-Sp	15902.9 ± 2782.8	446.3 ± 158.3	5.16	35.64	1.9e-28	1.6e-26	35	1
LNnT-[240]	Gal(_1-4)GlcNAc(_1-3)Gal(_1-4)Glc(_-Sp	76.6 ± 51.2	137.5 ± 39.0	-0.84	0.56	0.266	0.533	36	1
Globoside-P-[730]	GalNAc(_1-3)Gal(_1-4)Gal(_1-4)Glc(_-Sp	591.5 ± 227.0	1094.8 ± 142.5	-0.89	0.54	0.004	0.020	37	1
Globoside-P-[1030]	GalNAc(_1-3)Gal(_1-4)Gal(_1-4)Glc(_-Sp	1515.7 ± 219.0	860.2 ± 364.5	0.82	1.76	0.004	0.019	38	1
P1 tetra-[970]	GalNAc(_1-3)Gal(_1-4)Gal(_1-4)GlcNAc(_-Sp	1719.2 ± 650.0	1903.3 ± 347.3	-0.15	0.90	0.550	0.799	39	1
P1 penta-[620]	Gal(_1-4)Gal(_1-4)GlcNAc(_1-3)Gal(_1-4)Glc(_-Sp	2229.5 ± 453.6	2224.5 ± 627.7	0.00	1.00	0.989	1.000	40	1
P1x penta-[620]	Gal(_1-3)GalNAc(_1-3)Gal(_1-4)Gal(_1-4)GlcNAc(_-Sp	214.8 ± 78.8	471.3 ± 149.3	-1.13	0.46	0.014	0.057	41	1
Tri-LN-[380]	(Galb1-4GlcNAcb1-3)3b-Sp	804.1 ± 169.7	898.9 ± 376.5	-0.16	0.89	0.595	0.838	42	1
Di-N3-[970]	Fuc(_1-2)Gal(_-Sp	654.2 ± 217.7	111.6 ± 51.3	2.55	5.86	6.4e-07	6.9e-06	43	1
H type 1-[700]	Fuc(_1-2)Gal(_1-3)GlcNAc(_-Sp	1303.0 ± 160.3	760.0 ± 276.2	0.78	1.71	0.007	0.031	44	1
H-type-1-COOH-[1350]	Fuca1-2Galb1-3GlcNAcb-COOH	2040.0 ± 436.1	1348.7 ± 260.6	0.60	1.51	0.015	0.059	45	1
H-type-2-COOH-[135]	Fuca1-2Galb1-4GlcNAcb-COOH	1069.3 ± 275.2	971.6 ± 313.3	0.14	1.10	0.620	0.858	46	1
H-type-2-COOH-[540]	Fuca1-2Galb1-4GlcNAcb-COOH	16602.1 ± 2131.6	981.6 ± 92.2	4.08	16.91	1.2e-25	5.2e-24	47	1
H-type-2-COOH-[1350]	Fuca1-2Galb1-4GlcNAcb-COOH	35643.0 ± 4030.8	1069.9 ± 95.3	5.06	33.32	2.9e-29	3.6e-27	48	1
H-type-3-COOH-[100]	Fuc_(1-2)Gal_(1-3)GalNAc-_-COOH	127.2 ± 86.9	108.5 ± 73.0	0.17	1.12	0.761	0.896	49	4
H-type-3-COOH-[500]	Fuc_(1-2)Gal_(1-3)GalNAc-_-COOH	128.2 ± 181.4	71.5 ± 103.1	-0.10	0.93	0.447	0.614	50	5
H-type-3-COOH-[1000]	Fuc_(1-2)Gal_(1-3)GalNAc-_-COOH	170.0 ± 206.4	97.4 ± 100.0	0.40	1.32	0.615	0.728	51	5
H-type-4-COOH-[100]	Fuc_(1-2)Gal_(1-3)GalNAc-_-COOH	152.2 ± 145.8	95.7 ± 92.4	0.18	1.13	0.476	0.674	52	5
H-type-4-COOH-[500]	Fuc_(1-2)Gal_(1-3)GalNAc-_-COOH	204.6 ± 295.9	52.7 ± 58.0	1.00	2.01	0.438	0.503	53	5
H-type-4-COOH-[1000]	Fuc_(1-2)Gal_(1-3)GalNAc-_-COOH	7.5 ± 8.2	3.6 ± 3.4	0.70	1.63	0.715	0.808	54	5

H-type-5-COOH-[100]	Fuc_(1-2)Gal_(1-3)Gal-_-COOH	571.7 ± 467.8	112.2 ± 89.1	1.83	3.56	0.202	0.210	55	5
H-type-5-COOH-[500]	Fuc_(1-2)Gal_(1-3)Gal-_-COOH	2897.5 ± 2771.6	89.8 ± 95.8	3.73	13.25	0.207	0.223	56	5
H-type-5-COOH-[1000]	Fuc_(1-2)Gal_(1-3)Gal-_-COOH	7611.4 ± 7291.4	211.4 ± 217.1	4.97	31.36	0.024	0.064	57	5
H type 6-[950]	Fuc(_1-2)Gal(_1-4)Glc(_-Sp	5430.8 ± 987.7	297.4 ± 61.1	4.19	18.26	7.0e-25	2.6e-23	58	1
H-type-6-COOH-[100]	Fuc_(1-2)Gal_(1-4)Glc-_-COOH	85.5 ± 70.3	102.8 ± 94.4	-0.18	0.88	0.622	0.796	59	5
H-type-6-COOH-[500]	Fuc_(1-2)Gal_(1-4)Glc-_-COOH	2867.2 ± 2722.9	124.0 ± 144.0	3.93	15.23	0.027	0.074	60	5
H-type-6-COOH-[1000]	Fuc_(1-2)Gal_(1-4)Glc-_-COOH	2728.7 ± 1624.7	40.3 ± 23.4	4.88	29.50	0.200	0.200	61	5
A type 1-[700]	GalNAc(_1-3)[Fuc(_1-2)]Gal(_1-3)GlcNAc(_-Sp	3522.5 ± 569.3	3886.6 ± 494.4	-0.14	0.91	0.519	0.770	62	1
A-type-1-COOH-[135]	GalNAca1-3[Fuca1-2]Galb1-3GlcNAcb-COOH	362.2 ± 484.7	401.6 ± 729.2	0.35	1.28	0.506	0.637	63	8
A-type-1-COOH-[540]	GalNAca1-3[Fuca1-2]Galb1-3GlcNAcb-COOH	2131.2 ± 1415.9	966.5 ± 736.6	1.30	2.46	5.8e-04	0.003	64	8
A-type-1-COOH-[1350]	GalNAca1-3[Fuca1-2]Galb1-3GlcNAcb-COOH	57.2 ± 141.8	40.0 ± 97.0	0.11	1.08	0.875	0.928	65	7
A type 2-[920]	GalNAc(_1-3)[Fuc(_1-2)]Gal(_1-4)GlcNAc(_-Sp	1445.2 ± 430.0	1306.2 ± 254.7	0.15	1.11	0.566	0.813	66	1
A-type-2-COOH-[135]	GalNAca1-3[Fuca1-2]Galb1-4GlcNAcb-COOH	618.4 ± 220.3	470.4 ± 106.4	0.39	1.31	0.268	0.533	67	1
A-type-2-COOH-[540]	GalNAca1-3[Fuca1-2]Galb1-4GlcNAcb-COOH	3188.6 ± 613.9	3003.4 ± 587.3	0.09	1.06	0.702	0.926	68	1
A-type-2-COOH-[1350]	GalNAca1-3[Fuca1-2]Galb1-4GlcNAcb-COOH	2309.9 ± 219.2	1897.0 ± 380.5	0.28	1.22	0.218	0.478	69	1
A-type-3-COOH[135]	GalNAca1-3[Fuca1-2]Galb1-3GalNAcb-COOH	133.8 ± 103.8	186.7 ± 147.8	-0.49	0.71	0.567	0.691	70	8
A-type-3-COOH[1350]	GalNAca1-3[Fuca1-2]Galb1-3GalNAcb-COOH	113.9 ± 123.6	129.5 ± 137.6	-0.20	0.87	0.572	0.752	71	9
A-type-3-COOH[540]	GalNAca1-3[Fuca1-2]Galb1-3GalNAcb-COOH	527.8 ± 1127.9	152.0 ± 130.2	0.51	1.42	0.557	0.672	72	8
A-type-4-COOH-[100]	GalNAc_(1-3)[Fuc_(1-2)]Gal_(1-3)GalNAc-_-COOH	107.5 ± 36.1	92.2 ± 46.2	0.33	1.26	0.616	0.833	73	5
A-type-4-COOH-[500]	GalNAc_(1-3)[Fuc_(1-2)]Gal_(1-3)GalNAc-_-COOH	274.3 ± 443.1	180.5 ± 306.5	0.05	1.03	0.535	0.645	74	5
A-type-4-COOH-[1000]	GalNAc_(1-3)[Fuc_(1-2)]Gal_(1-3)GalNAc-_-COOH	126.2 ± 136.4	119.0 ± 142.0	-0.13	0.91	0.738	0.907	75	5
A-type-5-COOH-[135]	GalNAca1-3[Fuca1-2]Galb1-4Galb-COOH	166.5 ± 69.8	392.0 ± 50.2	-1.24	0.42	0.013	0.056	76	1
A-type-5-COOH-[540]	GalNAca1-3[Fuca1-2]Galb1-4Galb-COOH	791.4 ± 235.2	519.1 ± 140.9	0.61	1.52	0.065	0.199	77	1
A-type-5-COOH-[1350]	GalNAca1-3[Fuca1-2]Galb1-4Galb-COOH	7952.7 ± 1096.7	3199.7 ± 319.3	1.31	2.49	6.2e-08	8.2e-07	78	1
A type 6-[590]	GalNAc(_1-3)[Fuc(_1-2)]Gal(_1-4)Glc(_-Sp	1751.7 ± 126.2	3126.9 ± 685.6	-0.84	0.56	5.9e-04	0.004	79	1
A-type-6-COOH-[135]	GalNAca1-3[Fuca1-2]Galb1-4Glcb-COOH	92.2 ± 90.1	119.4 ± 135.1	-0.25	0.84	0.623	0.770	80	9
A-type-6-COOH-[540]	GalNAca1-3[Fuca1-2]Galb1-4Glcb-COOH	104.4 ± 190.5	166.0 ± 272.1	-0.76	0.59	0.527	0.671	81	9
A-type-6-COOH-[1350]	GalNAca1-3[Fuca1-2]Galb1-4Glcb-COOH	147.4 ± 209.1	171.3 ± 228.1	-0.39	0.76	0.652	0.844	82	9
B type 1-[620]	Gal(_1-3)[Fuc(_1-2)]Gal(_1-3)GlcNAc(_-Sp	1625.2 ± 146.3	1473.9 ± 111.3	0.14	1.10	0.552	0.800	83	1
B-type-1-COOH-[135]	Gala1-3[Fuca1-2]Galb1-3GlcNAcb-COOH	988.3 ± 429.1	1915.9 ± 519.8	-0.96	0.52	7.3e-04	0.004	84	1
B-type-1-COOH-[540]	Gala1-3[Fuca1-2]Galb1-3GlcNAcb-COOH	402.3 ± 196.4	833.4 ± 121.5	-1.05	0.48	0.003	0.017	85	1
B-type-1-COOH-[1350]	Gala1-3[Fuca1-2]Galb1-3GlcNAcb-COOH	2.1 ± 0.0	2.1 ± 0.0	0.00	1.00	1.000	1.000	86	1
B type 2-[970]	Gal(_1-3)[Fuc(_1-2)]Gal(_1-4)GlcNAc(_-Sp	955.5 ± 127.0	652.3 ± 86.2	0.55	1.46	0.058	0.183	87	1
B-type-2-COOH-[540]	Gala1-3[Fuca1-2]Galb1-4GlcNAcb-COOH	29.6 ± 32.9	104.7 ± 54.2	-1.82	0.28	0.067	0.203	88	1

B-type-3-COOH-[100]	Gal_(1-3)[Fuc_(1-2)]Gal_(1-3)GalNAc-_-COOH	165.4 ± 94.6	117.7 ± 74.8	0.43	1.35	0.548	0.739	89	5
B-type-3-COOH-[500]	Gal_(1-3)[Fuc_(1-2)]Gal_(1-3)GalNAc-_-COOH	225.1 ± 208.3	181.2 ± 134.9	-0.40	0.76	0.460	0.692	90	4
B-type-3-COOH-[1000]	Gal_(1-3)[Fuc_(1-2)]Gal_(1-3)GalNAc-_-COOH	4.3 ± 4.6	2.1 ± 0.0	0.61	1.53	0.781	0.830	91	4
B-type-4-COOH-[135]	Gala1-3[Fuca1-2]Galb1-3GalNAcb-COOH	221.4 ± 197.4	149.9 ± 113.5	0.46	1.38	0.440	0.586	92	7
B-type-4-COOH-[540]	Gala1-3[Fuca1-2]Galb1-3GalNAcb-COOH	36.9 ± 36.6	64.2 ± 72.2	-0.54	0.69	0.455	0.535	93	7
B-type-4-COOH-[1350]	Gala1-3[Fuca1-2]Galb1-3GalNAcb-COOH	125.2 ± 270.6	129.1 ± 217.3	-0.02	0.99	0.538	0.649	94	9
B-type-5-COOH-[540]	Gala1-3[Fuca1-2]Galb1-3Galb-COOH	952.5 ± 285.5	1192.6 ± 367.6	-0.32	0.80	0.242	0.511	95	1
B-type-5-COOH-[1350]	Gala1-3[Fuca1-2]Galb1-3Galb-COOH	542.7 ± 110.9	125.0 ± 33.1	2.12	4.34	2.8e-05	2.3e-04	96	1
B type 6-[920]	Gal(_1-3)[Fuc(_1-2)]Gal(_1-4)Glc(_-Sp	695.2 ± 158.9	864.8 ± 258.1	-0.32	0.80	0.299	0.566	97	1
B-type-6-COOH-[135]	Gala1-3[Fuca1-2]Galb1-4Glcb-COOH	186.8 ± 184.1	258.3 ± 223.9	-0.54	0.69	0.508	0.653	98	8
B-type-6-COOH-[540]	Gala1-3[Fuca1-2]Galb1-4Glcb-COOH	108.7 ± 119.1	182.7 ± 152.7	-0.81	0.57	0.272	0.477	99	6
B-type-6-COOH-[1350]	Gala1-3[Fuca1-2]Galb1-4Glcb-COOH	157.3 ± 245.8	100.3 ± 71.0	-0.02	0.99	0.482	0.672	100	8
2'F-B type 2-[220]	Gal(_1-3)[Fuc(_1-2)]Gal(_1-4)[Fuc(_1-3)]GlcNAc(_-Sp	131.0 ± 80.3	276.0 ± 118.5	-1.07	0.47	0.067	0.202	101	1
2'F-B type 2-[760]	Gal(_1-3)[Fuc(_1-2)]Gal(_1-4)[Fuc(_1-3)]GlcNAc(_-Sp	611.0 ± 212.8	623.5 ± 113.5	-0.03	0.98	0.930	1.000	102	1
H2-[430]	Fuc(_1-2)Gal(_1-4)GlcNAc(_1-3)Gal(_1-4)GlcNAc(_1-Sp	9912.6 ± 1602.1	738.4 ± 267.4	3.75	13.42	5.4e-23	1.8e-21	103	1
H3-[190]	Fuc(_1-2)Gal(_1-4)GlcNAc(_1-3)Gal(_1-4)GlcNAc(_1-3)Gal(_1-4)GlcNAc(_1-Sp	3923.7 ± 1074.1	123.0 ± 95.0	5.00	31.90	7.4e-21	2.0e-19	104	1
Tri-AN3-[1080]	GalNAc(_1-3)[Fuc(_1-2)]Gal(_-Sp	504.2 ± 160.5	72.5 ± 59.9	2.80	6.95	3.4e-06	3.2e-05	105	1
LeA-[950]	Gal(_1-3)[Fuc(_1-4)]GlcNAc(_-Sp	3084.8 ± 678.9	333.0 ± 195.3	3.21	9.26	1.7e-10	2.7e-09	106	1
Lex-[810]	Gal(_1-4)[Fuc(_1-3)]GlcNAc(_-Sp	773.1 ± 382.1	421.6 ± 118.0	0.87	1.83	0.014	0.059	107	1
Gala3Lex-[620]	Gal(_1-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_-Sp	1297.6 ± 273.6	1295.8 ± 342.5	0.00	1.00	0.994	1.000	108	1
Di-Lex-[410]	Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-Sp	11078.4 ± 1219.7	1198.2 ± 369.2	3.21	9.25	1.9e-20	5.0e-19	109	1
Ley-Lex-[350]	Fuc(_1-2)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_-Sp	1003.6 ± 256.4	156.9 ± 111.5	2.68	6.40	6.6e-09	9.5e-08	110	1
LeALex-[350]	Gal(_1-3)[Fuc(_1-4)]GlcNAc(_1-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_-Sp	7182.7 ± 1324.0	1643.8 ± 444.5	2.13	4.37	1.6e-13	2.9e-12	111	1
Lex-LeA-[410]	Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-3)Gal(_1-3)[Fuc(_1-4)]GlcNAc(_-Sp	10053.2 ± 1476.1	1552.0 ± 467.8	2.70	6.48	1.8e-17	4.1e-16	112	1
Lec-LeX-[570]	Gal(_1-3)GlcNAc(_1-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_-Sp	9777.0 ± 1035.3	5791.8 ± 801.6	0.76	1.69	8.1e-04	0.005	113	1
Tri-Lex-[430]	Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-3)Gal(_1-4) [Fuc(_1-3)]GlcNAc(_1-Sp	4309.6 ± 1147.4	865.2 ± 97.2	2.32	4.98	1.6e-14	2.9e-13	114	1
Ley-Di-Lex-[510]	Fuc(_1-2)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-3) Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-Sp	6855.6 ± 1108.5	175.1 ± 59.8	5.29	39.14	1.2e-28	1.1e-26	115	1
aMan-[840]	Man(_-S6	4289.9 ± 554.6	534.8 ± 81.8	3.00	8.02	3.9e-19	9.4e-18	116	1
(Man)3-[410]	Man(_1-6)[Man(_1-3)]Man(_-S6	16813.9 ± 5976.6	609.0 ± 144.3	4.79	27.61	4.5e-27	2.4e-25	117	1
(Man)3-[1300]	Man(_1-6)[Man(_1-3)]Man(_-S6	99185.0 ± 19602.8	750.9 ± 83.9	7.05	132.08	3.4e-34	5.8e-31	118	1
(Man)3-[1730]	Man(_1-6)[Man(_1-3)]Man(_-S6	49510.1 ± 9837.8	493.3 ± 95.7	6.65	100.37	2.7e-33	2.3e-30	119	1

11-[50]	Man(_1-6)[Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	1832.9 ± 758.6	1512.6 ± 284.1	0.28	1.21	0.272	0.533	120	1
11-[150]	Man(_1-6)[Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	1019.7 ± 271.1	582.5 ± 256.4	0.81	1.75	0.011	0.047	121	1
11-[500]	Man(_1-6)[Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	12594.7 ± 8441.2	2176.1 ± 443.2	2.53	5.79	1.8e-04	0.001	122	1
11-[750]	Man(_1-6)[Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	20588.4 ± 729.2	1810.5 ± 464.4	3.51	11.37	3.5e-22	1.1e-20	123	1
11-[780]	Man(_1-6)[Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	170643.1 ± 27191.8	19623.6 ± 5353.7	3.12	8.70	1.3e-18	3.1e-17	124	1
6-[50]	GlcNAc(_1-2)Man(_1-6)[GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	11257.7 ± 1578.5	7246.3 ± 554.0	0.64	1.55	0.004	0.021	125	1
6-[150]	GlcNAc(_1-2)Man(_1-6)[GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	31883.5 ± 10140.8	4930.4 ± 720.6	2.69	6.47	6.5e-17	1.4e-15	126	1
6-[500]	GlcNAc(_1-2)Man(_1-6)[GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	67942.6 ± 15613.2	27025.4 ± 2256.8	1.33	2.51	2.7e-07	3.2e-06	127	1
6-[750]	GlcNAc(_1-2)Man(_1-6)[GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	110486.0 ± 28681.9	10784.6 ± 1443.4	3.36	10.24	2.2e-20	5.6e-19	128	1
6-[810]	GlcNAc(_1-2)Man(_1-6)[GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	205926.8 ± 30867.4	11671.1 ± 2123.3	4.14	17.64	5.2e-24	1.8e-22	129	1
6-[1000]	GlcNAc(_1-2)Man(_1-6)[GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	55000.2 ± 5167.2	30735.6 ± 2459.6	0.84	1.79	3.8e-04	0.002	130	1
10-[50]	Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4) GlcNAc(_1-4)GlcNAc(_1-Sp	48.5 ± 65.7	17.5 ± 21.9	0.76	1.70	0.547	0.631	131	2
10-[150]	Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4) GlcNAc(_1-4)GlcNAc(_1-Sp	8775.4 ± 3602.5	7452.6 ± 947.1	0.24	1.18	0.305	0.572	132	1
10-[500]	Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4) GlcNAc(_1-4)GlcNAc(_1-Sp	34796.4 ± 10072.7	11641.8 ± 2383.8	1.58	2.99	4.7e-09	6.9e-08	133	1
10-[540]	Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4) GlcNAc(_1-4)GlcNAc(_1-Sp	40295.3 ± 11395.4	12486.4 ± 3339.5	1.69	3.23	1.1e-09	1.6e-08	134	1
10-[750]	Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	1073.6 ± 169.3	192.0 ± 93.2	2.48	5.59	2.1e-09	3.2e-08	135	1
10-[1000]	Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4) GlcNAc(_1-4)GlcNAc(_1-Sp	32576.2 ± 1860.5	16710.2 ± 1307.2	0.96	1.95	4.4e-05	3.5e-04	136	1
9-[50]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2) Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	2630.1 ± 565.2	1789.5 ± 241.6	0.56	1.47	0.018	0.069	137	1
9-[160]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2) Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	2151.3 ± 397.0	2333.5 ± 400.5	-0.12	0.92	0.611	0.853	138	1
9-[510]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2) Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	5199.3 ± 550.9	8103.6 ± 1114.4	-0.64	0.64	0.004	0.021	139	1
9-[730]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2) Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	5732.8 ± 491.2	7521.9 ± 715.1	-0.39	0.76	0.071	0.211	140	1

9-[950]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2) Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	2420.1 ± 795.3	7174.6 ± 652.2	-1.57	0.34	3.9e-09	5.7e-08	141	1
9-[970]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2) Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	4839.1 ± 1787.8	5158.4 ± 719.3	-0.09	0.94	0.684	0.913	142	1
8-[50]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)] Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;Gal(_1-4)GlcNAc(_1-2)Man(_1-6) [Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4) GlcNAc(_1-Sp;Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6) [Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	6688.8 ± 1056.4	5749.0 ± 415.7	0.22	1.16	0.310	0.579	143	1
8-[140]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)] Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;Gal(_1-4)GlcNAc(_1-2)Man(_1-6) [Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4) GlcNAc(_1-Sp;Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6) [Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	4499.9 ± 895.7	4364.2 ± 1111.8	0.04	1.03	0.843	1.000	144	1
8-[490]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)] Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;Gal(_1-4)GlcNAc(_1-2)Man(_1-6) [Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4) GlcNAc(_1-Sp;Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6) [Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	684.0 ± 190.0	830.1 ± 298.7	-0.28	0.82	0.367	0.641	145	1
8-[760]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)] Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;Gal(_1-4)GlcNAc(_1-2)Man(_1-6) [Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4) GlcNAc(_1-Sp;Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6) [Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	15735.8 ± 2378.3	7169.3 ± 1944.2	1.13	2.19	4.0e-06	3.8e-05	146	1
8-[1000]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)] Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;Gal(_1-4)GlcNAc(_1-2)Man(_1-6) [Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4) GlcNAc(_1-Sp;Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6) [Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	73225.0 ± 25934.7	43446.1 ± 9877.1	0.75	1.69	0.003	0.014	147	1
7-[50]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)] Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;GlcNAc(_1-2)Man(_1-6) [Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-	1247.4 ± 1225.3	1669.3 ± 1593.0	-0.46	0.73	0.139	0.340	148	2

	2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp								
7-[140]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)] Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	1953.2 ± 512.8	1592.9 ± 375.3	0.29	1.23	0.229	0.494	149	1
7-[760]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)] Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	29.9 ± 20.7	2.1 ± 0.0	3.86	14.53	0.011	0.047	150	1
7-[970]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)] Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;GlcNAc(_1-2)Man(_1-6)[Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp;Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-2)Man(_1-6)[GlcNAc(_1-2)Man(_1-3)]Man(_1-4)GlcNAc(_1-4)GlcNAc(_1-Sp	182058.5 ± 42200.1	30621.7 ± 8787.1	2.57	5.95	4.8e-15	9.5e-14	151	1
GM1-[460]	Neu5Ac(_2-3)[Gal(_1-3)GalNAc(_1-4)]Gal(_1-4)Glc(_-Sp	505.4 ± 293.0	352.1 ± 432.9	1.41	2.65	0.357	0.466	152	2
GM2-[190]	Neu5Ac(_2-3)[GalNAc(_1-4)]Gal(_1-4)Glc(_-Sp	2.1 ± 0.0	2.1 ± 0.0	0.00	1.00	1.000	1.000	153	1
GM2-[460]	Neu5Ac(_2-3)[GalNAc(_1-4)]Gal(_1-4)Glc(_-Sp	29.5 ± 32.9	7.5 ± 11.6	1.99	3.96	0.145	0.355	154	1
GM2-[590]	Neu5Ac(_2-3)[GalNAc(_1-4)]Gal(_1-4)Glc(_-Sp	2431.1 ± 342.6	347.1 ± 82.4	2.81	7.00	3.4e-16	7.1e-15	155	1
GM2-[1190]	Neu5Ac(_2-3)[GalNAc(_1-4)]Gal(_1-4)Glc(_-Sp	2.1 ± 0.0	2.1 ± 0.0	0.00	1.00	1.000	1.000	156	1
CT|Sda-[350]	Neu5Ac(_2-3)[GalNAc(_1-4)]Gal(_1-4)GlcNAc(_-Sp	601.1 ± 91.5	628.4 ± 178.2	-0.06	0.96	0.846	1.000	157	1
GD1a-[110]	Neu5Ac(_2-3)[Neu5Ac(_2-3)Gal(_1-3)GalNAc(_1-4)]Gal(_1-4)Glc(_-Sp	1394.8 ± 294.6	707.7 ± 264.5	0.98	1.97	7.2e-04	0.004	158	1
3'S-Di-LeA-[160]	Neu5Ac(_2-3)Gal(_1-3)[Fuc(_1-4)]GlcNAc(_1-3)Gal(_1-3)[Fuc(_1-4)]GlcNAc(_1-Sp	295.4 ± 103.9	165.0 ± 96.7	0.84	1.79	0.126	0.325	159	1
3'SLeA-Lex-[570]	Neu5Ac(_2-3)Gal(_1-3)[Fuc(_1-4)]GlcNAc(_1-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_-Sp	2812.1 ± 1011.5	784.7 ± 538.4	1.84	3.58	0.015	0.060	160	1
3'S-Tri-LeX-[160]	Neu5Ac(_2-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-3)Gal(_1-4)[Fuc(_1-3)]GlcNAc(_1-Sp	196.9 ± 75.6	556.5 ± 75.5	-1.50	0.35	8.8e-04	0.005	161	1
3'SLec-[430]	Neu5Ac(_2-3)Gal(_1-3)GlcNAc(_-Sp	710.7 ± 198.3	2229.6 ± 250.4	-1.65	0.32	2.3e-08	3.1e-07	162	1
GM3-[190]	Neu5Ac(_2-3)Gal(_1-4)Glc(_-Sp	2.1 ± 0.0	4.7 ± 5.1	-1.19	0.44	0.520	0.770	163	1
GM3-[540]	Neu5Ac(_2-3)Gal(_1-4)Glc(_-Sp	1273.7 ± 404.5	1493.6 ± 244.2	-0.23	0.85	0.365	0.641	164	1
GM3-[1190]	Neu5Ac(_2-3)Gal(_1-4)Glc(_-Sp	11.2 ± 15.8	4.7 ± 5.1	1.25	2.38	0.451	0.721	165	1

3'SLDN-[460]	Neu5Ac(_2-3)GalNAc(_1-4)GlcNAc(_-Sp	15200.9 ± 6012.6	7571.4 ± 1245.4	1.01	2.01	5.1e-05	4.0e-04	166	1
3'S-Di-Lec-[270]	Neu5Ac(_2-3)Gal(_1-3)GlcNAc(_1-3)Gal(_1-3)GlcNAc(_-Sp	1244.1 ± 289.7	2491.2 ± 430.1	-1.00	0.50	1.0e-04	7.6e-04	167	1
3'SLecLN-[860]	Neu5Ac(_2-3)Gal(_1-3)GlcNAc(_1-3)Gal(_1-4)GlcNAc(_-Sp	1566.7 ± 349.9	2722.6 ± 527.5	-0.80	0.58	0.001	0.007	168	1
3'SLN-Lec-[410]	Neu5Ac(_2-3)Gal(_1-4)GlcNAc(_1-3)Gal(_1-3)GlcNAc(_-Sp	319.0 ± 55.9	589.6 ± 94.8	-0.89	0.54	0.023	0.088	169	1
3'S-Di-LN-[1350]	Neu5Ac(_2-3)Gal(_1-4)GlcNAc(_1-3)Gal(_1-4)GlcNAc(_1-Sp	1071.2 ± 266.4	989.1 ± 301.0	0.11	1.08	0.681	0.911	170	1
3'STri-LN-[160]	Neu5Ac(_2-3)Gal(_1-4)GlcNAc(_1-3)Gal(_1-4)GlcNAc(_1-3)Gal(_1-4)GlcNAc(_1-Sp	1166.0 ± 367.1	453.5 ± 109.7	1.36	2.57	2.2e-05	1.9e-04	171	1
3'SLec (Gc)-[350]	Neu5Gc(_2-3)Gal(_1-3)GlcNAc(_-Sp	478.7 ± 102.5	547.5 ± 255.9	-0.19	0.87	0.605	0.848	172	1
3'SL (Gc)-[490]	Neu5Gc(_2-3)Gal(_1-4)Glc(_-Sp	3936.1 ± 356.3	8183.1 ± 853.7	-1.06	0.48	7.3e-06	6.7e-05	173	1
3'SLN (Gc)-[570]	Neu5Gc(_2-3)Gal(_1-4)GlcNAc(_-Sp	682.3 ± 251.1	1088.6 ± 383.8	-0.67	0.63	0.028	0.102	174	1
3'-KDNLec-[760]	Kdn(_2-3)Gal(_1-3)GlcNAc(_-Sp	381.9 ± 56.8	481.5 ± 114.5	-0.33	0.79	0.388	0.661	175	1
3'KDNLN-[510]	Kdn(_2-3)Gal(_1-4)GlcNAc(_-Sp	603.1 ± 250.0	223.9 ± 65.6	1.43	2.69	8.8e-04	0.005	176	1
6'SL-[620]	Neu5Ac(_2-6)Gal(_1-4)Glc(_-Sp	1661.2 ± 422.9	3952.3 ± 860.8	-1.25	0.42	1.7e-06	1.7e-05	177	1
6'SL-[760]	Neu5Ac(_2-6)Gal(_1-4)Glc(_-Sp	5373.5 ± 679.4	3641.2 ± 634.1	0.56	1.48	0.012	0.051	178	1
6'SLN-[350]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_-Sp	5672.0 ± 811.0	7152.8 ± 1159.5	-0.33	0.79	0.128	0.326	179	1
6'SLN (Gc)-[430]	Neu5Gc(_2-6)Gal(_1-4)GlcNAc(_-Sp	94.7 ± 32.4	111.7 ± 56.2	-0.24	0.85	0.751	0.952	180	1
6'SLDN-[620]	Neu5Ac(_2-6)GalNAc(_1-4)GlcNAc(_-Sp	38.2 ± 48.2	36.2 ± 27.1	0.08	1.05	0.943	1.000	181	1
6'SLN-Lec-[300]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-3)Gal(_1-3)GlcNAc(_-Sp	3305.5 ± 868.6	1038.0 ± 185.5	1.67	3.18	1.1e-09	1.7e-08	182	1
6'S-Di-LN-[300]	Neu5Ac(_2-6)Gal(_1-4)GlcNAc(_1-3)Gal(_1-4)GlcNAc(_1-Sp	428.9 ± 73.6	335.1 ± 65.1	0.36	1.28	0.383	0.654	183	1
GD2-[460]	Neu5Ac(_2-8)Neu5Ac(_2-3)[GalNAc(_1-4)]Gal(_1-4)Glc(_-Sp	627.0 ± 250.8	1026.2 ± 322.1	-0.71	0.61	0.025	0.093	184	1
GD3-[320]	Neu5Ac(_2-8)Neu5Ac(_2-3)Gal(_1-4)Glc(_-Sp	633.1 ± 223.1	805.3 ± 154.6	-0.35	0.79	0.270	0.533	185	1
GT2-[160]	Neu5Ac(_2-8)Neu5Ac(_2-8)Neu5Ac(_2-3)[GalNAc(_1-4)]Gal(_1-4)Glc(_-Sp	853.6 ± 84.5	2022.6 ± 579.6	-1.24	0.42	1.0e-05	9.2e-05	186	1
GT3-[510]	Neu5Ac(_2-8)Neu5Ac(_2-8)Neu5Ac(_2-3)Gal(_1-4)Glc(_-Sp	174.5 ± 98.2	161.5 ± 31.5	0.11	1.08	0.852	1.000	187	1
GQ2-[140]	Neu5Ac(_2-8)Neu5Ac(_2-8)Neu5Ac(_2-8)Neu5Ac(_2-3)[GalNAc(_1-4)]Gal(_1-4)Glc(_-Sp	1778.3 ± 395.9	2855.1 ± 1291.1	-0.68	0.62	0.013	0.055	188	1
(Galf)4-[<8]	Galf(_1-5)Galf(_1-5)Galf(_1-5)Galf(_-S8	1544.4 ± 152.4	740.6 ± 349.5	1.06	2.09	3.1e-04	0.002	189	1
Mana-sp-[20]	Mana-sp	284.5 ± 56.7	428.0 ± 136.6	-0.56	0.68	0.270	0.472	190	5
Mana-sp-[50]	Mana-sp	171.9 ± 208.9	310.2 ± 329.7	-1.01	0.50	0.197	0.370	191	5
Mana-sp-[300]	Mana-sp	1119.6 ± 791.5	1108.0 ± 613.0	-0.11	0.93	0.541	0.747	192	5
Mana-sp-[500]	Mana-sp	2048.1 ± 973.1	871.7 ± 440.4	1.21	2.32	0.004	0.015	193	5
Mana-sp-[700]	Mana-sp	42.9 ± 64.6	24.9 ± 37.2	0.42	1.34	0.374	0.511	194	4
Mana-C4-COOH-[25]	Mana-C4-COOH	32.0 ± 22.4	57.7 ± 44.8	-0.62	0.65	0.268	0.343	195	4
Mana-C4-COOH-[50]	Mana-C4-COOH	54.1 ± 47.9	47.3 ± 24.3	-0.06	0.96	0.684	0.908	196	2
Mana-C4-COOH-[150]	Mana-C4-COOH	49.9 ± 48.1	76.9 ± 70.1	-0.76	0.59	0.466	0.699	197	5

Mana-C4-COOH-[500]	Mana-C4-COOH	2514.1 ± 1204.0	282.0 ± 113.1	3.11	8.62	1.1e-11	1.9e-10	198	4
Mana-C4-COOH-[Max]	Mana-C4-COOH	30136.1 ± 36591.7	637.3 ± 798.5	6.00	64.15	3.4e-07	3.3e-06	199	9
Mana-PEG4-COOH-[25]	Mana-PEG4-COOH	192.2 ± 158.3	217.6 ± 141.4	-0.26	0.83	0.586	0.761	200	5
Mana-PEG4-COOH-[50]	Mana-PEG4-COOH	326.7 ± 192.9	321.8 ± 136.4	-0.09	0.94	0.551	0.802	201	5
Mana-PEG4-COOH-[150]	Mana-PEG4-COOH	141.2 ± 76.5	150.7 ± 81.4	-0.11	0.93	0.748	0.931	202	5
Mana-PEG4-COOH-[500]	Mana-PEG4-COOH	877.9 ± 290.3	155.9 ± 104.6	2.69	6.45	3.5e-06	3.1e-05	203	5
L-Mana-C8-COOH-[25]	L-Mana-C8-COOH	275.7 ± 285.3	352.4 ± 355.4	-0.23	0.86	0.370	0.637	204	5
L-Mana-C8-COOH-[50]	L-Mana-C8-COOH	198.7 ± 126.6	267.8 ± 151.7	-0.45	0.73	0.440	0.687	205	4
L-Mana-C8-COOH-[150]	L-Mana-C8-COOH	201.7 ± 168.1	146.1 ± 69.7	0.23	1.17	0.664	0.796	206	5
L-Mana-C8-COOH-[500]	L-Mana-C8-COOH	593.4 ± 391.3	556.8 ± 235.6	-0.18	0.88	0.290	0.514	207	4
L-Mana-C8-COOH-[Max]	L-Mana-C8-COOH	901.4 ± 1209.7	1284.8 ± 1688.4	-0.37	0.77	0.511	0.615	208	9
Man4-C5-N3-[20]	Man4-C5-N3	172.3 ± 168.2	281.4 ± 238.1	-0.60	0.66	0.380	0.524	209	4
Man4-C5-N3-[100]	Man4-C5-N3	780.1 ± 715.1	798.7 ± 735.7	0.23	1.17	0.777	0.932	210	5
Man4-C5-N3-[300]	Man4-C5-N3	1611.6 ± 580.6	498.5 ± 154.9	1.69	3.23	7.4e-04	0.004	211	5
Man4-C5-N3-[720]	Man4-C5-N3	13924.7 ± 16704.7	419.1 ± 493.8	4.12	17.36	0.009	0.035	212	4
Man4-C5-N3-[740]	Man4-C5-N3	3000.9 ± 3749.9	250.7 ± 323.9	2.49	5.62	0.265	0.297	213	4
Man4-P6-N3-[25]	Man4-P6-N3	971.6 ± 1305.3	815.5 ± 1066.2	0.18	1.13	0.558	0.776	214	5
Man4-P6-N3-[50]	Man4-P6-N3	13054.1 ± 11588.0	2032.0 ± 2213.5	2.03	4.09	0.171	0.228	215	4
Man4-P6-N3-[150]	Man4-P6-N3	4597.5 ± 6719.9	529.6 ± 778.6	3.15	8.90	0.003	0.013	216	4
Man4-P6-N3-[500]	Man4-P6-N3	5337.3 ± 2918.7	239.3 ± 120.7	4.32	19.96	1.6e-10	2.5e-09	217	5
Man4-P9-N3-[25]	Man4-P9-N3	369.0 ± 625.1	319.7 ± 422.9	-0.87	0.55	0.182	0.400	218	4
Man4-P9-N3-[50]	Man4-P9-N3	686.9 ± 849.5	657.7 ± 789.8	0.63	1.55	0.422	0.654	219	5
Man4-P9-N3-[150]	Man4-P9-N3	1374.0 ± 550.3	1046.4 ± 435.6	0.38	1.31	0.396	0.488	220	3
Man4-P9-N3-[500]	Man4-P9-N3	18341.1 ± 17487.1	981.3 ± 757.1	3.25	9.48	0.200	0.200	221	5
Man4-P12-N3-[25]	Man4-P12-N3	2736.2 ± 1221.0	2993.9 ± 1069.6	-0.17	0.89	0.458	0.657	222	4
Man4-P12-N3-[50]	Man4-P12-N3	576.0 ± 340.0	409.9 ± 220.3	0.43	1.35	0.336	0.532	223	4
Man4-P12-N3-[150]	Man4-P12-N3	5218.2 ± 4087.9	679.6 ± 495.7	2.80	6.95	3.2e-08	3.9e-07	224	5
Man4-P12-N3-[500]	Man4-P12-N3	22120.5 ± 20199.5	1515.3 ± 1272.1	3.70	12.96	2.9e-17	6.2e-16	225	4
L-Glca-C8-COOH-[25]	L-Glca-C8	315.0 ± 124.3	386.4 ± 150.3	-0.29	0.82	0.541	0.772	226	3
L-Glca-C8-COOH-[50]	L-Glca-C8	566.0 ± 606.3	482.8 ± 366.8	0.19	1.14	0.288	0.500	227	4
L-Glca-C8-COOH-[150]	L-Glca-C8	522.1 ± 87.0	505.7 ± 139.3	0.07	1.05	0.452	0.706	228	5
L-Glca-C8-COOH-[500]	L-Glca-C8	257.7 ± 256.4	249.9 ± 226.4	-0.19	0.88	0.706	0.857	229	5

L-Glcb-C8-COOH-[25]	L-Glcb-C8	3061.3 ± 2342.1	3639.9 ± 1788.5	-0.43	0.74	0.169	0.293	230	5
L-Glcb-C8-COOH-[50]	L-Glcb-C8	302.6 ± 262.3	370.6 ± 294.5	-0.49	0.71	0.293	0.511	231	4
L-Glcb-C8-COOH-[150]	L-Glcb-C8	582.2 ± 367.0	612.7 ± 285.6	-0.14	0.91	0.486	0.706	232	5
L-Glcb-C8-COOH-[500]	L-Glcb-C8	574.8 ± 453.2	436.5 ± 314.6	0.36	1.29	0.403	0.609	233	5
L-Rhaa-sp-[25]	L-Rhaa-sp	586.3 ± 353.9	919.2 ± 404.4	-0.70	0.61	0.095	0.236	234	3
L-Rhaa-sp-[50]	L-Rhaa-sp	474.5 ± 582.1	843.2 ± 1075.5	-0.73	0.60	0.183	0.277	235	3
L-Rhaa-sp-[150]	L-Rhaa-sp	1510.3 ± 1754.0	2726.8 ± 2901.2	-1.09	0.47	0.001	0.006	236	4
L-Rhaa-sp-[500]	L-Rhaa-sp	1028.1 ± 739.4	2242.4 ± 1597.1	-0.85	0.56	0.250	0.250	237	4
Man-6-P-P3-[25]	Man-6-P-P3	45.4 ± 53.3	19.0 ± 21.3	0.42	1.34	0.537	0.670	238	4
Man-6-P-P3-[150]	Man-6-P-P3	477.0 ± 656.4	613.1 ± 952.4	0.04	1.03	0.607	0.754	239	4
Man-6-P-P3-[500]	Man-6-P-P3	6549.2 ± 3539.3	5465.9 ± 2012.3	0.20	1.15	0.534	0.661	240	3
Man-6-P-P3-[1000]	Man-6-P-P3	48035.0 ± 27354.8	9224.6 ± 5076.2	2.33	5.04	3.2e-08	4.0e-07	241	4
L-Gala-C8-COOH-[20]	L-Gala-C8-COOH	54.2 ± 56.0	44.3 ± 26.3	-0.34	0.79	0.502	0.732	242	5
L-Gala-C8-COOH-[50]	L-Gala-C8-COOH	111.1 ± 72.3	124.1 ± 60.7	-0.25	0.84	0.611	0.824	243	4
L-Gala-C8-COOH-[150]	L-Gala-C8-COOH	1713.6 ± 1543.4	162.8 ± 168.7	3.25	9.50	0.031	0.080	244	4
L-Gala-C8-COOH-[500]	L-Gala-C8-COOH	9006.8 ± 2326.6	139.0 ± 92.1	6.27	76.92	1.7e-12	3.0e-11	245	4
L-Gala-C8-COOH-[Max]	L-Gala-C8-COOH	67764.9 ± 42191.5	876.1 ± 401.9	6.19	72.82	4.2e-30	5.6e-28	246	5
L-Galb-C8-COOH-[20]	L-Galb-C8-COOH	54.7 ± 34.5	43.9 ± 29.9	0.30	1.23	0.768	0.937	247	5
L-Galb-C8-COOH-[50]	L-Galb-C8-COOH	49.4 ± 41.6	48.7 ± 6.4	-0.46	0.72	0.334	0.554	248	4
L-Galb-C8-COOH-[150]	L-Galb-C8-COOH	600.4 ± 1051.1	136.4 ± 286.2	1.62	3.08	0.298	0.356	249	9
L-Galb-C8-COOH-[500]	L-Galb-C8-COOH	16476.6 ± 19451.5	572.5 ± 685.2	4.82	28.32	1.8e-04	0.001	250	9
L-Galb-C8-COOH-[Max]	L-Galb-C8-COOH	32499.6 ± 46651.4	915.5 ± 1298.6	5.33	40.29	9.6e-06	7.0e-05	251	9
Galb-C4-COOH-[20]	Galb-C4-COOH	1020.4 ± 1675.8	758.1 ± 1285.0	-0.45	0.73	0.348	0.536	252	9
Galb-C4-COOH-[50]	Galb-C4-COOH	874.8 ± 1293.2	810.5 ± 1223.6	0.79	1.72	0.512	0.688	253	9
Galb-C4-COOH-[150]	Galb-C4-COOH	796.7 ± 813.0	857.8 ± 852.2	0.31	1.24	0.532	0.700	254	9
Galb-C4-COOH-[500]	Galb-C4-COOH	771.2 ± 1103.7	939.0 ± 1442.8	0.02	1.02	0.583	0.745	255	9
Galb-C4-COOH-[Max]	Galb-C4-COOH	579.7 ± 779.5	577.8 ± 742.0	-0.14	0.91	0.541	0.742	256	9
Glcb-C4-COOH-[20]	Glcb-C4-COOH	855.8 ± 1088.3	743.6 ± 933.9	-0.48	0.71	0.582	0.747	257	9
Glcb-C4-COOH-[50]	Glcb-C4-COOH	546.0 ± 809.8	468.4 ± 699.7	0.58	1.50	0.541	0.685	258	9
Glcb-C4-COOH-[150]	Glcb-C4-COOH	935.8 ± 1127.5	692.1 ± 838.0	1.26	2.39	0.219	0.390	259	9
Glcb-C4-COOH-[500]	Glcb-C4-COOH	7329.4 ± 9757.5	672.7 ± 966.7	3.62	12.29	0.015	0.043	260	9

L-dimannan-(a-COOH-[500]	L-Man(a1-2)-L-Man(a-COOH	430.9 ± 409.2	367.9 ± 354.8	0.56	1.47	0.457	0.625	261	9
AzOH	AzOH	152.2 ± 112.3	180.9 ± 155.8	-0.12	0.92	0.559	0.731	262	14
blank	blank	2412.0 ± 1635.4	2399.3 ± 1874.2	0.35	1.27	0.206	0.392	263	8